PROPOSED CURRICULUM

(To be effective from Fall-2021 Semester)

DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING (CSE)

UNDERGRADUATE PROGRAM

Bachelor of Science in Computer Science and Engineering
(B.Sc. in CSE)

BANGLADESH ARMY INTERNATIONAL
UNIVERSITY OF SCIENCE & TECHNOLOGY

Cumilla Cantonment, Cumilla, Bangladesh.

PREFACE

Bangladesh Army International University of Science & Technology (BAIUST) offers various programs in engineering field. Department of Computer Science and Engineering (CSE) of BAIUST proposed this curriculum for CSE undergraduate students. This curriculum will guide CSE students in their four years of undergraduate program. Faculty members of CSE department will also find it valuable as their course reference.

This curriculum is divided into several chapters where it will provide general information about BAIUST, its historical background, faculties and departments. Different aspects of the course system, such as rules and regulations relating to admission, grading system, requirement for degrees will be available here. It also includes the course requirements, course objectives, detailed course outline and courses offered in different terms.

Importance of Computer Science and Engineering for the development and progress of society, and the need for specialists in this field has become obvious in our daily life. The University acknowledges this importance and has therefore introduced Department of Computer Science and Engineering by giving all academic, financial and administrative support to the department. In this regard the department prepares this curriculum including departmental and non-departmental courses for CSE students. Some fundamental computer courses are updated as those are compulsory in school level. As per the expert's opinion number of total courses is reduced to put more focus on a specific subject. Total credit of this program adjusted based on the UGC Curriculum Guideline for B.Sc. in Computer Science & Engineering published in 2019. Students will get more chances to do project based works as Integrated Project & Capstone Project are introduced.

The current curriculum of CSE Program of BAIUST was approved by UGC in 2015. The proposed curriculum follows the OBE (Outcome Based Education) format which is shared by UGC in 2020. Academic committee of CSE department along with external subject expert Dr. Md. Shahidur Rahman, Professor, CSE, SUST and Dr. Md. Mokammel Haque, Professor, CSE, CUET, revised existing curriculum. Department will place this proposed curriculum before the academic council, BAIUST for necessary approval. With necessary approval from UGC, this curriculum will be applicable for all students admitted in Fall-2021 and onwards.

As per UGC policy all engineering curriculum needs to be revised after every three years. CSE department will made necessary modification time to time.

Cumilla, Bangladesh
May 2021

Md. Asaduzzaman Khan
Associate Professor
Head of the Department
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CERTIFICATE

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CHAPTER 1

GENERAL INFORMATION

1.1 BAIUST in brief

Bangladesh Army International University Science & Technology (BAIUST), a private University of Bangladesh Armed Force, commenced its academic session on 14 February 2015 as a spring center of excellence for higher tanning, research and development in science, engineering, technology, business and humanities. The University has been raised in full compliance with the instruction issued by them and the Private University Act 2010.

BAIUST is located at a secured and serene environment of Cumilla Cantonment, adjacent to Dhaka Chittagong Highway. It offers highly secured and academic friendly environment. BAIUST permanent location has been selected at Syedpur, Adarsha Sadar, Cumilla and its Development Project Plan (DPP) is finalized. Hopefully, it will encompass all components to make the University a hub of teaching, learning and research.

BAIUST started its academic journey with Computer Science & Engineering (CSE) and Electrical & Electronic Engineering (EEE) as pioneer programs from Spring, 2015.

The BBA program under the School of Business was launched from Fall, 2015. The University has opened Civil Engineering Department under Faculty of Civil Engineering and Department of English under the Faculty of Science & Humanities from the spring 2016. Finally, the LLB program under the Department of Law. Launched from the spring, 2019.

Recently BAIUST has automated library with thousands of book and journal. BAIUST has Hostel Facilities for both male and female students from the very beginning of its academic activities. Currently BAIUST has three hostels for male and female which are fully furnished.

Though BAIUST is mainly a science and technology university, it has also opened up a new horizon for the business and humanities studies. Its motto is Knowledge, Wisdom and Technology and emphasizes on quality rather than quantity.

1.2 Mission and Vision of BAIUST

1.2.1. Vision of BAIUST

The University aspires to transform into a center of excellence in Science, Engineering and Technology programs by providing innovative, multi-disciplinary courses and extensive research facilities to the young generation of the country and beyond. It endeavors to make the University a hub of knowledge and be recognized as a leading university of the country.

1.2.2. Mission of BAIUST

The mission of the BAIUST is to provide state of the art education and research in science, engineering and other relevant programs to develop human resources for the socio-economic wellbeing of the society.

1.3 Objectives

1. To offer need based programs in important disciplines of Science, Technology, Social Science, and Business for both undergraduate and graduate students.
2. To strive for excellence by providing quality teaching through wide varieties of instructional approaches.
3. To produce highly competent, and global standard manpower in Science and Technology; who will be proficient in communication skill, critical thinking and analytical abilities.
4. To generate adequate resources to support the University's instructions, research and extension pursuits.
5. To provide a timely, friendly, and accessible support system that promotes partnership and sense of shared responsibilities between students, faculty, alumni, and other stakeholders.
6. To strive for the gradual expansion of faculties, departments, and programs in accordance with the need, and in abidance with the Private University Act 2010.
7. To develop joint collaboration with similar universities, institutes and organizations, both at home and abroad and sign agreements and memorandum of understandings (MOU) to this effect.
8. To sensitize and infuse amongst the students and the graduates the core values of BAIUST i.e. Integrity, Openness, Intellectual Curiosity, Commitment to Excellence, Self-Discipline, Sense of Responsibility, Diversity and Respect for Others.
9. To carry out other necessary works of the University to meet the need of the time.

1.4 Core Values

Integrity: It is a commitment to uphold the highest standard of moral, ethical and professional conduct.

Openness: One should think clearly, study freely with open mind, willing to learn and examine the complex problems with reasoning.

Intellectual Curiosity: One must strive to broaden his/her knowledge, stretch intellectual horizon, and push beyond the comfort zones. As one gains knowledge, it will give him/her the power, authority and confidence.

Commitment to Excellence: The commitment to produce highly competent and global standard manpower on science and technology who will be proficient in communication skills, critical thinking, and analytical abilities. The aim is to search for truth and delivery of socially relevant, scientific and technological contributions. This is a place where hard work will be rewarded and talents will be nurtured.

Self-Discipline: One must impose self-discipline to abide by the rules, and regulations of the University and beyond.

Sense of Responsibility: All must inculcate the sense of responsibility, and discharge the assigned duties judiciously.

Diversity: In due course of time, the University aims to have students, faculty, and researchers from various disciplines, nationalities, and cultural backgrounds.

Respect for others: As members are from devise culture, race and religion, all must show respect for others.

1.5 Faculties

1.5.1 Faculty of Electrical & Computer Engineering (ECE):

Department of Electrical and Electronic Engineering (EEE)

Department of Computer Science and Engineering (CSE)

1.5.2 Faculty of Civil Engineering (CE):

Department of Civil Engineering (CE).

1.5.3 School of Business:

Department of Business Administration

1.5.4 Faculty of Science & Humanities:

Department of English.

Department of Law.

Department of Science and Humanities.

1.6 Admission Policy:

Admission wing is responsible for all admission matters under the supervision of the Registrar and through lateral cooperation with the faculties. Responsibilities of admission wing are as follows:

- Circulation, Admission form distribution and Short listing of candidates;
- Seat Plan, Invigilators duty plan and Conduct the Admission test;
- Result compilation and result publication;
- Maintenance of the Secrecy of the admission process and results; and
- Admission/ Enrollment/Registration of students

1.7 Eligibility of Students to Apply for Admission In Undergraduate Programs

1.7.1 Engineering Programs:

1.7.1. 1 SSC/Dakhil and HSC/Alim Examinations:

HSC/Alim or equivalent examination in Science Group with Mathematics, Physics and Chemistry scoring minimum GPA 3.0 for each. The sum total GPA of SSC/Dakhil and HSC/Alim should be minimum 7.0 (with additional subject).

1.7.1. 2 GCE Applicants:

O Level: Minimum C Grade in 5 subjects including, Mathematics, Physics and Chemistry (in the scale of A=5, B=4, C=3, D=2 and E=1).

A Level: Minimum C Grade in 2 subjects including Mathematics, Physics/ Chemistry.

The sum total of GPA in GCE A and O level should be 6.

The candidates with E grade in any subject will not be considered.

1.7.2 Non-Engineering Programs:

1.7.2.1 SSC/Dakhil and HSC/Alim Examination:

Minimum GPA of 3.00 in HSC or equivalent examination. The sum total of GPA of HSC/Alim and SSC/Dakhil should be minimum 6.5(with additional subject).

1.7.2.2 GCE Applicants:

O Level: Minimum C Grade in five subjects including, Mathematics, Physics and Chemistry (in the scale of A=5, B=4, C=3, D=2 and E=1).

A Level: Minimum C Grade in 2 subjects including Mathematics, Physics/ Chemistry.

The sum total of GPA in GCE A and O level should be 6.

The candidates with E grade in any subject will not be considered.

Note: 3% of the total seats are reserved for the children of Freedom Fighters with tuition fee waiver.

1.8 Syllabus for Admission Test:

1.8.1 Engineering Programs:

A Written and MCQ test will be conducted on mathematics, Physics, Chemistry and English (Comprehension and Functional) as per the syllabus of HSC. The test will be of 100 marks and 2 hour's duration.

The distribution of marks is given below:

SI.	**Subjects**	**Marks**
1.	Mathematics	40
2.	Physics	30
3.	Chemistry	20
4.	English	10
	Total	100

1.8.2 Non-Engineering Program:

Bachelor of Business Administration (BBA): A written and MCQ test will be conducted on Bangla, English (as per the syllabus of HSC), ICT, General Knowledge and Analytical Ability. The test will be of 100 marks and 2 hour's duration. The distribution of marks is given below for students of different majors.

1.9 Question Pattern:

Business		**Science**		**Arts**	
Bangla:	10	Bangla:	10	Bangla:	10
English:	30*	English:	30*	English:	30*
ICT:	10	ICT:	10	ICT:	10
Prin. of Business:	25	Math:	40	General Knowledge:	40
Accounting:	25	Analytical Ability:	10	Analytical Ability:	10

*Thirty marks for English will be divided into two parts i.e. MCQ-10 and Written-20. In the written part, students will answer two questions. The written questions will consist of an essay, paragraph on any recent issue. MCQ pattern will be followed in all other parts of the test.

1.9.1 Bachelor of Arts (Hons) in English:

A written and MCQ test will be conducted on Bangla, English (as per the syllabus of HSC) and General Knowledge. The test will be of 100 marks and 2-hour duration. The distribution of marks is given below:

SI.	**Subjects**	**Marks**
1.	English (MCQ & Descriptive)	65
2.	Bangla	25
3.	General Knowledge	10
	Total	**100**

1.9.2. Bachelor of Laws (LLB):

The test will be of 100 marks and 2 hour's duration. Total 100 Marks will be divided into two parts i.e. MCQ 50 marks and Written 50 marks.

The distribution of marks is given below:

SI.	**Category**	**Marks**
1.	History of Bangladesh	10
2.	General knowledge (Current BD Affairs)	15
3.	General knowledge (Current Int' | Affairs)	15
4.	Basic Legal and Constitutional Concepts	20
5.	English Grammar and Vocabulary	20
6.	English Paragraph Writing (within 200 words)	20
	Total	100

1.10. Final Selection:

Selection is exclusively based on merit in the Admission Test:

Selection Basis	Percentage
Descriptive and MCQ Admission Test	100%

1.11. Cancellation of Admission:

The cancellation of admission is not permissible. However, in special circumstances, such cancellation may be considered as per the following guidelines.

S.I	Description	Refundable
1.	Before commencement of the classes	100% of the Establishment Fee and 100% of the Semester Fee.
2.	Within 1st Week of commencement of the classes	The Caution money, 75% of the Establishment Fee and the Semester Fee.
3.	Within 2 Weeks of commencement of the classes	The Caution money, 50% of the Establishment Fee and the Semester Fee.
4.	After 2 Weeks of commencement of the classes	Caution money only.

1.12. Withdrawal Policy:

1.12.1. Term Withdrawal:

Following circumstances and rules will apply:

In case of severe sickness, which lead to missing of more than 40% classes (supported by requisite medical document), student(s), may be allowed to withdraw the term and repeat the whole term with re-registration with the regular term in the next academic term, subject to the approval of Academic Council, BAIUST. She/he will get a W grade in case of such term withdrawal.

If a student does not attend more than 40% classes without notification and prior approval, his/her term registration will be cancelled and His/her courses will be treated as "Incomplete". She/he will have to repeat the term with re-registration in the next academic session, subject to the approval of Academic Council, BAIUST. She/he will get an "I" grade in case of such term withdrawal.

However, if a student appears in more than 60% classes but doesn't appear the exam after collecting the Admit Card of Exam, she/he will get an "F" grade in such cases.

If a student does not attend classes for two consecutive terms without prior approval and is interested to continue from the next academic session, she/he will be penalized for each term and may be allowed to repeat the terms with re-registration, subject to the approval of Academic council, BAIUST. She/he will get an F grade in case of such withdrawal of terms.

However, she/he will have to complete the whole undergraduate program within 06 (Six) academic year from the date of his/her registration.

The mentor/student advisor must have to recommend such a term withdrawal.

1.12.2. Program Withdrawal:

Following circumstances and rules will apply:

Failure to secure/achieve minimum CGPA of 2.25 in two consecutive Level will also lead to withdrawal of the student from the program.

If a student does not attend classes for two consecutive terms without notification and prior approval, she/he will be permanently withdrawn from the program.

In case of Term Repeat, students repeating a particular term (for example L-1, T-1) and if she/he fails again in three or more courses in that term will require the approval of the Academic Council for further consideration.

However, she/he will have to complete the whole undergraduate program within 06(Six) academic year from the date of his/her registration. Otherwise, she/he will be withdrawn from the program.

The mentor/student advisor has to recommend such a program withdrawal.

1.12.3. Withdrawal On Own Accord:

Following circumstances and rules will apply:

1.12.4. Permanent Withdrawal:

A student who has already completed some course and has not performed satisfactorily may apply for a permanent withdrawal.

1.12.5. Temporary Withdrawal:

A Student, if she/he applies, may be allowed to withdraw temporarily from the program, subject to the approval of Academic Council of the University, but she/he has to complete the whole program within 06 (Six) academic years from the date of his/her registration.

Note: The mentor/student advisor must recommend such a withdrawal.

1.13. Conduct and Discipline

During the stay in the University all students are required to abide by the existing rules, regulations and code of conduct. Students are strictly forbidden to form or be members of student organizations or political party, club, society etc. Other than those set up by university

authority in order to enhance student's physical, intellectual, moral and ethical development. Zero tolerance will be shown in cases of sexual abuse and harassment in any forms and drug addiction in the campus.

1.13.1. Expulsion/Withdrawal on Disciplinary Ground:

1.13.1.1. Unfair means: Adoption of unfair means may result in expulsion of a student from the program and so from the university. The Academic Council of the University will approve such expulsion on the basis of recommendation of the Disciplinary Committee. Following would be considered as unfair means adopted during examinations and other contexts:

Communicating with fellow students for obtaining help in the examination.

Copying from another student's script/report/paper.

Copying from desk or palm of hand or from other incriminating documents including mobiles.

Possessing of any unauthorized document whether used or not.

1.13.1.2. Influencing Grades: Academic Council of the University may expel/withdraw any student for approaching directly or indirectly in any form to influence a Teacher or University authority for grades.

1.13.1.3. Other Indiscipline Behaviors: Academic Council of the University may withdraw/expel any student on disciplinary grounds, if any form of indiscipline or unruly behavior is seen in him/her which may disrupt the academic environment/ program or is considered detrimental to University's image.

1.13.2. Immediate Action by the Disciplinary Committee of the University: The Disciplinary Committee may take immediate disciplinary action against any student of this institution. In case of withdraw/expulsion, the matter will be referred to the Academic Council of the University for post-facto approval.

CHAPTER 2

The Department of Computer Science & Engineering

2.1 Introduction to the CSE Program

The department of CSE in BAIUST is the leading department that is fully equipped with most updated lab facilities; carried by a skilled and energetic team of faculties. Its eight batches of students are continuing the academic learning with excellence. The department of computer science and engineering strives to build excellence in the competitive market of CSE by organizing, applying and imparting practical and comprehensive educational programs.

In the aim of facilitating technological and technical basis, studious supervision and leadership skills to its students in the upcoming years the department of CSE is promising to a great extent. To fulfill the aim and make the students familiar with the real engineering world the department is focused on the following activities:

- Arranging various training programs.
- Organizing contests and workshops.
- Participating in national and international Computer Programming Contests.
- Software Development Projects.
- Gaming and Robotic Competitions.
- Mobile Apps Development and so on.

With all the realistic activities and the ruthless efforts and group effort of the qualified, sincere and enthusiastic faculty members, the department is progressing to the way of becoming a distinctive and desirable one in the field of Computer Science and Engineering and Information Technology.

This department is fully committed to provide quality services and facilities to the students to ensure that the students are being introduced to the recent technology trends. The globalization of technology is spreading so fast and the impact is reflected on the national economic status of every promising nation. There is no alternative to industry oriented learning mechanisms to strengthen the learning mechanism in an effective manner. The department of CSE of BAIUST is focused on collaborating with the top industry people both from the national and international arena. Several programs, seminars, camps, workshops have already been organized fruitfully

with the coordination of Bangladesh Open Source Network (BdOSN) and ICT division of the peoples' republic of Bangladesh. BdOSN chapter of BAIUST is working actively with the direct cooperation of the department of CSE. Industry oriented learning, problem solving capability building, and interactive collaboration with ICT fellows are leading the department of CSE to a center of excellence in the field of innovation and R&D.

2.2. Vision of the Program:

To become a prominent department of Computer Science and Engineering producing competent professionals with research and innovation skills, including moral values and social concerns.

2.3. Mission of the Program:

The mission of the Computer Science and Engineering Department is to prepare students for careers as software professionals and for advanced studies in Computer Science and Engineering Field. We develop and work with a range of technologies to build systems and applications that help solve today's problems, and share skills and knowledge with the community.

We are committed to providing the opportunity to master the computer science and Engineering discipline to students from all backgrounds. Our program combines strong fundamentals, project- and team- oriented activities, and soft skills, leading to a well-rounded professional education.

2.4. Intended Learning Outcome:

The field of Computer Science and Engineering plays a vital role in the progress of the country. Thus our graduates are able to seek out opportunities to provide innovative solutions to real-life problems using their expertise, and contribute towards the economic development of the country. Our graduates have the foresight to drive industry in new directions capable of exploiting their expertise to the fullest.

A B.Sc. graduate in Computer Science and Engineering should be able to:

- Demonstrate the understanding of Computer Science and Engineering fundamentals.
- Carry out scientific research, and effectively communicate findings.
- Solve technical and mathematical problems by the use of computer programming.
- Analyze, design and develop software for different needs.
- Able to develop Android and IOS based Applications for future needs.
- Possess interdisciplinary expertise in Computer science and engineering fields.
- Exhibit strong analytical and problem solving skills.

- Develop software in high level languages to solve engineering problems, including PC- and microprocessor-based ones for real-time monitoring and control systems and digital signal processing applications.
- Be able to pursue graduate education in Computer science and engineering and related fields.
- Operate basic and complex instruments which are used in computer networks.
- Design, test, install, supervise, diagnose and maintain different computer networks.
- Identify, analyze, plan and manage different communications systems and projects.
- Critically observe safety rules and occupational hazard prevention techniques.
- Exhibit apposite disposition towards science and technology to positively influence others.

Students graduating from CSE department should have gained the following abilities, skills, knowledge, and understanding:

- Ability to communicate with others; with listening, speaking, reading, writing, and non-verbal communication skills sufficient to offer and receive complex information.
- Ability to identify, deconstruct and solve complex problems; with skills sufficient to: obtain and analyze relevant information, including numerical information; use logical reasoning in constructing arguments; consider options and alternative viewpoints; and propose and implement holistic solutions.
- Research skills sufficient to obtain information from diverse sources.
- Ability to use contemporary information technology for communication, research, and analysis.
- Specific skills sufficient to meet a specific national workforce need.
- Life-skills sufficient to live and work in both urban and rural workplaces with international colleagues.
- Knowledge of the requirements of ethical behavior in the workplace.
- Commitment to independent learning and the ability to self-teach and adapt to change in careers and workplace.
- Ability to cope with the continuous change of a rapidly evolving society.
- Self-confidence and the ability to accept and provide constructive criticism.
- Understanding of gender equity issues and the processes for addressing gender based discrimination.

- Understanding of the fundamentals of leadership, and the management of people, projects, time, and money.
- Team-working and conflict resolution skills.
- Appreciation and valuing of cultural and intellectual diversity and an ability to function in a multi-cultural or global environment.
- Ability to interpret local issues and events within a global perspective.
- Literacy in science, technology, environment, economics, and law sufficient to understand national issues.
- Understanding of the needs of the general population in terms of health and education; including the impacts of alcohol and other drugs, violence and sexually transmitted infections.
- Understanding of the needs of the general population in terms of livelihood maintenance and the requirements of sustainability in development.
- Understanding of the social origin of crime and the processes for addressing it.
- Familiarity with social and ethical responsibilities of citizenship, and the functions of representative democracy.

2.5. Graduate Profile:

2.5.1. Specialized knowledge

• Developing basic and applied skills in the major fields of study
• An understanding and appreciation of contemporary issues and developments in the major fields of knowledge studied

2.5.2. Practical and Problem solving skills

• Enabling the graduates to handle the practical cases
• Understanding and developing hands-on skills

2.5.3. Numeracy and Analytical Skills

• Ability to recognize when information is incomplete
• Ability to process and interpret data
• Ability to develop analytical skill

2.5.4. Entrepreneurship and innovative skills

• Developing skills to contribute in the IT field.
• Able to participate in research and development (R&D)

2.5.5. Communication and ICT skills

• Exchange of ideas and information with others effectively
• Logical expression of ideas and subject matter in communication
• Linguistic skills (English) in writing and speaking
• Negotiation skills
• Ability to use technology to access, analyzes, organize, process and share

information.
- Ability to use internet facilities and general software like Word, PowerPoint and Excel for professional purposes

2.5.6. Interpersonal and Teamwork and Leadership skills

- Respect for the values of other individuals and groups, and an appreciation of human and cultural diversity
- Ability to work in a team respecting and accommodating others' opinion

2.5.7. Self-management and Personal development skills

- Adaptability in a diverse cultural setting
- Time management
- Awareness of the requirements of ethical issues and behavior
- Self-motivation, discipline and confidence
- An intellectual openness – curious and questioning mind

2.5.8. Commitment to Community, Country and Humanity

- Recognizing the professional code of conduct
- Commitment to job
- Sense of responsibility

2.6. Teaching Strategy:

Teachers of BAIUST use lecture, reading assignment, group assignment, role playing, presentation, students centered learning/ interactive teaching, seminar in theory courses and for practical courses they use demonstration, project / assignment, lab work. The teaching aids that are used by the teachers are: multimedia projector, whiteboard, overhead projector but now-a-days mostly multimedia projector.

2.7. Assessment Strategy:

In the common examination ordinance of BAIUST there is a provision of at least 4(four) class tests for three credit course, class performance and final examination. Class tests are considered as continuous assessments. There are also provisions of Lab performance viva in practical examinations. However, the teachers follow classroom assessment, written examination (essay question. short question, objective question in the form of MCQ, T/F, fill in the blanks, matching, yes/no etc.), viva voce, dissertation, assignment, practical exam etc.

Beside all theory courses CSE department uses Rubric for Design Project Evaluation

Evaluation Criteria	Beginning (4)	Developing (6)	Proficient (8)	Exemplary (10)
Identification of Problem or Definition of Project	Insufficient identification of problems; inadequately objectives.	Partial identification of the problem; lack of specifics does impair solution of design.	Adequate identification of the problem; any lack of specifics does impair solution of design.	Clear and complete identification of design goals and objectives.

Complexity of Design	Below typical technical complexity for course level.	Average typical complexity for course level.	Meets typical technical complexity for course level.	Exceeds typical technical complexity for course level.
Research	No evidence of use of outside information	One research source cited.	Two research sources are cited.	Three or more research sources are cited; evidence of variety
Meeting Design Requirements	Few design requirements are met. <u>Design Areas:</u> Physical Features Operation and Use Safety Cost	Only basic requirements are met. <u>Design Areas:</u> Physical Features Operation and Use Safety Cost	Design requirements are met. <u>Design Areas:</u> Physical Features Operation and Use Safety Cost	All design requirements are met and exceeded. <u>Design Areas:</u> Physical Features Operation and Use Safety Cost
Engineering Analysis	Most analysis is skipped or does not contribute to creating and effective design. Analyses are performed as a separate process after design choices are made. Analyses are likely incorrect and are difficult to understand due to poor setup.	Analyses performed are done after the fact to prove that choices made were valid. Analyses performed after discovering that trial and error is not working well. Analyses are haphazard and do not follow a logical flow.	Analyses performed are needed for design effectiveness and meeting design parameters. Analyses performed to prevent trial and error when the prototype is built. Analyses are performed correctly and contain many elements of a quality engineering analysis.	Analyses performed are used to enhance design effectiveness and choose design parameters. Analyses performed to save significant effort by preventing trial and error when the prototype is built. Analyses are performed correctly and follow steps for quality.
Alternative Solutions	No evidence of alternative designs.	One alternative presented as the project solution.	At least two alternatives presented.	Three or more alternatives presented.
Optimizing, Testing and Evaluation	No reason provided for the selected solution. Project does not include evaluation of results.	Reasoning for the solution based on opinion only. Results evaluated but without any focus.	Reasoning for the solution is supported by facts. Results evaluated; reflects the project's need.	Solution selected by the use of comparative data. Evaluation data are collected to support needs.
Design Documentation and Presentation	Reports may have poor quality writing and mix jargon with engineering language. Reports miss many important topics and are not easy to read.	Reports attempt appropriate language/format for the engineering field. Reports are fairly informative and generally easy to read.	Reports used mostly appropriate language/format for the engineering field. Reports are mostly informative and easy to read. Information in reports is well	Reports used appropriate language/format for the engineering field. Reports are informative and easy to read.

	Information in the report is not organized. Data or design features explanations are very difficult to locate. Demonstration may not work and derails the purpose of the presentation. Project results are not shared with others. Evidence of plagiarism.	Information in reports organized into sections with data or design features explanation present. Demonstration is present but breaks up flow of presentation. Project results shared with others. The team did not quote all the sources of information that they used.	organized. All data and design features can be found without difficulty. Demonstration enhances understanding during presentation. Project results shared; both positive and negative results presented. The team quoted nearly all the sources of information that they used.	Information in reports is well organized so that data or design features explanations are easy to find. Demonstration is effectively incorporated into the presentation. Project results shared; improvements presented. Avoid plagiarism, do not use information without giving credit to the appropriate source.
Time Management	No evidence of planning. Missed significant milestones or project not completed	Plan was made but not followed; some goals accomplished; inconsistent use of time.	Plans and procedures followed during the project. Goals accomplished; most milestones met; misses some deadlines.	Project plan, procedures followed and documented. Identify plan and timeline; consistently met deadlines.
Teamwork Dynamics	Students build cohesion in groups through verbal and non-verbal behavior. Takes an active role to encourage participation of all members.	Student behavior brings sometimes cohesion in groups. Sporadic jokes and pranks encourage participation of team members.	Student jokes and pranks bring cohesion in the group but meetings become very time consuming.	Student apathy negatively affects group performance. Group cohesion is broken by non-verbal behavior.

2.8. Category of courses

•Language: 3 Theory courses (7 credit hours)
Composition, writing and Communication in English, Functional Bengali Language

•General Education (GED): 6 Theory Courses (15 credit hours)
Students will have to complete six GED Courses in the following academic areas: Social Science, Arts and Humanities and Business.

•Basic Sciences: 3 Theory and 2 Sessional Courses (11credit hours)
Courses in the following academic areas qualify as science credits: Chemistry and Physics. Total basic science courses are 11 credit hours.

•Mathematics: 4 Courses (12 credit hours)

Courses in the following academic areas qualify as mathematics credits: Differential and Integral Calculus, Coordinate Geometry and Vector Analysis, Ordinary and Partial Differential Equations, Linear Algebra, Probability and Statistics.

•Other Engineering Courses: 3 Theory courses and 3 Sessional Courses (11credit hours)

Students will have to complete six interdisciplinary Engineering Courses in the following academic areas: Electrical & Electronic Engineering and Civil Engineering.

•Program Core courses: 21 Theory courses and 17 Sessional Courses (89credit hours including Sessional)

Students will have to complete 21 Courses including 17Sessional courses in the following areas: Programming, Hardware Systems, Logics & Algorithms, Systems and Software Systems and Engineering.

•Technical Electives: 3Theory courses and 1 Sessional Courses (9 credit hours including Sessional)

Students will have to complete 4 Courses including Sessional courses in the field of communication, Data Science, Software Engineering, Hardware and ICT.

•Project/Thesis: 2 Projects (8 credit hours)

Students will have to complete 2 Projects, one on Integrated Design and another one as Capstone Project.

CHAPTER 3

RULES AND REGULATIONS FOR UNDERGRADUATE PROGRAM

3.1. Terms and Duration

There will be two terms in an academic year.

1. Term I
2. Term II

Term I will be of 24 weeks and Term II will be of 28 weeks (Total 52 weeks). An Academic Calendar will be provided to all the enrolled students to make them familiar to all academic events. The holidays will also be included in the calendar.

3.2. Annual Terms Plan:

Duration Term I (Spring) and Term II (Fall) will be as follows:

Term I:

Sl No	Events	Duration
1.	Classes before Mid Term	7 weeks
2.	Mid Term Break	1 week
3.	Classes after Mid Term	7 weeks**
4.	Lab Test Week	1 week*
5.	Preparatory leave	2 weeks
6.	Term Final Examination	3 weeks
7.	Result Publication and Term End Vacation	3 weeks
	Total	**24 weeks**

Term II:

Sl No	Events	Duration
1.	Classes before Mid Term	7 weeks
2.	Mid Term Break	1 week
3.	Classes after Mid Term	7 weeks**
4.	Lab Test Week	1 week*
5.	Preparatory leave	2 weeks
6.	Term Final Examination	3 weeks
7.	Result Publication and Term End Vacation	3 weeks
8.	Industrial Training and Survey Practical	4 weeks*
	Total	**28 weeks**

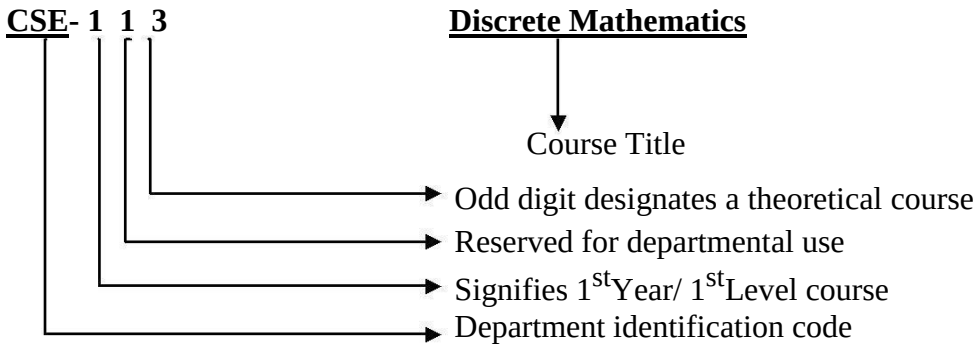
3.3. Course Pattern and Credit Structure

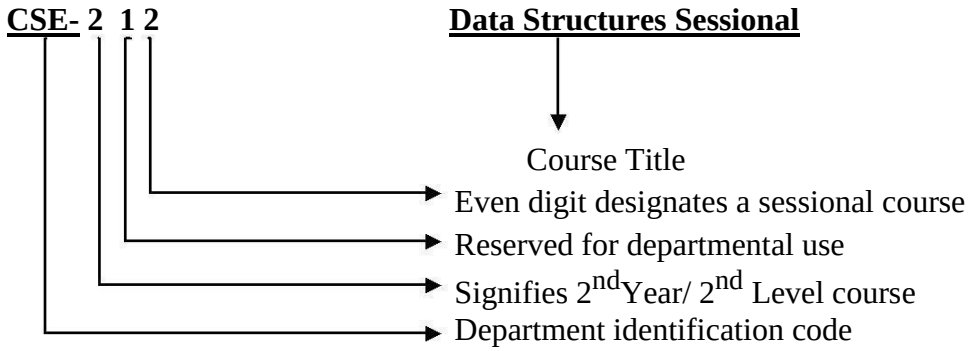
The undergraduate program is covered by a set of theoretical courses along with a set of laboratory (sessional) courses to support them. Various related guidelines are as follows:

3.3.1. Course Designation System

Each course is designated by a maximum of four letter code identifying the department offering the course followed by a three-digit number having the following interpretation:

- The first digit corresponds to the year/level in which the course is normally taken by the students.
- The second digit is reserved for departmental use. It usually identifies a specific area/group of study within the department.
- The last digit is an odd number for theoretical courses and an even number for sessional courses.
- The course designation system is illustrated as follows:





3.3.2 Assignment of Credits

The assignment of credits to a theoretical course follows a different rule from that of a sessional course.

- ✓ For theoretical Courses, one lecture per week per term is equivalent to one credit.

- ✓ For sessional Courses, 2 sessional contact hours per week per term is equivalent to one (01) credit.
- ✓ Credits are also assigned to project and thesis work taken by the students. The amount of credits assigned to such work varies from one discipline to another.

3.3.3 Types of Courses

The types of courses included in the undergraduate curricula are divided into the following groups:

- **Core Courses:** In each discipline, a number of courses are identified as core courses, which form the nucleus of the respective bachelor's degree program. A student has to complete all the designated core courses of his/her discipline.

- **Prerequisite Courses:** A prerequisite course is one of the core courses, which should be completed before taking some other course (s).

- **Optional Courses:** Apart from the core courses, the students can choose from a set of optional courses. A required number of optional courses from a specified group have to be chosen.

3.4 Student Adviser

One adviser is normally appointed for a group of students by the Board of Undergraduate Studies (BUGS) of the concerned department. The adviser advises each student about the courses to be taken in each term by discussing the academic program of that particular term with the student.

However, it is also the student's responsibility to keep regular contact with his/her adviser who will review and eventually approve the student's specific plan of study and monitor subsequent progress of the student.

For a student of second and subsequent terms, the number and nature of courses for which he/she can register is decided on the basis of academic performance during the previous term. The adviser may permit the student to drop one or more courses based on previous academic performance.

3.5 Course Registration Procedures:

Registration will done at the beginning of each Term/Semester.
The guidelines are as follows:

- The date and time of registration will be announced in advance by the Registrar's Office and students have to complete their registration through online.
- Students have outstanding dues to University or hall of residence shall not be permitted to register. They have to clear their dues and get a clearance certificate, on

production of which, they will be given the necessary Registration Forms to complete registration.

- All students submitted registration form will be approved by his/her advisor. The student will contact with treasurer branch to complete their payments in order to complete their registration procedure.
- It is absolutely necessary that all students register for the courses within the specified time. Late registration is, however, permitted during the first week on payment of a late registration fee.

3.6 Withdrawal from a Term

Following circumstances and rules will apply:

- In case of severe sickness, which lead to missing of more than 40% classes (supported by requisite medical documents), student (s) may be allowed to withdraw the term and repeat the whole term with re-registration with the regular term in the next academic term, subject to the approval of Academic Council, BAIUST. S/he will get a W grade in case of such term withdrawal.
- If a student does not attend more than 40% classes without notification and prior approval, his/her term registration will be cancelled and his/her courses will be treated as "Incomplete". S/he will have to repeat the term with re-registration in the next academic session, subject to the approval of Academic Council, BAIUST. S/he will get "I"grade in case of such term withdrawal.
- However, if a student appears more than 60% classes but doesn't appear the exam after collecting the Admit Card of Exam, s/he will get "F" grade in such cases.
- If a student does not attend classes for two consecutive terms without prior approval and is interested to continue from the next academic session, s/he will be penalized for each term and may be allowed to repeat the terms with re-registration, subject to the approval of Academic Council, BAIUST. S/he will get "F" grade in case of such withdrawal of terms.
- However, s/he will have to complete the whole undergraduate program within 06 (Six) academic year from the date of his/her registration.
- The mentor/student advisor must have to recommend such kind of term withdrawal.

3.7 Program Withdrawal:

Following circumstances and rules will apply:

- Failure to secure/achieve minimum CGPA of 2.25 in two consecutive Level will also lead to withdrawal of the student from the program.
- If a student does not attend classes for two consecutive terms without notification and prior approval, s/he will be permanently withdrawn from the program.

- In case of Term Repeat, students repeating a particular term (for example L-1 T-1) and if s/he fails again in three or more courses in that term will require the approval of the Academic Council for further consideration.
- However, s/he will have to complete the whole undergraduate program within 06 (Six) academic year from the date of his/her registration. Otherwise, s/he will be withdrawn from the program.
- The mentor/student advisor must have to recommend such kind of term withdrawal.

3.8 Withdrawal On Own Accord:

Following circumstances and rules will apply:

- **Permanent Withdrawal:** A student who has already completed some courses and has not performed satisfactory may apply for a permanent withdrawal.
- **Temporary Withdrawal:** A student, if s/he applies, may be allowed to withdraw temporarily from the program, subject to the approval of Academic Council of the University, but s/he has to complete the whole program within 06 (Six) academic years from the date of his/her registration.
- The mentor/student advisor must have to recommend such kind of term withdrawal.

3.9 The Grading System

The total performance of a student in a given course is based on a scheme of continuous assessment. For theoretical courses, this continuous assessment is made through a set of quiz, class evaluation, class participation, presentation, homework assignment, term paper and a term final examination. The assessments for sessional courses are made by evaluating performance of the student at work during the class, viva-voce during laboratory hours and quiz. Besides that, at the end there will be a final lab test. Each course has a certain number of credits, which describes its corresponding weightages. A student's performance is measured by the number of credits completed satisfactorily and by the weighted average of the grade points earned. A minimum grade point average (GPA) is essential for satisfactory progress. Letter grades and corresponding grade points will be given as follows (following the UGC specified standard):

Numerical Markings	Grade	Grade Points
80% and above	A+	4.00
75% to below 80%	A	3.75
70% to below 75%	A-	3.50
65% to below 70%	B+	3.25
60% to below 65%	B	3.00
55% to below 60%	B-	2.75
50% to below 55%	C+	2.50
45% to below 50%	C	2.25
40% to below 45%	D	2.00
below 40%	F*	0.00
Incomplete	I	-
Withdrawal	W	-
Project/ Thesis continuation	X	-

*Subject in which the student gets F grade shall not be regarded as earned credit hours for the calculation of Grade Point Average (GPA).

3.10 Distribution of Marks:

3.10.1 For Theoretical Courses:

Thirty percent (30%) marks of theoretical course shall be allotted for continuous assessment, i.e. quiz, class tests, home assignments, class evaluation and class participation and 70% shall be allotted to the Term Final Examination. The final examination is conducted centrally by the University.

Distribution of marks for a given course is as follows:

Category	Marks %
Class Participation/Presentation skill	5
Class Attendance	5
Home work/Assignment	5
Class Test0	15
Final Examination (maximum 3 hours)	70
Total	**100**

Note: The number of class tests of a course shall be *n*+1 where n is the number of credits of the course. Evaluation of performance in class tests will be on the basis of the best *n* class test. The theme of continuous assessment that a particular teacher wishes to follow for a course will be explained by course appraisal on the first day of the Term.

3.10.2 For Sessional/Practical Courses:

The marks distribution of sessional courses will be distributed according to the type of the sessional course. The distribution of marks for three types of sessional is given below:

I. Marks distribution of Lab/ Project based sessional

Category	Marks
Lab test/Project	30
Written test/Quiz	20
Viva voce	10
Attendance	10
Data Sheet/Home Assignment/Report	10
Class Performance/ Continuous Evaluation	20
Total	**100 %**

II. Marks distribution of programming based Sessional

Category	Marks
Onlie-1	25
Online-2	25
Viva voce	10
Attendance	10
Continuous Evaluation	10
Class Performance	20
Total	**100 %**

III. Marks distribution of programming based sessional

Category	Marks
Project	40
Quiz	20
Viva / Presentation	10
Class Participation	10
Report	10
Class Assessment	10
Total	**100%**

3.10.3 Engineering Programs:

- All students are required to attend 80% of the class for all courses.
- In case of sickness 70% attendance may be considered by the VC with proper medical documents provided by the student.
- Students failing to attend 80% classes are liable to pay a fine of Tk. 2000/-per course.
- Students not eligible to sit for exam of a particular course due to poor attendance; their concerned course will be deleted from the Admit Card.

- **The guidelines for attendance marks are as follows:**

Category	Marks
For Theory Courses	5
For Sessional Courses	10

- **The attendance marks distribution for the final assessment:**

Attendance (100)	Theory (05) Marks
90% and above	05
85% - 89%	4.5
80% - 84%	04
75% - 79%	3.5
70% - 74%	03
Below 70%	0%
Sessional (10) Marks	As per earned percentage (%) of attendance.

3.11. Calculation of CGPA

Grade Point Average (GPA) is the weighted average of the grade points obtained of all the courses passed/completed by a student. For example, if a student passes/completes *n* courses in a term having credits of C_1, C_2, ..., C_n and his grade points in these courses are G_1, G_2,....,G_n, respectively, then

$$\text{GPA} = \frac{Grade\ points\ earned\ in\ the\ semester}{Credits\ completed\ in\ the\ semester}$$

$$= \frac{Summation\ of\ (credit\ hours\ in\ a\ course \times Grade\ point\ earned\ in\ that\ course)}{Total\ number\ credit\ hours\ completed}$$

$$= \frac{\sum_{i=1}^{n} C_i * G_i}{\sum_{i=1}^{n} C_i}$$

The Cumulative Grade Point Average (CGPA) is the weighted average of the GPA obtained in all the term passed/completed by a student. For example, if a student passes/completes *n* terms having total credit of TC_1, $TC_{2,.....,}$ TC_n and his/her GPA in these terms are GPA_1, GPA_2,....., GPA_n, respectively, then

$$CGPA = \frac{\sum_{i=1}^{n} TC_i * GPA_i}{\sum_{i=1}^{n} TC_i}$$

3.12 Minimum Earned Credit and GPA Requirement for Obtaining Degree

Minimum credit hour requirements for the award of bachelor's degree in engineering (B.Sc. Engineering) and other discipline will be decided as per existing rules. The minimum GPA requirement for obtaining a Bachelor's degree in Engineering is 2.25.

3.13 Types of Different Examination

3.13.1 <u>Re-Examination/ Referred Examination:</u>

- ✓ Students failing in maximum of two theoretical courses in the final examination in any term will be promoted to the next term and s/he will be allowed to appear at the re-examination on failed course(s). S/he has to apply to appear at the re-examination.
- ✓ Students at all levels will be allowed to appear at the referred/re-examination on maximum two regular courses excluding the "Backlog" one.
- ✓ When a student will appear at re-examination in which s/he previously obtained F grade, s/he will be eligible to get a grade as earned in such a course. A fee has to be paid by the students to register and appear at the re-examination.
- ✓ If a student fails in a sessional course(s) then s/he will have to appear at the sessional classes and exams to clear/pass the concerned course(s) in any of the subsequent terms, his/her performance will be evaluated as per the marks distribution policy of sessions.
- ✓ The student(s) must have to take recommendation of the mentor/student advisor to sit for the re-examination of failed course (s).
- ✓ Re-examination will be conducted from 3rd week of the semester.

3.13. 2. **Backlog:**

Following circumstances and rules will apply for backlog:

- ✓ Students failing in maximum of two theoretical courses in a Term Final Examination will be promoted to the next Term and will appear the re-examination of the failed course(s) at the end of that Term as per Academic Schedule.

- ✓ If the student fails again in the re-examination of the failed course (s) will be termed as "Backlog" course(s).

- ✓ The Backlog student will be allowed to appear at the re-examination of the Backlog course (s) in the following Term only for once and have to clear/pass the "Backlog" student (s) in the Backlog exam. Otherwise, s/he will have to retake the course in any of the subsequent terms.

- ✓ A student appearing at the re-examination of any backlog course (s) will be eligible to get a grade as earned. A scheduled fee has to be paid by the student to register and appear at the re-examination of Backlog course (s).

- ✓ A student will be allowed to appear at the re-examination of maximum 4 theoretical courses (Backlog or Regular failed course(s)) in a single term End re-examination program.

- ✓ The student (s) must have to take recommendation of the student advisor to sit for the re-examination of backlog course(s).

3.13.3 **Term Repeat:**

Following circumstances and rules will apply:

- ✓ If a student fails in three or more theoretical courses in the Term Final Examination, then s/he will repeat the same Term with registration.

- ✓ Students repeating a Term may seek exemption for that/those course(s) in which they have passed in the previous Term. If applied, then the previous grade of a particular course will be carried forward for his/her onward GPA/CGPA calculation. It implies that the student will not attend classes or appear at tests/examinations of the concerned course (s) in the Repeat Term.

- ✓ Students repeating a particular Term (for example l-1 T-1) and failing again in three or more theoretical course (s) in that Term will require approval from the Academic Council for the further consideration.

3.13.4 **Improvement:**

Following circumstances and rules will apply:

- ✓ If a student obtains grade B or below B in a course, s/he will be allowed to appear in the re-examination only once for that course for the purpose of grade improvement and s/he will be eligible to get grade as earned. A student can appear in the improvement examination for a particular course within next three consecutive semesters and can appear at maximum 6 courses in 4 years graduation program. These will be shown in the transcript.

- ✓ In case of improvement of a particular course, the better/higher grade will be considered as the final grade for GPA calculation of the particular term.

- ✓ The student (s) must have to take recommendation of the student advisor to sit for the improvement exam of the course (s).

3.14. Mapping of CO's with PO

Course	Program Outcomes											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
	Knowledge	Analysis	Design	Investigation	Tool Usages	Society	Environment	Ethics	Individual & Teamwork	Communication	Project Management & Finance	Life Long Learning
ENG 111	X	X	X	X								
ENG 121						X	X	X				
HUM 211						X	X					
GED 121						X	X					
GED 211												
GED 221												
GED 411												
GED 421												
GED 423												
CHE 111	X	X	X									
CHE 112	X	X	X	X								X
PHY 121	X	X										
PHY 124						X						
PHY 211	X	X										
MATH 111	X											
MATH 121	X	X										
MATH 211	X	X										
MATH 221	X		X	X								
CE 311						X			X	X		
CE 312						X			X	X		
EEE 111	X	X							X			
EEE 112						X	X					
EEE 121	X	X							X			
EEE 122						X	X					
CSE 112					X					X		X
CSE 113	X	X				X						
CSE 121	X	X							X			
CSE 122						X			X			

CSE 211	X	X	X									
CSE 212						X			X			
CSE 213	X	X							X			
CSE 214								X	X		X	
CSE 221	X	X	X	X								X
CSE 222						X			X			
CSE 223	X		X									
CSE 224						X	X		X			
CSE 225	X	X										
CSE 226						X			X		X	
CSE 311	X	X	X									
CSE 312							X		X	X	X	
CSE 313	X	X	X						X			
CSE 314									X			
CSE 315	X	X	X									
CSE 317	X	X	X	X							X	
CSE 318												
CSE 319	X	X	X	X								
CSE 321	X	X										
CSE 322						X			X			
CSE 323												
CSE 324												
CSE 325	X	X										
CSE 326						X		X		X		
CSE 327	X	X	X									
CSE 328									X		X	
CSE 329												
CSE 301	X	X	X	X	X	X			X			
CSE 411	X	X										
CSE 412						X			X	X		
CSE 413	X	X	X									
CSE 414						X			X		X	
CSE 415												
CSE 421	X	X	X	X		X	X		X	X		
CSE 422						X		X	X			
CSE 401			X				X	X	X	X	X	X
CSE 402			X				X	X	X	X	X	X
CSE 423												
CSE 424												
CSE 427												
CSE 428												
CSE 443												
CSE 444												
CSE 441												
CSE 442												

CHAPTER 4

COURSE REQUIREMENTS FOR THE STUDENTS OF UNDERGRADUATE PROGRAM (B. Sc. in CSE) OF THE DEPARTMENT OF CSE, BAIUST

Curriculum for B.Sc. in Computer Science and Engineering (Proposed)

Summary of Courses

(To be effective from Fall-2021 semester)

Types of Courses	No. of Courses	Credit Hours	Remarks
Language Courses	3T	07	T – Theory L – Laboratory/ Sessional P – Project
General Education Courses	6T	15	
Basic Science Courses	3T+2L	09+02	
Mathematics Courses	4T	12	
Other Engineering Courses	3T+3L	07+04	
Program Core Courses	21T+17L	59+22	
Elective Courses	3T+1L	08+01	
Project	2P	08	
Total	**43T+23L+2P**	**154**	

List of Courses

Language Courses (7 credit hours)

Course Code	Course Title	Credit Hours	Contact Hours/ Week	Prerequisite Courses
ENG 111	English Reading and Speaking	2	2	N/A
ENG 121	English Writing and Listening	3	3	ENG 111
HUM 211	Bangla Language	2	2	N/A
Total		**7**	**7**	

General Education Courses (15 credit hours)

Course Code	Course Title	Credit Hours	Contact Hours/ Week	Prerequisite Courses
GED 121	History of the Emergence of Bangladesh	2	2	N/A
GED 211	Finance and Investment	3	3	N/A
GED 221	Engineering Economics	3	3	N/A
GED 411	Business Communications	3	3	N/A
GED 421	Sociology for Engineers	2	2	N/A
GED 423	Engineering Ethics	2	2	N/A
Total		**15**	**15**	

Basic Science Courses (11 credit hours)

Course Code	Course Title	Credit Hours	Contact Hours/ Week	Prerequisite Courses
CHE 111	Chemistry	3	3	N/A
CHE 112	Chemistry Sessional	1	2	N/A
PHY 121	Mechanics, Wave, Heat and Thermodynamics	3	3	N/A
PHY 124	Basic Physics Sessional	1	2	N/A
PHY 211	Electromagnetism, Optics and Modern Physics	3	3	N/A
Total		**11**	**13**	

Mathematics Courses (12 credit hours)

Course Code	Course Title	Credit Hours	Contact Hours/ Week	Prerequisite Courses
MATH 111	Differential and Integral Calculus	3	3	N/A
MATH 121	Coordinate Geometry and Vector Analysis	3	3	N/A
MATH 211	Linear Algebra and Complex Analysis	3	3	N/A
MATH 221	Probability and Statistics	3	3	N/A
Total		**12**	**12**	

Other Engineering Discipline Courses (11 credit hours)

Course Code	Course Title	Credit Hours	Contact Hours/ Week	Prerequisite Courses
CE 311	Engineering Drawing	1	1	N/A
CE 312	Engineering Drawing & CAD Sessional	1	2	N/A
EEE 111	Electrical Circuit Analysis	3	3	N/A
EEE 112	Electrical Circuit Analysis Sessional	1.5	3	N/A
EEE 121	Electronic Devices and Circuits	3	3	EEE 111
EEE 122	Electronic Devices and Circuits Sessional	1.5	3	N/A
Total		**11**	**15**	

Program Core Courses with Project/Thesis (89 credit hours)

Course Code	Course Title	Credit Hours	Contact Hours/ Week	Prerequisite Courses
CSE 112	Introduction to Computing Sessional	1.5	3	N/A
CSE 113	Discrete Mathematics	3	3	N/A
CSE 121	Structured Programming Language	3	3	N/A
CSE 122	Structured Programming Language Sessional	1.5	3	N/A
CSE 211	Data Structures	3	3	CSE 121
CSE 212	Data Structures Sessional	1.5	3	N/A
CSE 213	Object Oriented Programming Language	3	3	CSE 121
CSE 214	Object Oriented Programming Language Sessional	1.5	3	N/A
CSE 221	Computer Algorithms and Complexity	3	3	CSE 211
CSE 222	Computer Algorithms and Complexity Sessional	1.5	3	N/A
CSE 223	Digital Logic Design	3	3	EEE 121
CSE 224	Digital Logic Design Sessional	1.5	3	N/A
CSE 225	Numerical Methods	3	3	N/A
CSE 226	Numerical Methods Sessional	1	2	N/A
CSE 311	Operating System	3	3	N/A
CSE 312	Operating System Sessional	1	2	N/A
CSE 313	Database Management System	3	3	CSE 213

CSE 314	Database Management System Sessional	1.5	3	N/A
CSE 315	Computer Architecture and Design	3	3	N/A
CSE 317	Software Engineering and Design Patterns	3	3	N/A
CSE 318	Software Engineering and Design Patterns Sessional	1	2	N/A
CSE 319	Theory of Computation	3	3	N/A
CSE 321	Data Communications	3	3	N/A
CSE 322	Data Communications Sessional	1	2	N/A
CSE 323	Web Technologies	2	2	N/A
CSE 324	Web Technologies Sessional	1.5	3	N/A
CSE 325	Microprocessor, Assembly Language and Computer Interfacing	3	3	CSE 223
CSE 326	Microprocessor, Assembly Language and Computer Interfacing Sessional	1.5	3	N/A
CSE 327	Computer Networks	3	3	N/A
CSE 328	Computer Networks Sessional	1.5	3	N/A
CSE 329	Management Information System	2	2	N/A
CSE 301	Integrated Design Project	2	4	N/A
CSE 411	Machine Learning	2	2	N/A
CSE 412	Machine Learning Sessional	1	2	N/A
CSE 413	Compiler Design and Construction	3	3	N/A
CSE 414	Compiler Design and Construction Sessional	1	2	N/A
CSE 415	Research Methodology and Complex Engineering Activities	2	2	N/A
CSE 421	Artificial Intelligence	3	3	N/A
CSE 422	Artificial Intelligence Sessional	1	2	N/A
CSE 401	Capstone Project (Part 1)	2	4	N/A
CSE 402	Capstone Project (Part 2)	4	8	N/A
Total		**89**	**119**	

List of Specialized Elective Courses

(Any three theories with sessional courses (if any) to be taken)

Course Code	Course Title	Credit Hours	Contact Hours/ Week	Prerequisite Courses
CSE 423	Computer Graphics	2	2	CSE 319
CSE 424	Computer Graphics Sessional	1	2	N/A
CSE 425	Bioinformatics	2	2	N/A
CSE 426	Bioinformatics Sessional	1	2	N/A
CSE 427	Digital Signal Processing	2	2	N/A
CSE 428	Digital Signal Processing Sessional	1	2	N/A
CSE 443	Digital Image Processing	2	2	CSE 427
CSE 444	Digital Image Processing Sessional	1	2	N/A
CSE 429	Computer and Cyber Security	3	3	N/A
CSE 431	Cellular Mobile and Satellite Communication	3	3	N/A
CSE 433	Cloud Computing	3	3	N/A
CSE 435	Internet of Things	3	3	N/A
CSE 437	VLSI Design	3	3	N/A
CSE 439	Neural Network and Fuzzy Logic	2	2	N/A
CSE 440	Neural Network and Fuzzy Logic Sessional	1	2	N/A
CSE 441	Data Mining	2	2	N/A

CSE 442	Data Mining Sessional	1	2	N/A
CSE ***	Contemporary Course for Computer Science & Engineering	3	3	N/A

Semester-wise Course Distribution

First Semester (Level 1 Term I)

Course Code	Course Title	Credit Hours	Contact Hours/ Week	Prerequisite Courses
CSE 113	Discrete Mathematics	3	3	N/A
CSE 112	Introduction to Computing Sessional	1.5	3	N/A
CHE 111	Chemistry	3	3	N/A
CHE 112	Chemistry Sessional	1	2	N/A
ENG 111	English Reading and Speaking	2	2	N/A
MATH 111	Differential and Integral Calculus	3	3	N/A
EEE 111	Electrical Circuit Analysis	3	3	N/A
EEE 112	Electrical Circuit Analysis Sessional	1.5	3	N/A
Total		**18**	**22**	

Second Semester (Level 1 Term II)

Course Code	Course Title	Credit Hours	Contact Hours/ Week	Prerequisite Courses
CSE 121	Structured Programming Language	3	3	N/A
CSE 122	Structured Programming Language Sessional	1.5	3	N/A
GED 121	History of the Emergence of Bangladesh	2	2	N/A
ENG 121	English Writing and Listening	3	3	N/A

MATH 121	Coordinate Geometry and Vector Analysis	3	3	N/A
EEE 121	Electronic Devices and Circuits	3	3	EEE 111
EEE 122	Electronic Devices and Circuits Sessional	1.5	3	N/A
PHY 121	Mechanics, Wave, Heat and Thermodynamics	3	3	N/A
PHY 124	Basic Physics Sessional	1	2	N/A
Total		**21**	**25**	

Third Semester (Level 2 Term I)

Course Code	**Course Title**	**Credit Hours**	**Contact Hours/ Week**	**Prerequisite Courses**
CSE 211	Data Structures	3	3	CSE 121
CSE 212	Data Structures Sessional	1.5	3	N/A
CSE 213	Object Oriented Programming Language	3	3	CSE 121
CSE 214	Object Oriented Programming Language Sessional	1.5	3	N/A
MATH 211	Linear Algebra and Complex Analysis	3	3	N/A
PHY 211	Electromagnetism, Optics and Modern Physics	3	3	N/A
HUM 211	Bangla Language	2	2	N/A
GED 211	Finance and Investment	3	3	N/A
Total		**20**	**23**	

Fourth Semester (Level 2 Term II)

Course Code	Course Title	Credit Hours	Contact Hours/ Week	Prerequisite Courses
CSE 221	Computer Algorithms and Complexity	3	3	N/A
CSE 222	Computer Algorithms and Complexity Sessional	1.5	3	N/A
CSE 223	Digital Logic Design	3	3	EEE 121
CSE 224	Digital Logic Design Sessional	1.5	3	N/A
CSE 225	Numerical Methods	3	3	N/A
CSE 226	Numerical Methods Sessional	1	2	N/A
MATH 221	Probability and Statistics	3	3	N/A
GED 221	Engineering Economics	3	3	N/A
Total		**19**	**23**	

Fifth Semester (Level 3 Term I)

Course Code	Course Title	Credit Hours	Contact Hours/ Week	Prerequisite Courses
CSE 311	Operating System	3	3	N/A
CSE 312	Operating System Sessional	1	2	N/A
CSE 313	Database Management System	3	3	N/A
CSE 314	Database Management System Sessional	1.5	3	N/A
CSE 315	Computer Architecture and Design	3	3	N/A
CSE 317	Software Engineering and Design Patterns	3	3	N/A
CSE 318	Software Engineering and Design Patterns Sessional	1	2	N/A

CSE 319	Theory of Computation	3	3	N/A
CE 311	Engineering Drawing	1	1	N/A
CE 312	Engineering Drawing and CAD Sessional	1	2	N/A
Total		**20.5**	**25**	

Sixth Semester (Level 3 Term II)

Course Code	Course Title	Credit Hours	Contact Hours/ Week	Prerequisite Courses
CSE 321	Data Communications	3	3	N/A
CSE 322	Data Communications Sessional	1	2	N/A
CSE 323	Web Technologies	2	2	N/A
CSE 324	Web Technologies Sessional	1.5	3	N/A
CSE 325	Microprocessor, Assembly Language and Computer Interfacing	3	3	CSE 223
CSE 326	Microprocessor, Assembly Language and Computer Interfacing Sessional	1.5	3	N/A
CSE 327	Computer Networks	3	3	N/A
CSE 328	Computer Networks Sessional	1.5	3	N/A
CSE 301	Integrated Design Project	2	4	N/A
CSE 329	Management Information System	2	2	N/A
Total		**20.5**	**28**	

Seventh Semester (Level 4 Term I)

Course Code	Course Title	Credit Hours	Contact Hours/ Week	Prerequisite Courses
CSE 411	Machine Learning	2	2	N/A

CSE 412	Machine Learning Sessional	1	2	N/A
CSE 413	Compiler Design and Construction	3	3	N/A
CSE 414	Compiler Design and Construction Sessional	1	2	N/A
CSE 415	Research Methodology and Complex Engineering Activities	2	2	N/A
CSE ***	Elective 1	3	3	N/A
GED 411	Business Communications	3	3	N/A
CSE 401	Capstone Project (Part-1)	2	4	N/A
Total		**17**	**21**	

Eighth Semester (Level 4 Term II)

Course Code	**Course Title**	**Credit Hours**	**Contact Hours/ Week**	**Prerequisite Courses**
CSE 421	Artificial Intelligence	3	3	N/A
CSE 422	Artificial Intelligence Sessional	1	2	N/A
GED 421	Sociology for Engineers	2	2	N/A
GED 423	Engineering Ethics	2	2	N/A
CSE ***	Elective 2	3	3	N/A
CSE ***	Elective 3	2+1	2+2	N/A
CSE 402	Capstone (Part-2)	4	8	N/A
Total		**18**	**24**	

Credit Distribution Summary

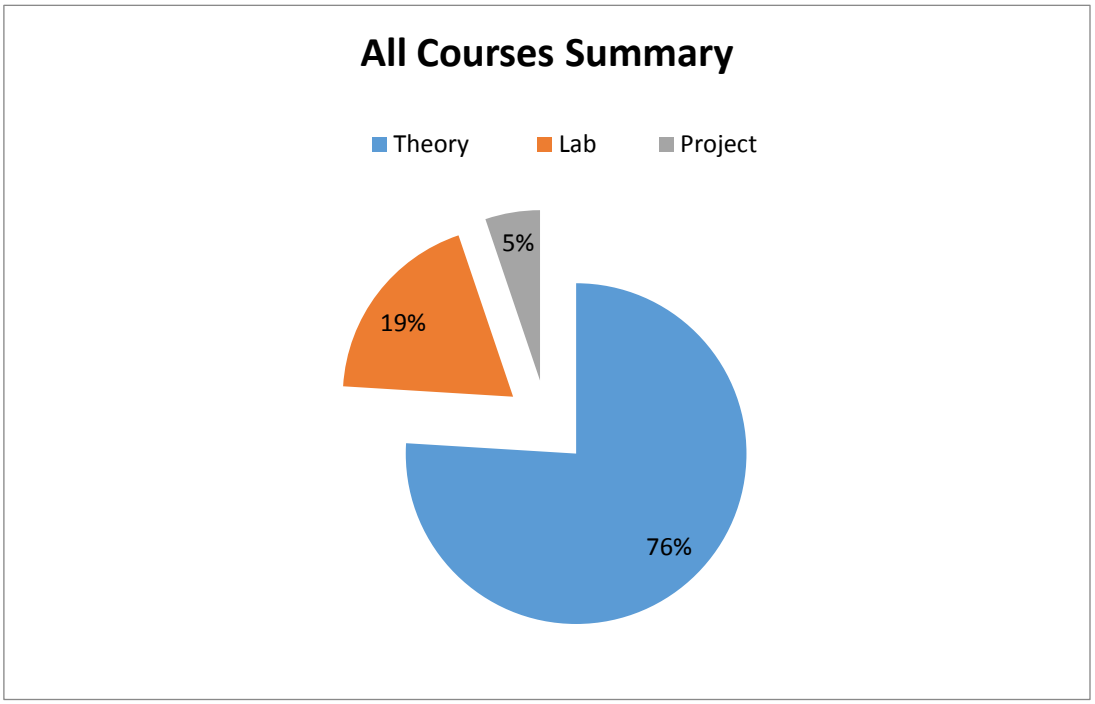
Level / Term	Total Credit	Contact Hours/ Week
1-1	**18**	**22**
1-2	**21**	**25**
2-1	**20**	**23**
2-2	**19**	**23**
3-1	**20.5**	**25**
3-2	**20.5**	**28**
4-1	**17**	**21**
4-2	**18**	**24**
Total	**154**	

Credit Distribution Summary (Departmental, non-departmental & Project)

Level / Term	Total Credit (Theory + Lab + Project)	Contact Hours/ Week
1-1	**14+4+0=18**	**22**
1-2	**17+4+0=21**	**25**
2-1	**17+3+0=20**	**23**
2-2	**15+4+0=19**	**23**
3-1	**16+4.5+0=20.5**	**25**
3-2	**13+5.5+2=20.5**	**28**

4-1	**13+2+2=17**	**21**
4-2	**12+2+4=18**	**24**
Total	**117+29+8=154**	

Pie Chart:

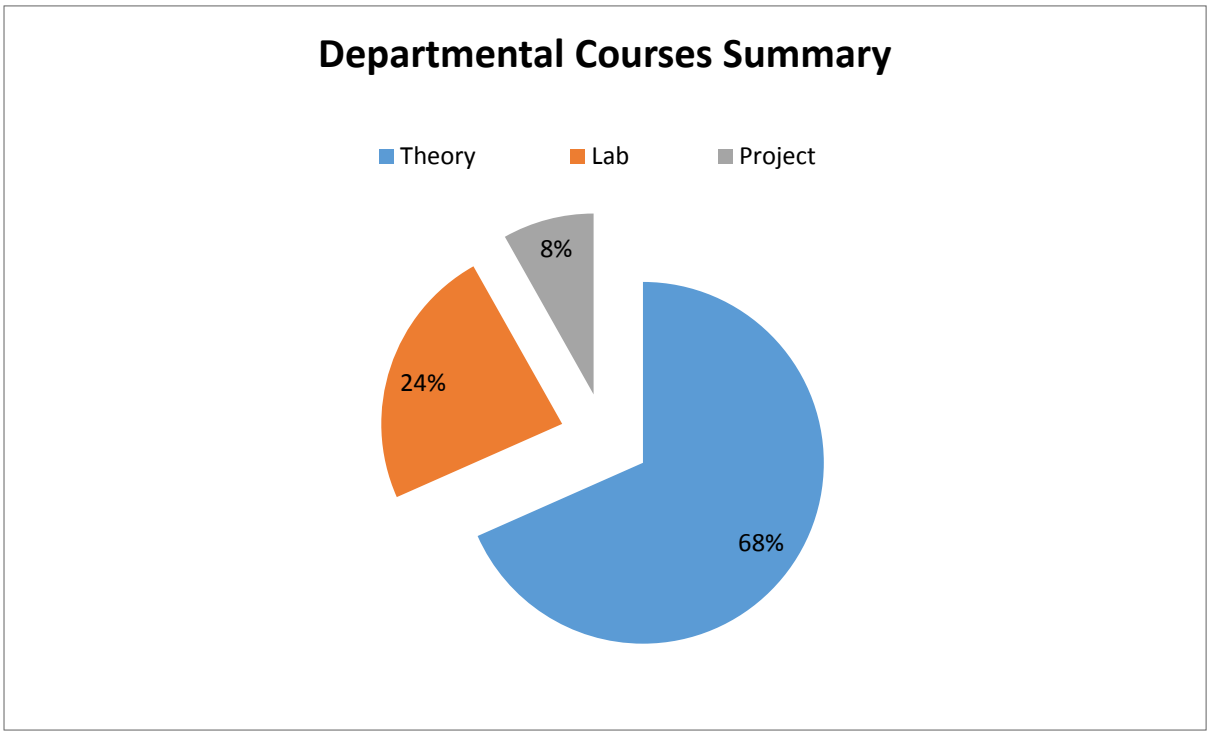


Credit Distribution Summary (Departmental only)

Level / Term	Total Credit (Theory + Lab + Project)
1-1	**3+1.5+0=4.5**
1-2	**3+1.5+0=4.5**
2-1	**6+3+0=09**
2-2	**9+4+0=13**
3-1	**15+3.5+0=18.5**

3-2	**13+5.5+2=20.5**
4-1	**10+2+2=15**
4-2	**8+2+4=18**
Total	**67+23+8=98**

Pie Chart:



CHAPTER 5

DETAIL COURSE OUTLINE OF ALL COURSES OFFERED BY DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

LEVEL -1 TERM –I

Course Title: Discrete Mathematics
Course Code: CSE 113, Credit: 3.0, Total hours: 42

Course Description:

1. Mathematical Logic: propositional calculus and predicate calculus
2. Set theory: sets, relations, partial ordered sets, functions.
3. Mathematical reasoning and proof techniques.
4. Counting: permutations, combinations, principles of inclusion and exclusion.
5. Discrete Probability.
6. Relations and closures.
7. Recurrence relations and recursive algorithms.
8. Growth of functions.
9. Graph theory: graphs, paths, trees, cycles.
10. Algebraic structures: rings and groups.

Course Outcome (CO) Matrix:
At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Solve and verify** logical reasoning problem using different concepts of logical operation and proof.	C5		A2		PO2
CO 2	**Apply** concept of set theory, function and relation in terms of representing discrete entities.	C3				PO1
CO 3	**Use** the concept of number theory for various problem solving and security purposes in the field of Computer Science.	C3		A2		PO6
CO 4	**Apply** different counting techniques for various types of events.	C3				PO1
CO 5	**Apply** the concept of probability in different problem solving process in various domains.	C3				PO1
CO 6	**Analyze** various types of problems in context with graphs and trees for solving purposes	C4				PO2

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		S										
CO2	S											
CO3						M						
CO4	S											
CO5	S											
CO6		S										

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Over view on Syllabus, orientation of lectures and topics of the subject, Discrete Mathematics Applications, Purpose of studying Discrete mathematics.	1	1	
2	Propositional Logic	1	1	
3	Propositional Equivalences	1	1	
4	Predicates and Quantifiers	1	1	
5	Rules of Inference	1	1	
6	Introduction to proofs, proofs methods and strategies.	1,3	1	
7	Sets	2		
8	Set operations	2		
9	Functions	2	1	
10	Sequences and Summations	2	1	
11	Review		1	Lecture 2-10
12	Class Test -1			Lecture 2-10
13	Algorithms	5		
14	The growth of functions, Complexity of algorithms	5	1	
15	The Integers and Division, Primes	5	1	
16	Applications of number theory, Matrices	5	1	
17	Mathematical Induction	3	1	
18	Strong induction and well ordering	3		
19	Recursive definitions and structural Induction	3		
20	Recursive algorithms	3		
21	The Basics of Counting	4	1	
22	The Pigeonhole principle	4	1	
23	Permutations	4	1	
24	Combinations	4	1	
25	Generalized and Generalizing permutations and Combinations	4	2	
26	Class Test-2		2	Lecture 13-25
27	An introduction to Discrete probability, Probability Theory	5	2	
28	Bayes' theorem, Expected value and variance	5		
29	Relations and their properties			
30	n-array relations and their applications, closures of relations	3	1	
31	Graphs and Graph Models,	6	1	
32	Representing Graphs ,Graphs isomorphism	6	1	
33	Connectivity, Graph Coloring	6	1	
34	Euler and Hamilton paths	6	1	

35	Shortest path algorithms, planner graphs	6	1	
36	Introduction to trees, Applications of tress	6	1	
37	Tree Traversal	6	1	
38	Spanning Trees	6	1	
39	Minimum Spanning tress	6	1	
40	Review	6	1	Lecture 27-39
41	Class Test-3		1	Lecture 27-39
42	Review Class, Problem Solutions and Suggestion			Lecture 1-40

Reading Reference:

1. Discrete Mathematics and Its Applications- Kenneth H.Rosen

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Introduction to Computing Sessional
Course Code: CSE 112, Credit: 1.5, Total hours: 42

Course Description:

Introduction to computations; history of computing devices; Computers; Major components of a computer; Hardware: processor, memory, I/O devices; Software: Operating system, application software; Report writing and Presentation; Basic architecture of a computer; Basic Information Technology; Number system: binary, octal, hexadecimal, binary arithmetic; Basic programming concepts; Program development stages: flow charts; Programming constructs: data types, operators, expressions, statements; Introduction to Object Oriented Programming; An Introduction to Python, Reserved Words, Naming Conventions, Basic Python Syntax, Language Components, Indenting Requirements, Control Statements, Bit Manipulation, Python Collections, Functions, Keyword and Optional Parameters, Lambda, Inner Functions, Modules, Exceptions, Input and Output, Classes in Python, Regular Expressions, Writing GUIs in Python, Python and CGI Scripts, The OS Module, Network Programming, Synchronization, Data Compression, Python and Databases.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Recognize the most-up-to-date technology in an ever-changing discipline.

2. Demonstrate the fundamentals of computers and computer nomenclature, particularly with respect to personal computer hardware and software, the Web, and enterprise computing.
3. Learn a handful of various Python advanced topics including high level data structures, network programming.

Mapping of CO and Program Outcomes:

Course Outcomes(CO) of the Course		Program Outcome (PO)											
		1	2	3	4	5	6	7	8	9	10	11	12
CO 1	**Recognize** the most-up-to-date technology in an ever changing discipline.												✓
CO 2	**Demonstrate** the fundamentals of computers and computer nomenclature, particularly with respect to personal computer hardware and software, the Web, and enterprise computing										✓		
CO 3	**Learn** a handful of various Python advanced topics including high level data structures, network programming					✓							

Lecture Outline:

Week	Lecture	Topics
1	Lab 1	MS Word 2010, MS Excel 2010, MS Power Point 2010, Making a Good Power Point Presentation, Grouping for Presentation, Selection of Topics for Presentation
2	Lab 2	Introduction to Computations, History of Computing Devices, Identification of Hardware Parts of PC (Monitor HDD, FDD, CDROM, Data Cord, Motherboard, Microprocessor, Cooling Fan, BIOS Battery, RAM, BIOS Chip, Ports, Keyboard, Mouse etc.)
3	Lab 3	Assemble of Hardware (Starts from Processor Connection to Full Computer Assembling) and Trouble Shootings
4	Lab 4	**Practical Exam** on Lab 02, 03 (Hardware Parts)
5	Lab 5	Introduction to Python, Data Types, Variables, Indentation, Comment, Basic Input/ Output, Naming Convention, Operators and Expressions in Python.
6	Lab 6	Decision Control Statement, Loop Control Statements in Python.
7	Lab 7	Lists in Python, Manipulating Strings in Python
8	Lab 8	Function and Recursion in Python, Keyword and Optional Parameters, Lambda, Inner Functions, Modules
9	Lab 9	Introduction to OOP, Classes in Python, Regular Expressions, Writing GUIs in Python
10	Lab 10	Python and CGI Scripts, The OS Module, Network Programming
11	Lab 11	Synchronization, Data Compression, Python and Databases
12	Lab 12	Exception Handling, Bit Manipulation, Number System: Binary, Octal, Decimal and Hexadecimal Systems, Flow Charts
13	Lab 13	**Final Exam**
14	Lab 14	**Quiz+Group Presentation**

Reading Reference:

1. Computer Fundamentals (7th Edition) – Peter Norton, McGraw Hill Education (2017).
2. The Complete PC Upgrade and Maintenance Guide (16th Edition) – Mark Minasi, Sybex (2005).
3. How to Think Like a Computer Scientist: Learning with Python (2nd Edition) - Allen B. Downey, Samurai Media Limited (2016).
4. Core Python Programming (2nd Edition) - Wesley J Chun, Prentice Hall (2006).

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Chemistry
Course Code: CHE 111, Credit: 3.0, Total hours: 42

Course Description:

1. Atomic structure: Bohr's atomic model, the hydrogen spectrum, quantum mechanical model: Heisenberg's uncertainty principle, de-Broglie equation, Schrödinger's wave equation, quantum numbers, types of quantum numbers: principal quantum number, orbital quantum number, Magnetic Quantum Number, Spin Quantum Number, orbit, orbitals (s, p, d, f), rules for filling the orbitals with electrons and electronic configuration of elements: Aufbau principle, Pauli's exclusion principle, Hund's rule of maximum multiplicity, ground state electronic configuration of elements.
2. Periodic Table: Mendeleev's periodic law and periodic table, Modern periodic law and periodic table, Most modern periodic law and periodic table, rules for the determination of position of the elements in the most modern periodical table, types of elements on the basis of electronic configurations: s-block, p-block, d-block, f-block elements, periodic properties of elements: atomic radius, ionization energy, electron affinity, electro negativity, properties of noble gases.
3. Chemical bonding: Definition, formation and characteristics of different types of bond: ionic bond, covalent bond, metallic bond, coordinate bond, hydrogen bond and its application, nature of covalent bond and shapes of molecules: valence bond theory (VBT).
4. Acid-Base Chemistry: Arrhenius concept of acid base, its utilities and limitations; Bronsted -Lowry concept of acid base, its utilities and limitations, conjugate acid-base pairs, amphiprotonic substances; Lewis concept of acid base, its utilities and limitations; Lux-flood concept of acid base.
5. Thermodynamics and thermo chemistry: (a) Scope and limitations of thermodynamics. System, Surrounding, Intensive and extensive properties. Thermodynamic processes:

isothermal, adiabatic, isobaric, reversible, irreversible processes; first law of thermodynamics, (b) Thermo chemistry: reaction enthalpies; thermo chemical laws: Lavoisier and Laplace's law, Hess's Law (c) Second law of thermodynamics, its necessity; enthalpy, entropy.
6. Nomenclature and Selective Organic Reactions: Alkanes: Structure, nomenclature, conformation, preparation. Alkenes: Structure, nomenclature, conformation, preparation, Alkynes': Structure, nomenclature, conformation, preparation. Alcohol: Structure, nomenclature, conformation, preparation. Aldehyde and Ketones: Structure, nomenclature, conformation, preparation. Carboxylic acids: Structure, nomenclature, conformation, preparation. Amines: Structure, nomenclature, conformation, preparation.
7. Solution and Colloidal Chemistry: Solute, solvent, solution, colloid, suspension, different types of solution: gaseous solution, liquid solution (liq. in liq., solid in liq., gas in liq.), solid solution (liq. in solid, solid in solid, gas in solid); dispersion medium, dispersion phase; saturation: saturated solution, unsaturated solution, supersaturated solution; miscible solution, immiscible solution, properties of colloidal solution and its application; dilute solutions: colligative properties: lowering of vapour pressure, elevation of boiling point, depression of freezing point, osmotic pressure.
8. Chemical Equilibrium: The law of mass action. The equilibrium constant, La-Chatelier principle, application of La-Chatelier principle.
9. Phase Equilibrium: Phase rule and its derivation (phase, component, degree of freedom), one component system (water).
10.Electrochemistry: Faraday's first & second laws of electrolysis with explanation, oxidation, reduction, cathode, anode, electrolyte, electrolysis, conductor, cell reaction, EMF calculation, electrolytic cell (NaCl, PdCl2, KBr), Voltaic cell: Dry cell- construction, reaction, working principle, uses; Lead acid cell- construction, reaction, working principle, uses; Fuel cell- construction, reaction, working principle, uses;
11. Chemistry of Cements: Types of cements, composition of different types of cements and their preparations, chemistry of ceramics, preparation of ceramic material.

12. Water Chemistry: Sources and classification of solid wastes. Water quality parameters: DO, BOD, pH, TDS, Characteristics and perspective of municipal, industrial and hazardous wastes, public health aspects, methods of collection, Environmental problems in Bangladesh

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Explain** different elements, structure of compound, behavior of electron, characteristics of material and hardware based compounds.	C1				PO1
CO 2	**Interpret** different types of material required for engineering propose and understanding of suitable use of chemicals and their purifications for practical uses in engineering sectors.	C2	P1		CT	PO2
CO 3	**Apply** knowledge of basic chemistry on engineering knowledge, electrochemical property, and behavior of engineering materials.	C3	P2	A1	CT	PO3
CO 4	**Analyze** hardware based materials and associated electrochemical property.	C4	P2	A2	CT	PO3

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

Cos	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2		S										
CO3			S									
CO4				M								

Lecture Outline:

Lec. No.	Topic(s) scheduled to be covered	COs	Text Book and Reference	Lecture outcomes/Activities
01	**Atomic Structure:** Definition of Atom, History of Atomic model (Democritus, Aristotle, Dalton, J.J. Thomson, Rutherford, Niels Bohr)	1	1	Knowledge on structure of atom
02	Quantum Mechanical Model (Werner Heisenberg Uncertainty Principle, Louis de Broglie, Erwin Schrodinger)	1,2	1	Knowledge on behavior of electrons
03	Quantum numbers, types of quantum numbers: principal quantum number, orbital quantum number, Magnetic Quantum Number, Spin Quantum Number, orbit, orbitals (s, p, d, f).	1	1	Understanding Behavior of electron
04	Rules for the electronic configuration of elements: Aufbau principle, Pauli's Exclusion Principle, Hund's Rule.	1	1	Electronic configuration of atom
05	**Periodic Table:** Importance of periodic table, Periodic Laws and Table: Mendeleev's, Mosley's and Most modern periodic table.	1	1	Introduction with periodic table
06	Classification of elements in the Periodic Table: (1) Based on Electronic Configuration and (2) Electron losing and gaining property	1	1	Periodic properties/characteristics of element

07	Periodic trends of element: Melting points and boiling points; Atomic Size & Covalent Radius; Ionisation Energy; Electron affinity; Electronegativity.	1	1	Periodic properties of element
08	**Chemical bond:** Ionic bond, covalent bond, formation and characteristics of these bonds, octet rule, polar and nonpolar covalent bond.	1	1	Formation of chemical bond
09	Coordinate Covalent Bond, Hydrogen Bonding, Applications for Hydrogen Bonds, Metallic bond, Shape of molecule.	1	1	Types of covalent bond
10	**CT-01: Atomic Structure and Periodic Table**			
11	Molecular Orbital Theory (MOT)	1	1	Understanding MOT
12	**Acid-Base Chemistry:** Arrhenius and Bronsted Lowry Theory of acid-base.	1	1	Acid Base knowledge
13	Lewis and Lux–Flood Theory of acid-base, pH, Basic pH scale calculation, Different uses of pH in daily life	1	1	Uses of calculation of acid base.
14	Ionization of water, pH calculation and acid rain	1,2	1	Uses of pH concept.
15	**Solution Chemistry:** Solution, Components of a solution, miscible & immiscible liquids, Saturation.	1	2	Learning of solution
16	Colloids, characteristics of a colloid, Classification of colloid.	1	2	Application of solution concept.
17	**CT-02: Chemical Bond**			
18	Properties of colloid, its destruction and applications	1	2	Application of solution concept.
19	Colligative properties and its Classification	1	2	Application of solution concept
20	**Thermodynamics and thermos chemistry**: Thermochemistry, enthalpy change, Mathematical problems	1,2	4	Knowledge and application of thermochemistry
21	The language of TD, two laws of thermodynamics, thermodynamic processes.	2,3	4	Application of thermodynamics

22	Thermochemical laws: Lavoisier and Laplace law and Hess's Law	3.4	4	Problem solution on thermodynamics
23	**Chemical Equilibrium:** Forward reaction, Backward reaction, Reversible reaction, Chemical Equilibrium, Dynamic nature of equilibrium, Reversible Reactions and Rate, Magnitude of K_{eq} , Law of Mass Action, Le Châtelier's Principle.	1	4	Knowledge of Chemical Equilibrium
24	Effect of Various Changes on Equilibrium, Le Châtelier's Principle.	3	4	Problem solving on chemical equilibrium
25	**Phase Equilibrium:** Phase of a System, Component of a System, Variables of a system, Extensive property, Intensive property, Gibbs Phase Rule, Degree of freedom, Phase diagram(One Component System), Triple point, Critical point	3,4	4	Understanding of phase transition with graph
26	**Chemistry of Cements**: Definition of Cement, Composition of Cement & manufacturing of cement.	1,2	4	Basic knowledge of metallurgy or engineering material.
27	Types of Cement, quality test of cement.	3,4	4	Application of knowledge of construction material
28	Definition of Ceramics, raw materials, types	1,2	4	Ceramic production and use.
29	**CT-03: Chemical Equilibrium and Phase Equilibrium**			
30	Types of Ceramics and their uses	3,4	4	Uses of synthetic material
31	**Water Chemistry**: Importance of water quality; Water Quality Variables: Temperature; Dissolved Oxygen (DO); Biochemical oxygen demand (BOD).	3,4	4	Water quality
32	Water quality parameter: Chemical Oxygen Demand (COD); Threshold limit value (TLB); Alkalinity.	1,2	4	Water purification

33	**Electrochemistry:** Electrochemical cells, electrochemical conventions, types of electrochemical cells, reversible & irreversible cell, types of electrode.	3.4	4	Electrochemical knowledge
34	Dry Cell Battery, Ni-Cad Battery, Rechargeable battery, Rust formation**.**	1,2	2	Application of chemical knowledge in electricity production.
35	Faraday's Law of Electrolysis	2,3	2	Construction and working principle of electrochemical cell.
36	Fuel Cells	3,4	2	Problem solving with electrolysis
37	**CT-04: Electrochemistry / Acid base**	1,2		Alternative sources of electricity.
38	**Organic Chemistry:** Organic Naming Rules	1	1	Synthetic products
39	Preparation of Alkane, Alkene & Alkyne Compounds, phenolic group.	1	1	Polymeric compound
40	Confirmation tests of functional groups.	1	1	Functional group
41	Confirmation Tests for Alcohol, Carbonyl, Carboxylic acid and amino group	1	1	Functional group
42	Review (Whole Syllabus)			

Reading Reference:

1. Selected Topics in Inorganic Chemistry-Wahid U. Malik; G.D. Tuli; R.D. Madan;
2. Physical Chemistry-Haque and Nawab;
3. Advance physical chemistry, Monimul Huque
4. Industrial chemistry, BK Sharma

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Chemistry Sessional
Course Code: CHE 112, Credit: 1.0, Total hours: 28

Rationale:

This course is a laboratory course for the basic chemistry in the field of inorganic and physical chemistry. The course will be emphasized by fundamental experiments on different fields of chemistry which can be applicable in a wide spectrum of engineering disciplines. This laboratory course will enable students to understand basic chemistry practically as well as do work with team or individual.

Objective:

- To develop basic chemistry knowledge practically
- To practice the use of basic scientific instrument.

Course Content:

Quantitative chemical analysis in the field of inorganic and physical chemistry such as: Acid-base titration, Redox titration, Iodometric and Iodometric titration, Complexometric titration. Na2-EDTA) Solution by using Eriochrome black T indicator.

Course Outcomes and Skill Mapping:

No.	COURSE OUTCOMES (COs)	PROGRAMME OUTCOMES (POs)											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
1	**Define** the different parameters regarding inorganic and physical Chemistry.	✓											
2	**Describe** the different phenomena regarding acid-base, iodo-iodimetric, complexometric and redox titration etc.	✓											
3	**Construct** Experiments by an individual or by a group to determine different phenomena regarding acid-base, iodo-iodimetric, complexometric and redox titration etc.									✓			

Course Outcomes and Generic Skills:

No.	Course Outcomes	Corresponding POs	Bloom's Taxonomy	CP(WP)	CA(EA)	KP(WK)	Assessment Methods
CO1	**Define** the different parameters regarding inorganic and physical chemistry.	1	C1	-	-	1	Quiz
CO2	**Describe** the different phenomena regarding acid-base, iodoiodimetric, complexometric and redox titration etc.	1	C1	-	-	1	Test, Final Exam
CO3	**Construct** Experiments by an individual or by a group to determine different phenomena regarding acid-base, iodo-iodimetric, complexometric and redox titration etc.	9	C3	-	-	2	Test, Final Exam
WP= Washington Accord Complex Problem Solving/ CP= Complex Problem Solving; EA= Engineering Activities/ CA= Complex Activities; WK= Washington Accord Knowledge Profile/ KP= Knowledge Profile							

Lecture Outline:

Week	Topics	Remarks
1	Orientation and Introductory lecture	Quiz, Test, Final Examination, Report
2	Standardization of Sodium Hydroxide (NaOH) Solution with Standard Oxalic Acid dihydrate ($C_2H_2O_4.2H_2O$) Solution	
3	Standardization of Hydrochloric Acid (HCl) Solution with Standard Sodium Hydroxide (NaOH) Solution.	
4	Standardization of Hydrochloric Acid (HCl) Solution with Standard Sodium Carbonate (Na_2CO_3) Solution	
5	Determination of Calcium (Ca) Content in a Calcium Chloride dihydrate ($CaCl_2.2H_2O$) Solution with Standard Di-Sodium Ethylenediaminetetraacetic Acid (Na_2EDTA) Solution.	
6	Mid Term	

7	Standardization of Sodium Thiosulphate Pentahydrate ($Na_2S_2O_3.5H_2O$) Solution with Standard Potassium Dichromate ($K_2Cr_2O_7$) Solution.	
8	Estimation of Copper (Cu) Content in a Copper Sulphate Pentahydrate ($CuSO_4.5H_2O$) (Blue Vitriol) Solutions by Iodometric Method with Standard Sodium Thiosulphate Pentahydrate ($Na_2S_2O_3.5H_2O$) Solution	
9	Standardization of Potassium Permanganate ($KMnO_4$) Solution with Standard Oxalic Acid dihydrate ($C_2H_2O_4.2H_2O$) Solution	
10	Determination of Ferrous (Fe) Content in a Ammonium Ferrous Sulphate (Mohr`s Salt) [$FeSO_4(NH_4)_2SO_4.6H_2O$] Solution with Standard Potassium Permanganate ($KMnO_4$) Solution	
11	Revision class and final lecture	-
12	Exam	-
13	Viva	-
14	Reserved for exam (if required)	-

Reading Reference:

1. G.H. Jeffery, J. Bassett, J. Mendham, R.C. Denney, Vogel's Textbook of Quantitative Chemical Analysis, 5th Edition, Longman Scientific & Technical, 1989
2. G. D. Christian., Analytical Chemistry, 6th Edition, Wiley India Pvt. Limited, 2007

Assessment Methods:

Components	**Marks %**	**CO**	**Bloom's Taxonomy**
Continuous Assessment Class performance Report Writing	 10% 30%	CO1,CO4	C1,C2
Final Examination Lab Test Viva Quiz	 30% 10% 20%	CO1,CO2,CO3	C1,C3
Total	**100%**		

Course Title: English Reading & Speaking
Course Code: ENG 121, Credit: 2.0, Total hours: 28

Rationale:

To provide fundamental knowledge on how to improve spoken English skills and to enable to write more effectively in English.

Objective:

1. To give the students exposure to different types of texts in English in order to make them informed and critical reader.
2. To gain an understanding of the underlying writing well-organized paragraphs and also to teach how to edit and revise their own as well as peer's writing.
3. To teach grammar and vocabulary in a contextualized way.
4. To teach how to write formal letters for a range of academic purposes.
5. To develop skills to communicate effectively and professionally.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Students will develop their reading skills through various reading techniques which will enable them to understand, analyze and evaluate reading texts.
2. By practicing writing different types of paragraphs, students are going to acquire a good command over structure and techniques of paragraph writing.
3. Vocabulary and grammatical structure from extensive reading will help to develop student's ability to express themselves through writing for academic and other purposes.
4. Students will acquire skills in summarizing, paraphrasing, synthesizing and explaining information from different sources.
5. Employ strategies of pre-writing, drafting, and revising, taking into consideration rhetorical purpose, the knowledge and needs of varied writing contexts, and the feedback of instructors and peers.
6. Demonstrate the knowledge of genre conventions and structure (e.g., introductions, paragraphing, transitions) in ways that serve the development and communication of information and ideas.
7. Edit such that choices in style, grammar, spelling, and punctuation contribute to the clear communication of information and ideas.
8. Reflect on what contributed to their writing process and evaluate their own and their peers' work.
9. Become an academic writer and critic of any text written in English.
10. Participate successfully in the discourse of the specific community and beyond.

Course Content:

Reading: Reading techniques: scanning, skimming, predicting, inference, analysis, summarizing and interpretation of texts; Academic reading: comprehension from subject related passages, article reading, research paper reading, newspaper reading, reading selected short stories, Reading for book review, report review, case study review, Reading from departmental text for literature review.

Writing: Introductory discussion on writing, prewriting, drafting, topic sentence, paragraph development, paragraph structure, describing a person/scene/picture, narrating an event, paragraph writing, Compare-contrast and cause- effect paragraph, Essay writing: writing steps, principles and techniques, outlining, revising, editing, proofreading etc., narrative and descriptive essays, comparison-contrast and cause – effect essays, argumentative essay,

summary, paraphrase, E-mail, Report: purpose of a report, classification of report, organizing a report, writing short report, analysis and illustration of report, problems in writing reports, term paper writing: introduction to writing of a term paper and its methodologies.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes (CO) of the Course	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
Students will develop their reading skills through various reading techniques which will enable them to understand, analyze and evaluate reading texts.	√											
By practicing writing different types of paragraphs, students are going to acquire a good command over structure and techniques of paragraph writing.	√											
Vocabulary and grammatical structure from extensive reading will help to develop student's ability to express themselves through writing for academic and other purposes.		√										
Students will acquire skills in summarizing, paraphrasing, synthesizing and explaining information from different sources.	√											
Employ strategies of pre-writing, drafting, and revising, taking into consideration rhetorical purpose, the knowledge and needs of varied writing contexts, and the feedback of instructors and peers.			√									
Demonstrate the knowledge of genre conventions and structure (e.g., introductions, paragraphing, transitions) in ways that serve the development and communication of information and ideas.		√										
Edit such that choices in style, grammar, spelling,												
and punctuation contribute to the clear communication of information and ideas.		√										
Reflect on what contributed to their writing process and evaluate their own and their peers' work.			√									
Become an academic writer and critic of any text written in English.				√								
Participate successfully in the discourse of the specific community and beyond.			√									

Lecture Schedule:

Week	Lecture	Topics	Class Test
1	Lec 1 Lec 2	Introduction to Language, Methodologies/Approaches of English Teaching-Learning Process	
2	Lec 3 Lec 4	Introduction to Phonetics, IPA Transcription and variation of accents	**Class Test 1**
3	Lec 5 Lec 6	Types of Sentences, Introduction to Tense	
4	Lec 7 Lec 8	Use of Main Verbs, Auxiliaries	
5	Lec 9 Lec 10	Modals & Semi modals, Causatives, Conditionals	**Class Test 2**
6	Lec 11 Lec 12	Active / Passive Sentences, Forming Questions	
7	Lec 13 Lec 14	Adverbs and Adverb related structures	
8	Lec 15 Lec 16	Problems with Adjectives, Conjunction, Prepositions	
9	Lec 17 Lec 18	Reading: Skimming, and Scanning	**Class Test 3**
10	Lec 19 Lec 20	Reading Comprehension, Selected short Stories	
11	Lec 21 Lec 22	Writing: Writing Strategies. General Correspondence & Communication/Tech	
12	Lec 23 Lec 24	Writing: Introduction to General Correspondence	
13	Lec 25 Lec 26	Formal Communication	**Class Test 4**
14	Lec 27 Lec 28	Report Writing, how to write a curriculum Vitae	

Reading Reference:

1. Langan, J. (2005). College Writing Skills with Readings (6th Ed). McGraw-Hill Publication
2. Interactions 1 (Reading), John Langan, Latest edition, McGraw-Hill Publication
3. Hand-out will be provided by the instructors.

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Differential and Integral Calculus
Course Code: MATH 111, Credit: 3.0, Total hours: 42

Course Description:

Differential Calculus: Limits, continuity and differentiability. Successive differentiation of various types of functions. Leibnitz's theorem. Rolle's theorem, Mean value theorem, Taylor's and Maclaurin's theorems in finite and infinite forms. Lagrange's form of remainders. Cauchy's form of remainders. Expansion of functions, evaluation of indeterminate forms of L' Hospital's rule. Partial differentiation. Euler's theorem. Tangent and normal. Sub tangent and subnormal in Cartesian and polar co-ordinates. Determination of maximum and minimum values of functions. Curvature, Asymptotes.
Integral Calculus: Integration by the method of substitution. Standard integrals. Integration by successive reduction. Definite integrals, its properties and use in summing series. Wallis's formulae. Improper integrals. Beta function and Gamma function. Area under a plane curve and area of a region enclosed by two curves in Cartesian and polar co-ordinates. Volumes and surface areas of solids of revolution.

Course Outcome (CO) Matrix:
At the end of the course, the students should be able to:

		Level of Domain				
		C	P	A	S	PO
CO 1	Introduced to differential and integral calculus of functions of one or more variables, an involving continuous change of variables and some basic tools which are useful in modelling and analyzing physical phenomena.	C1				PO1
CO 2	Identify the extrema of a function on an interval and classify them and implement knowledge of partial derivative to find the tangent-normal , curvature and areas and volumes using integrals geometrically and mathematically.	C2				PO1

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2	S											

Lecture Plan:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Discussion of Course Details	1	1	Book/ Slides
2	Limit	1	1	Calculate the limit of a function at a point numerically and algebraically using appropriate techniques.
3	Fundamental Properties of Integration.	2	2	Use the basic integration rules to find indefinite integrals
4	Limit	1	1	Identify situations where limit fails to exist.
5	Continuity	1	1	Use the limit concept to determine where a function is continuous.
6	Method of Substitution	2	2	Use substitution techniques to determine the definite and indefinite integrals of functions.
7	Continuity and Discontinuity	1	1,3	Find points of discontinuity for functions.
8	Differentiability	1	1,3,4	Interpret the derivative as a rate of change.
9	Method of Substitution	2	2,3	Use substitution techniques to determine the definite and indefinite integrals of functions.
10	**CT-01**			Limit, Continuity and Discontinuity
11	Differentiability	1	1,2	Compute the expression for the derivative of a function using the rules of differentiation Including the power rule, product rule, and quotient rule and chain rule.
12	Differentiability	1	4	Differentiate exponential, logarithmic, and trigonometric and inverse trigonometric functions.

13	Integration By Parts	2	2	Apply integration by parts find anti-derivatives.
14	Successive differentiation	1	1, 3	Obtain expressions for higher order derivatives of a function using the rules of differentiation.
15	Successive differentiation	1	4	Obtain expressions for higher order derivatives of a function using the rules of differentiation.
16	**Assignment-01** Integration By Parts	1	1	Differentiability, Successive differentiation, Fundamental Properties of Integration.
				Use integration by parts in conjunction with other techniques (substitution, basic anti-derivatives, etc.).
17	Leibnitz's theorem	1	1,3,4	Apply the Leibnitz's theorem for finding nth derivative of product of two functions
18	Rolle's theorem, Mean value theorem, Taylor's and Maclaurin's theorems in finite and infinite forms	1	3	Understand the consequences of Rolle's, Mean Value, Taylor's and Maclaurin's theorems for differentiable functions.
19	Rational Fraction	2	3	Understand the basic operations of rational number.
20	**CT-02**			Method of Substitution, Integration By Parts.
21	Expansion of Function	1	1	Expand functions using Taylor's and Maclaurin's series.
22	Rational Fraction	2	2	Use the basic operations of rational number to find indefinite integrals
23	Definite Integral	2	2	Evaluate a definite integral using an anti-derivative
24	Expansion of Function	1	1	Expand functions using Taylor's and Maclaurin's series and use their application.
25	Beta and Gamma	2	3	Beta and Gamma functions are used to evaluate integrals.
26	Areas of Plane Curve	2	2	Interpret the definite integral geometrically as the area under a curve

27	Expansion of Function			Expand functions using Taylor's and Maclaurin's series and use their application.
28	Areas of Plane Curve	2	2,3	Compute the area of regions whose boundaries are defined by equations.
29	Length & Volume of Plane Curve	2	2	Find arc length and curvature of space curves.
30	Maxima and Minima	1	1	Identify the extrema of a function on an interval and classify them as minima or maxima using the first derivative test.
31	**CT-03**			Rational Fraction, Definite Integral ,Beta and Gamma
32	Maxima and Minima	1	3	Establish the working rule for finding the maxima and minima of a function using the first and the second derivatives of the function.
33	Partial Differentiation	1	4	Calculate the partial derivatives of the functions of several variables.
34	Partial Differentiation	2	1	Evaluate partial derivatives, including higher order derivatives.
35	Tangent	2	1	Need to find Tangent of curves for analyzing forces acting on a moving body.
36	Normal	2	3	Need to find Normal of curves for analyzing forces acting on a moving body.
37	**Assignment-2** Tangent and normal			Rolle's theorem, Mean value theorem, Expansion of Function, Areas of Plane Curve, Length & Volume of Plane Curve.
		2	3	Need to find Tangent and Normal of curves for analyzing forces acting on a moving body.
38	Curvature	2	1	To understand the concept of curvature.
39	Curvature	2	4	The curvature measures how fast a curve is changing direction at a given point.
40	**CT-04**			Partial Differentiation, Tangent and normal, Maxima and Minima.
41	Asymptote	2	1,4	Find the vertical, horizontal and diagonal asymptotes of a function, if they exist.

42	Review			----------

Reading Reference:

1. Differential Calculus: Das and Mukherjee
2. Integral Calculus: Das and Mukherjee.
3. Calculus-1(Differential & Integral Calculus): Abu Yusuf
4. Differential Calculus: Shanti Narayan.

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Electrical Circuit Analysis
Course Code: EEE 111, Credit: 3.0, Total hours: 42

Course Description:

1. Fundamental electrical concepts and measuring units; Direct current (dc): Current, voltage, resistance, power and energy; Series/Parallel Circuits;
2. Methods of network analysis and Network Theorems; Capacitors, Inductors and introduction to magnetic circuits.
3. Alternating current (ac): Instantaneous current, voltage and power for various combinations of R, L and C circuits; Effective current and voltage, average power;
4. Phasor representation of sinusoidal quantities; Sinusoidal Single-Phase Circuit Analysis; Introduction to three phase circuits; Power factor and power equation (Δ and Y circuits).

Course Outcome (CO) Matrix:
At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Explain** different circuit elements and electrical circuitries characteristics	C2				PO1
CO 2	**Interpret** simplification of complex circuit to speed up solving process by applying different circuit theorems	C2			CT	PO1
CO 3	**Apply** basic circuit principles to the analysis and design of DC and AC electrical circuits	C3			CT	PO2
CO 4	**Demonstrate** knowledge of electrical circuit analysis in a cyber-physical entity		P3		TW	PO9

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2	S											
CO3		S										
CO4									M			

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Introduction to circuit and circuit elements	1	1	Name basic electrical components and use
2	Basic concepts of basic electric circuit elements (Charge, current, voltage, power, sources, and their types, resistor etc.), Sign convention for passive components, Sign convention for passive components and sources	1	1	Relate different D.C quantities of electrical networks
3	Ohm's Law, Nodes, Branch and loops,	1	1	Solve simple circuits to find current and voltages
4	Kirchhoff's laws, basic circuit solving	1	1	Solve simple circuits to find current and voltages
5	Equivalent resistance (series, parallel & mixed)	1	1	Differentiate resistance orientations and application
6	Equivalent resistance (Voltage division and current division)	1	1	Differentiate resistance orientations and application
7	Review			Lecture 1 - 6
8	Class Test 1			Lecture 1 - 6
9	Wye-delta transformations	2, 3	1, 3	Solve large circuits
10	Nodal analysis	2, 3	1, 3	Solve large circuits
11	Mesh analysis	2, 3	1, 3	Solve large circuits
12	Problem Solving			Lecture 9 – 12
13	Superposition Theorem	2, 3	1, 3	Simplify complex circuit with resistors and sources
14	Thevenin's theorem	2, 3	1, 3	Simplify complex circuit with resistors and sources
15	Norton's theorem	2, 3	1, 3	Simplify complex circuit with resistors and sources
16	Maximum power transformation	2, 3	1, 3	Explain constraints in power transfer
17	Problem Solving			Lecture 13 – 16
18	Review			Lecture 9 – 16
19	Class Test 2			Lecture 9 – 16

20	Introduction to Alternating currents, sketching waves, Analog value, RMS value, peak factor, Form factor…etc.	3	2	Relate different A.C. quantities of electrical networks
21	Capacitor I-V equations, A capacitor integrates current, Capacitor i-v equation in action	3	2	Natural and forced responses
22	Problem Solving			Lecture 20 – 21
23	Inductor equations, Inductor, Inductor i-v equation in action, RL natural response	3	2	Recognize Natural and forced responses
24	RC natural response, RC step response	3	2	Recognize Natural and forced responses
25	Sketching exponentials, LC natural response, RLC natural response	3	2	Formulate RLC networks
26	Problem Solving			Lecture 23 – 25
27	Review			Lecture 20 – 25
28	Class Test 3			Lecture 20 – 25
29	Phasor diagram	3	2	Explain phasor calculation
30	Phasor Algebra	3	2	Explain phasor calculation
31	Basic Theorems for AC circuit	3	2	Analyze simple A.C. circuits
32	**Series and Parallel AC Circuits**	3	2	Analyze simple A.C. circuits
33	Power calculation of AC-Circuits	3	2	Solve Power calculation
34	Sinusoidal Single-Phase Circuit Analysis	3	2	Analyze Single-Phase Circuit
35	Project Assignment and Presentation	4		
36	Introduction to three phase circuits	3	2	Analyze three phase circuit
37	Magnetic circuits and device operation	3	2	Interpret the functionality of different devices
38	Power factor and power equation (Δ and Y circuits)	3	2	Solve Power calculation
39	Problem Solving			Lecture 29 – 34 and Lecturer 36 – 38
40	Review			Lecture 29 – 34 and Lecturer 36 – 38
41	Class Test			Lecture 29 – 34 and Lecturer 36 – 38
42	Review Class, Problem Solutions and Suggestion			Lecture 1 – 41

Reading Reference:

1. Introductory Circuit Analysis - Robert L. Boylestad.
2. Alternating Current Circuits - Russel M Kerchner and George F Corcoran.
3. Fundamentals of Electric Circuits - Charles K Alexander, Mathew N O Sadiku.
4. Introduction to Electric Circuits - Richard C. Dorf, James A. Svoboda.

Assessment Methods:

Category	Marks %
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15

Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Electrical Circuit Analysis Sessional
Course Code: EEE 112, Credit: 1.5, Total hours: 42

Rationale:

To perform several experiments to verify practically the theories and concepts learned and understand the difference between theory and application.

Objective:

1. To understand the basic electrical engineering principles and abstractions on which the design of electronic systems is based.
2. To analyze and design simple electronic circuits.
3. To understand the relationship between the mathematical representation of circuit behavior and corresponding real-life effects.

Course Description:

Construction and operation of simple electrical circuits, verification of KVL, verification of KCL, verification of superposition principle, verification of Thevenin's theorem, familiarization with Alternating Current (Ac) waves, study of R-L-C series circuit, different types of filters and its characteristics with different input frequency, series resonance and parallel resonance.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Recognize how to develop and employ circuit models for elementary electrical components, e.g. resistors, sources, inductors, capacitors etc.
2. Generate sine, square and triangular waveforms with required frequency and amplitude using function generator.
3. Compare to use various methods of circuit analysis both for DC and AC networks and their solution methods.
4. Appraise the characteristics of different electronic devices.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes (CO) of the Course	**Program Outcome (PO)**											
	1	2	3	4	5	6	7	8	9	10	11	12
Recognize how to develop and employ circuit models for elementary electrical components, e.g. resistors, sources, inductors, capacitors etc.						√						
Generate sine, square and triangular waveforms with												

required frequency and amplitude using function generator.						√						
Compare to use various methods of circuit analysis both for DC and AC networks and their solution methods.							√					
Appraise the characteristics of different electronic devices.							√					

Lecture Plan:

Week	Lecture	Topics
1	Lab 1	Construction & Operation of Simple Electrical Circuits
2	Lab 2	Verification of KVL
3	Lab 3	Verification of KCL
4	Lab 4	Verification of Superposition Theorem
5	Lab 5	Verification of Thevenin's Theorem
6	Lab 6	**Lab Test- 01**
7	Lab 7	Familiarization with alternating current (ac) waves
8	Lab 8	Study of R-L-C series circuit
9	Lab 9	Different types of filters and its characteristics with different input frequency
10	Lab 10	Series Resonance and Parallel Resonance
11	Lab 11	**Lab Test- 02**
12	Lab 12	**Quiz**
13 + 14	Lab 13 + 14	**Viva**

Reading Reference:

1. Introductory Circuit Analysis (12th ed)- Robert L. Boylestad, Pearson Education India (2013).
2. Alternating Current Circuits (3rd ed) - Russel M Kerchner and George F Corcoran, J. Wiley & Sons Inc. (1951).
3. Fundamentals of Electric Circuits (6th ed)- Charles K Alexander, Mathew N O Sadiku, McGraw-Hill Education (2016).
4. Introduction to Electric Circuits (6^{9h} ed)- Richard C. Dorf, James A. Svoboda, Wiley (2013).

Assessment Methods:

Category	Marks %
Lab Test	40
Quiz	20
Viva	10
Class Participation	10
Home Assignment/Report	10

Class performance/observation	10
Total	**100**

LEVEL -1 TERM -II

Course Title: Structured Programming Language
Course Code: CSE 121, Credit: 3.0, Total hours: 42

Course Description:

1. Introduction to the fundamental programming concept;
2. Understand basic data types and various operation;
3. Introduction to decision making, various looping statements;
4. Understand different data structure;
5. Understanding structure, union and enum and there uses.
6. Demonstrating various file operation in C.

Course Outcome (CO) Matrix:
At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	1. **Illustrate** the fundamental concept of computer programming.	C1				PO1
CO 2	2. **Explain** various data types and operation in C.	C2				PO1
CO 3	3. **Apply** decision making and looping statements in problem solving**.**	C3				PO2,PO9
CO 4	4. **Illustrate** the advantages of using array and modular programming in problem solving.	C1				PO1
CO 5	5. **Explain** different file operations on text and binary files.	C2				PO2

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2		S										
CO3		S							M			
CO4	S											
CO5		S										

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Introduction to programming language	1	1	What is a Computer? What is inside the computer? What programming language?
2	Introduction to C programming language	1	1	C language, Compiler. C program structure.
3	Introduction to C programming language	1,2	1	Basic data types, Variable.
4	Introduction to C programming language	1,2	1	Header file, return type, main method, reserve words.
5	Operator and expression	1,2	1	Basic arithmetic operation,
6	Operator and expression	1,2	1	Basic arithmetic operation,
7	Decision making in C	3	1	If statement, if – else statement,
8	Decision making in C	3		Nested if – else, problem solving using if – else.
9	Class Test 1	1,2		Lecture 1 – 5
10	Repetition control structure	3	1	Define loop, importance of loop, for loop
11	Repetition control structure	3	1	While loop, do…. While loop.
12	Repetition control structure	3	1	For, while and do… while loop with break and continue, switch statement. Practice some problem with repetition.
13	Modular programing.	4	1	Define function, importance of using function. User define function and library function.
14	Modular programing.	4	1	Declared function, call function with no

				parameter, with parameter.
15	Modular programing.	4	1	Function with void return types, Function with return types.
16	Modular programing.	4	1	Recursion function.
17	Modular programing.	4	1	Practice some problem related to function.
18	Static variable.	2,4	1	Define static variable, pass static variable to function.
19	Class Test 2	3,4		Lecture 10 – 17
20	Array structure	4	1	Define array, importance of array. Declare array.
21	Array structure.	4	1	Access array, Pass array to Function.
22	Array structure.	4	1	Multidimensional Array. Pass multidimensional array to functions.
23	Dynamic memory allocation.	2,4	1	Define what dynamic memory allocation is. Define pointer and its advantage.
24	Dynamic memory allocation.	2,4	1	Declared pointer, pass pointer to function.
25	Dynamic memory allocation.	2,4	1	Operation on pointer.
26	String	2	1	Fundamentals of Strings and Characters. Fundamentals of Strings and Characters.
27	String	2	1	Standard Input/Output Library Functions, String Conversion Function, Comparison Functions.
28	String	2	1	String Manipulation Functions.
29	Class Test 3	2,4		Lecture 20– 25

30	User-defined Data types.	2	2	Introduction to user define data types, introduction to Structure. Declared Structure. Practice with Structure.
31	User-defined Data types.	2	2	Introduction to union, declared Union, practice with union.
32	User-defined Data types.	2	2	Introduction to enum, Declared Enum , Practice with enum.
33	File in C	5	2	Introduction to File in C, importance of file in C.
34	File in C	5	2	Declared File pointer, open file in different mode.
35	File in C	5	2	Sequential file processing, random access file processing.\ Read a file, write into a file and update a file.
36	File in C	5	2	Coding practice with file access.
37	Problem solving	3	2	Competitive programming and problem solving.
38	Problem solving	3	2	Competitive programming and problem solving.
39	Review Class, Problem Solutions and Suggestion	5		Lecture 1 – 14
40	Class Test 4			Lecture 33 – 36
41	Review Class, Problem Solutions and Suggestion			Lecture 15 – 28
42	Review Class, Problem Solutions and Suggestion			Lecture 29 – 38

Reading Reference:

1. Teach Yourself C - Herbert Schidlt
2. Programming in ANSI C by Balagurusamy.

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Structured Programming Language Sessional
Course Code: CSE 122, Credit: 1.5, Total hours: 42

Rationale:

To introduce the fundamental principles, mechanism of programming skills and develop basic programming skills to program design and development.

Objective:

1. To learn basic idea of programming languages.
2. To learn how to program with C.
3. To learn how to think about the problems, their solutions and translating it to programming language.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Discuss algorithm and solve problems using computers.
2. Practically analyze the fundamental principles, typical characteristics and mechanisms of a structured programming language.
3. Apply practical knowledge to develop basic programming skills with respect to program design and development.

Course Content:

Basic Knowledge: Mathematical problems using printf, scanf, Operators, If, Else if, Switch; Programming Concept: Loop, Nested Loop, Function, Recursion; Dynamic Concept: Arrays and Strings, Multidimensional Array and Pointers, User Defined Data Types: Structures, Unions, Enumerations, File I/O, Header files, Preprocessor. Bitwise Operations & Dynamic memory allocation.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
Discuss algorithm and solve problems using computers.									√			
Practically analyze the fundamental principles, typical characteristics and mechanisms of a structured programming language.						√						
Apply practical knowledge to develop basic programming skills with respect to program design and development.						√						

Lecture Outline:

Week	Lab	Topics
1	Lab 1	Mathematical problems using printf, scanf
2	Lab 2	Number System, memory allocation
3	Lab 3	Control Structure: “if else”, “else if”, “switch”
4	Lab 4	Control Structure: Problem on Loop- For, Do While
5	Lab 5	Problem on Nested Loop
6	Lab 6	Function, parameter passing convention
7	Lab 7	**Lab Test-1**
8	Lab 8	Problem on Recursion
9	Lab 9	Problem on Pointer, Array, String
10	Lab 10	Problem on Multidimensional Array, Dynamic Memory Allocation
11	Lab 11	Problem on File IO, Problem on Structure, Union
12	Lab 12	Problem on Bitwise Operation
13	Lab 13	**Lab Test-2**
14	Lab 14	Quiz/Viva + Project

Reading Reference:

1. Teach Yourself C - Herbert Schidlt
2. Programming in Ansi C - E Balagurusamy
3. C: The Complete Reference - Herbert Schildt
4. C Programming Language – Dennis M. Ritche
5. Sober Jonno Computer Programming Language C- Md Kamruzzaman Niton

Assessment Methods:

Category	Marks %
Lab Test-1	25
Lab Test-2	25
Viva	20
Class Participation	10
Home Assignment/Report	10
Class performance/observation	10
Total	**100**

Course Title: History of the Emergence of Bangladesh
Course Code: GED 121, Credit: 2.0, Total hours: 28

Course Description:

Introduction: Introduction of people and country including Land, Rivers, Hills, Tribes, Religion, Weather & Climate, Regional language etc., Importance of learning history of the emergence of independent Bangladesh 2. Brief Background History: The Mughal Bengal, Bhuyan's Bengal, Siraj-Ud-Daulah & Plassey's battle, British Bengal, Mughal Structure and Architecture etc. 3. Bilingualism, Pakistan state structure and Discrimination: Sectarian politics, Lahore Proposition in 1940, Banga vanga and its causes, Discrimination between east and west Pakistan 4. Language Movement and Trial for Establishing Democratic Government: Different stages in language movement, Establishment of Juktofront in 1954, 21 point commitment of Juktofront election and its results, Juktofront government and its activities. 5. Martial Law of Ayub Khan and Development of Nationalism: Ayub martial law in 1958-1962, Reform activities of Ayub government, Revolution against Ayub, Presidential election of 1965, Resistance against Bangali cultural aggression, 6 point movement of Sheikh Mujibur Reahman, Agartala conspiracy 6. Mass Orchestration of 1969 and 11 Point Movement 7. Election of 1970 and Banghobondhu's Declaration of Independence 8. Liberation War of 1971: 25th March of 1971, Mujibnagar Government, Stages of war, Publicity of liberation war, Role of world superpower, Activities of Shanti committee; Al-badr; Al-shams and Razakar's. Killing of Bengali Intellectuals.
Government and politics of Bangladesh: Democratic Practice, constitution and its major amendments, Bureaucracy and its functions, Political economy of Bangladesh, Bangladesh agriculture, Bangladesh industry, Unemployment, Role of NGO, Foreign policy of Bangladesh, Regional and international organizations: SAARC, ASIAN, UNO etc. 10. Major Tourist Places in Bangladesh: The world's largest sea beach in Cox's Bazar, St. Martin's Island, Mangrove Forest in Sundarban, Sixty Dome Mosque, Lalbagh Fort, Ahsan Manjil, Sonargoan Museum, Jatiya Sangsad Bhaban, Jatiya Sriti Shoudho etc.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Bloom's Taxonomy Level

Sl. No.	COs By the end of this course, students are expected to:	POs	Bloom's Taxonomy		
			C	A	P

1	Understand importance and develop proper knowledge about independence of their nation	6	2		
2	Gain insight about the major historical events that lead to independence and the sacrifices of freedom fighters in liberation war	7	3		
3	Gather knowledge about the history of different aspects of Bangladesh	7	2		

Mapping of COs with Knowledge Profiles, Complex Engineering Problem Solving and Complex Engineering Activities

Course Outcome	Knowledge Profile	Complex Problem Solving	Complex Engineering Activities
CO1	K7		
CO2	K7		
CO3	K7		

Week wise distribution of contents and assessment methods

Week	Topics	Assessment Method(s)
1	Introduction of people and country including Land, Rivers, Hills, Tribes, Religion, Weather & Climate, Regional language etc. Importance of learning history of the emergence of independent Bangladesh.	
2	The Mughal Bengal, Bhuyan's Bengal, Siraj-Ud-Daulah & Plassey's battle, British Bengal, Mughal Structure and Architecture etc	
3	Sectarian politics, Lahore Proposition in 1940, Banga vanga and its causes, Discrimination between east and west Pakistan	**Quiz-1**
4-5	Different stages in language movement, Establishment of Juktofront in 1954, 21 point commitment of Juktofront election and its results, Juktofront government and its activities	
6-7	Ayub martial law in 1958-1962, Reform activities of Ayub government, Revolution against Ayub, Presidential election of 1965, Resistance against Bangali cultural aggression, 6 point movement of Sheikh Mujibur Reahman, Agartala conspiracy	**Quiz-2**
8	Mass Orchestration of 1969 and 11 Point Movement	
10	Election of 1970 and Banghobondhu's Declaration of Independence	**Quiz-3**
11	25th March of 1971, Mujibnagar Government, Stages of war, Publicity of liberation war, Role of world superpower, Activities of Shanti committee; Al-badr; Al-shams and Razakar's. Killing of Bengali Intellectuals.	
12	Democratic Practice, constitution and its major amendments, Bureaucracy and its functions, Political economy of Bangladesh, Bangladesh agriculture, Bangladesh industry, Unemployment, Role of NGO, Foreign policy of Bangladesh, Regional and international organizations: SAARC, ASIAN, UNO etc	**Quiz-4**
13	The world's largest sea beach in Cox's Bazar, St. Martin's Island, Mangrove Forest in Sundarban, Sixty Dome Mosque, Lalbagh Fort, Ahsan Manjil, Sonargoan Museum, Jatiya Sangsad Bhaban, Jatiya Sriti Shoudho etc.	
14	Review classes	

References

Required :

1. History of the Emergence of Independent Bangladesh by Muntassir Mamoon, Md. Mahbubur Rahman. Published by the University Grants Commission of Bangladesh. Dhaka. December 2017.

Recommended

1. Lonely Planet Bangladesh by R. Plunkett, A. Newton, and B. Wagenhauser

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: English Writing and Listening
Course Code: ENG 121, Credit: 3.0, Total hours: 42

Rationale:

The course is designed to help students develop writing and listening skills for academic Excellence as well as for their real-life context. Writing is undeniably one of the most important skills required for almost all other tasks. Writing ability opens doors to any place, situation, and opportunity. In fact, Writing module helps students to learn how to write effective diverse types of paragraphs and essays. While listening module enables students to listen carefully and thoroughly with a critical mind and respond accordingly.

Course Objectives:

This course has been designed
To develop students' creative skills in writing tasks
To enable the students to receive input-listening basics

Matrix of Course Learning Outcomes (CLOs):

Upon successful completion of this course, students should be able:		Learning Domains			
		C	A	P	S
CLO 1	to write different types of essays	√			√
CLO 2	to write different types of paragraphs	√			√
CLO 3	to evaluate the quality of diverse types of essays	√	√		√
CLO 4	to use different writing strategies and	√			√
CLO 5	to listen and respond appropriately by picking up the main ideas, key details and inferred meanings,	√	√		

	etc. from short, authentic news clips or in person communication				

*C: Cognitive; A: Affective; P: Psychomotor; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CLOs with PLOs

PLOs⇒	**PO 2**	**PO 4**	**PO 9**	**PO 10**	**PO 12**
CLOs ⇓					
CO *1*	**L**	**L**	**M**	**S**	**M**
CO *2*			**M**	**S**	**M**
CO *3*		**M**	**M**	**S**	**L**
CO *4*				**S**	**L**
CO 5		**L**		**S**	**L**

Legends:
S = Strong**, M** = Medium**, L** = Low

Course Contents:

Listening, paragraph, style in written English, punctuation, vocabulary, Listening, different types of essays, sentence variety, vocabulary

Lecture Outline:

Sl./ Class	**Topics/ Assignment**	**Specific Outcomes**	**Teaching Strategies & Suggested Activities**	**Teaching Aids/Tools / Materials**	**Assessment Strategies**	**Alignment with CLOs**
1.	1.General discussion on two modules of writing and Listening 2.Basic listening tips and techniques 3.Listening to conversations between two persons on social contexts To describe fundamental techniques of Listening To apply their skills in listening test To locate and select relevant information		Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5
2.	Listening: conversation between two people set in an everyday social context	To Identify missing information To make use of contextual clues	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5

3.	Listening: conversation between two people set in an everyday social context	To Identify missing information To make use of contextual clues	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5
4.	Listening: conversation between up to four people set in an educational or training context	To make use of contextual clues To locate and select relevant information	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5
5.	Listening: conversation between up to four people set in an educational or training context	To make use of contextual clues To locate and select relevant information information;	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5
6.	**Class Test 01** (*Written)*					
7.	1. What is a paragraph? 2. Different kinds of 3. paragraphs 4. Paragraph 5. Marking Criteria 6. development by example	to make a list of different parts of a paragraph , ,and write example paragraphs	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO2
8.	i.Paragraph development by listing ii.use of linkers	to write listing paragraphs	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO2
9.	1.Paragraph development by comparison ii.use of linkers	to write contrast paragraphs to write comparison paragraphs	Lecture, Discussion	PPT Slides, Online resource , Text, "Historicization: Questioning the Present through the	Q/A Feedback	CLO2

10.	i.Paragraph development by contrast ii.uses of punctuation	to write generalization paragraphs	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO2
11.	Paragraph development by cause and effect ii.uses of punctuation					
12.	i.Paragraph development by classification ii.uses of punctuation	to write compare -contrast paragraphs	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO2
13.	1.Paragraph development by generalization					
14.	Paragraph development by space and time					CLO 2
15.	1.Style in Written English-i 2.vocabulary-i	Solve stylistic problems in English	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO2
16.	1.Style in Written English-ii 2. vocabulary-ii	Solve stylistic problems in English	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO1 & CLO2
17.	1.Paragraph development by process 2.vocabulary-iii		Lecture, Discussion			CLO2
18.	Quiz Test 03 (*Short Answer)*					
19.	Listening: conversation between two people set in an everyday social context	To Identify missing information To make use of contextual clues	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5
20.	Listening: conversation between up to four people set in an	To make use of contextual clues To locate and	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5s

	educational or training context					
21.	Review class on Writing & Listening		Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	
	Midterm					
22.	Style in Written English-iii	Solve stylistic problems in English	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO1 & CLO2
23.	Style in Written English-iv	Solve stylistic problems in English	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO1 & CLO2

24.	Argumentative Essay-i	To write argumentative essays	Lecture, Discussion	PPT Slides, Online resource , Text, "Tolstoy and the Twenty-first Century" from Gary Saul Morson's Anna Karenina *in Our Time Seeing More Wisely* [See Recommended Reading.]	Q/A Feedback	CLO1
25	i.Argumentative Essay-ii ii. sentence variety-i	To write argumentative essays	Lecture, Discussion	PPT Slides, Online resource , Text,	Q/A Feedback	CLO1
26	Argumentative Essay-iii Sentence variety-ii	To write argumentative essays	Lecture, Discussion	PPT Slides, Online resource , Text,	Q/A Feedback	CLO 1
27	1.Argumentative Essay-iv 2. Sentence variety-iii	To write argumentative essays	Lecture, Discussion	PPT Slides, Online resource , Text,	Q/A Feedback	CLO 1

28.	Listening: conversation between up to four people set in an educational or training context	To make use of contextual clues To locate and select relevant information	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5
29.	Listening: conversation between up to four people set in an educational or training context	To make use of contextual clues To locate and select relevant information	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5
30.	Listening: conversation between up to four people set in an educational or training context	To make use of contextual clues To locate and select relevant information	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5
31	Listening: conversation between up to four people set in an educational or training context	To make use of contextual clues To locate and select relevant information	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5

32.	Listening: conversation between up to four people set in an educational or training context	To make use of contextual clues To locate and select relevant information	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5
33.	Problem-solution Essay-i	To write problem-solution essays	Lecture, Discussion	PPT Slides, Online resources	Q/A Feedback	CLO 1
34.	1.Problem-solution Essay-2. Sentence variety-iv	To write problem-solution essays	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 1
35.	1.Problem-solution Essay-iii 2.Sentence variety-v	To write problem-solution essays	Lecture, Discussion	PPT Slides, Online resources	Q/A Feedback	CLO 1

36.	1.Problem-solution Essay-iv 2. vocabulary-iv	To write problem-solution essays	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 1
37.	Listening: conversation between up to four people set in an educational or training context	To make use of contextual clues To locate and select relevant information	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5
38.	Listening: conversation between up to four people set in an educational or training context	To make use of contextual clues To locate and select relevant information	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5
39.	Class Test					

40.	Listening: conversation between up to four people set in an educational or training context	To make use of contextual clues To locate and select relevant information	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5
41.	Listening: conversation between up to four people set in an educational or training context	To make use of contextual clues To locate and select relevant information	Lecture, Discussion	PPT Slides, Online resource , Text	Q/A Feedback	CLO 5
42.	Review					

Prescribed Reading:

Cullen, P., French, A., & Jakeman, V. (2014). The Official Cambridge Guide to IELTS Student's Book with Answers with DVD-ROM. Cambridge University Press.

Jakeman, V., & McDowell, C. (2008). New insight into IELTS: Student's book with answers. Ernst Klett Sprachen.

Maurice, I. and Herman, H. *From Paragraph to Essay*

Dorothy E. Zemach, Lisa Academic Writing

Pinker , S. The Sense of Style Penguine Claasics

Clark, P. R Writing Tools

Connelly, M. (2006) Get Writing : Paragraphs & Essay

Brandon , L. Brandon, K. E- Book : Paragraphs & Essays , with integrated Readings

Latimer , K. Kimberly, H. C. Beyond the five Paragraph Essay

Wishon, G. E., & Burks, J. M. (1968). Let's Write English, Complete Book; for Students of English as a Second Language.

Recommended Reading:

Colonna, M. R., Miller, J. L., Cohen, R. F., & Gilbert, J. E. (2006). *Reason to Write: Strategies for Success in Academic Writing: Advanced*. Oxford University Press.

Robitaille, J. Robert C. Books. Writer's Resources: From Paragraph to Essay (9781413021028).

S.A. Hornby .Oxford Writing Tutor. In Oxford Advanced Learner's Dictionary (8th Edition)

OUP WT1-WT32

Wyatt, R. (2001). *Check your vocabulary for English for the IELTS examination*. Peter Collin Publishing.

Assessment Methods:

Category	Weight	Percentage
Class Participation	5	5
Class Attendance	5	5
Class Test	5	5
Tutorial/Spot Test/Quiz Test	5	5
Presentation/Class Test	5	5
Term Paper	5	5
Mid Term Examination (1.0 hours)	20	20
Final Examination (3 hours)	50	0
Total	**100**	**100**

Course Title: Coordinate Geometry and Vector Analysis
Course Code: MATH 121, Credit: 3.0, Total hours: 42

Course Description:

Ordinary Differential Equations:

Degree and order of ordinary differential equations Formulation of Differential Equations; Solution of first order differential equations by various methods, Solution of differential equation of first order but higher degrees; Solution of general linear equations of second and higher orders with constant co-efficient.

Partial Differential Equations:
Introduction, Linear and nonlinear first order differential equations; Standard forms; linear equations of higher order; Equation of second order with variable coefficients.

Coordinate Geometry:

Transformation of coordinates, axes and its uses; Equation of conies and its reduction to standard forms; Pair of straight lines; Homogeneous equations of second degree; Angle between the pair of straight lines; Circles; System of circles; Orthogonal circles: Radical axis, radical center, properties of radical axes; Coaxial circles and limiting points: Equations of parabola, ellipse and hyperbola in Cartesian and polar coordinates; Chord of contact; Chord In terms of Its middle points; Pole and polar parametric co-ordinates.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Understand** the order, degree and various standard forms of differential equations.	C1				PO1
CO 2	Develop the ability to **Apply** differential equations to significant applied problems.	C3			CT	PO2
CO3	**Analyze** characteristics and properties of two- and three-dimensional geometric shapes and develop mathematical arguments about geometric relationships	C4			CT	PO2

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2	S											
CO3		M										

Lecture Outline:

Lec. no	Topics/ Assignment	COs	Reading Reference	Lecture Outcomes/Activities
1	Discussion about course details & Degree and Order of ordinary differential equations Formulation of differential equations.	1	1	Describe differential equations

2	Formation of differential equations	1, 2	1, 3, 4	Understand the order, degree and various standard forms of differential equations
3	Solution of first order differential equations by various methods(Variable Separation)	2	1, 3, 4	Determine solutions to first order separable differential equations
4	Solution of first order differential equations by various methods (Equation Reducible).	2	4	Solve linear homogeneous equation when one solution is given.
5	Solution of first order differential equations by various methods (Homogeneous)	2	1, 3	Find the general solution of first order linear homogeneous equations with constant coefficients
6	Solution of first order differential equations by various methods (Homogeneous)	2	1, 3	Solve the general solution of second order linear homogeneous equations with constant coefficients
7	Solution of differential equation of first order but higher degrees.	3	1, 4	Evaluate and apply linear differential equations of second order (and higher)
8	**CT-01**			Lec 1-6
9	Solution of differential equation of first order but higher degrees.	3	1,3	Evaluate and apply linear differential equations of second order (and higher)
10	Solution of differential equation of first order but higher degrees.	3	1, 4	Evaluate and apply linear differential equations of second order (and higher)
11	Solution of general linear equations of second and higher orders with constant co-efficient.	3	1, 4	Determine solutions to second order linear non-homogeneous differential equations with constant coefficients
12	Solution of general linear equations of second and higher orders with constant co-efficient.	3	1	Solve the solutions of second order linear non-homogeneous differential equations with constant coefficients
13	Cauchy-Euler equation	4	1	Simplify the equation of second order with variable coefficients
14	Solution of Euler's Homogeneous linear equations	4	1	Identify the equation of second order with variable coefficients

15	**CT-02**			Lec 7-13
16	Partial Differential Equation (Introduction)	1	1	Identify Partial Differential Equation
17	Partial Differential Equation (Introduction)	1, 2	3	Distinguish Differential Equations
18	Linear first order differential Equation	3	1	Acquire the idea of Lagrange's method for solving the first order linear partial differential equations
19	Linear first order differential Equation	3	1	Apply Lagrange's method for solving the first order linear partial differential equations
20	**Assingment-021** Linear equations of higher order	3	1	Define the problems of linear equations of higher order
21	Linear first order differential Equation	3	1, 3	Solve linear first order differential Equation
22	Equation of second order with variable coefficients	3, 4	1	Be familiar with the modeling assumptions and derivations that lead to PDEs
23	Equation of second order with variable coefficients	3, 4	1	Simplify the equation of second order with variable coefficients
24	**CT-03**			Lec 16-21
25	Non-Linear first order differential Equation	4	1, 3	Resolve non-Linear first order differential Equation
26	Non-Linear first order differential Equation	4	1, 3	Resolve non-Linear first order differential Equation
27	Standard forms	2, 4	1	Identify and obtain the solution of Clairaut's equation
28	Standard forms	4	1	Describe the origin of partial differential equation and distinguish the integrals of first order linear partial differential equation into complete, general and singular integrals
29	Transformation of coordinates, axes and its uses;	5	2, 5	Demonstrate understanding of the principles and concepts of coordinate geometry
30	Transformation of coordinates, axes and its uses;	5	5	Identify the coordinate of geometry and its applications

31	Pair of straight lines; Angle between the pair of straight lines.	5	2	Identify and define points, lines, and planes.
32	Pair of lines joining the origin lo the point of intersection of two given curves.	5	5	Identify parallel and perpendicular lines.
33	Angle between the pair of straight lines; Pair of lines joining the origin lo the point of intersection of two given curves.	5	5	Find the measurement of angles.
34	circles; System of circles	2, 5	2, 5	
35	Orthogonal circles: Radical axis, radical center, properties of radical axes; Coaxial circles and limiting points:	5	5	Explain different type of circles
36	Equation of conies and its reduction to standard forms.	7	2	Understand the basic ideas of conics
37	Equation of conies and its reduction to standard forms.	7	5	Explain the ideas of conics and their various applications
38	**CT-04**			Lec 29 – 35
39	Equation of conies and its reduction to standard forms.	5	5	Apply the properties of conics to solve problems
40	Equation of conies and its reduction to standard forms.	5	5	Apply equation of conies and its reduction to standard forms.
41	Review			----------
42	Review			----------

Reading Reference:

1. Ordinary and Partial Differential Equations – M.D.Raisenghania.
2. A Text Book on of Coordinate Geometry with Vector Analysis Rahman and Bhattacharjee
3. Differential Equations – B. D. Sharma
4. Differential Equations – Md. Abu Yousuf
5. Analytic and Vector Geometry-Dr. M. F Rahman Dr. Md. Elias.

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Electronic Devices and Circuits
Course Code: EEE 121, Credit: 3.0, Total hours: 42

Course Description:

1. Introduction to semiconductors, p-n junction diode, I–V characteristics; Diode applications: half and full wave rectifiers, clipping and clamping circuits, regulated power supply;
2. Bipolar Junction Transistor (BJT): principle of operation, Transistor circuit configurations (CE, CB, CC), BJT biasing, BJT Transistor modeling, small-signal analysis of single and multi- stage amplifiers, frequency response of BJT amplifier.
3. Field Effect Transistors (FET): Principle of operation of JFET and MOSFET, Depletion and enhancement type MOSFETs, Switching circuits using FETs, CMOS, biasing of FETs, FET small signal analysis, Low and high frequency response of FETs;
4. Study of Operational amplifiers and its applications; Feedback and oscillators circuits;
5. Operation, characteristics and application of SCR, TRIAC, DIAC and UJT;
6. Introduction to IC fabrication processes

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Comprehend** the basic concepts and characteristics of the electronic devices and circuits, IC fabrication	C2				PO1
CO 2	**Exemplify** basic operations and applications of diode, different transistors, op-amp, thyristors	C2			CT	PO1
CO 3	**Implement** different transistor, amplifier circuits	C3			CT	PO2
CO 4	**Demonstrate** knowledge of electronics devices and circuit analysis in a cyber-physical entity		P3		TW	PO9

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
CO1	S											
CO2	S											
CO3		S										
CO4									M			

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Revision on Semiconductor Physics, Intrinsic semiconductors, Extrinsic semiconductors, Important semiconductor Parameters	1	1	Explain basic principles behind semiconductor use in Electronic devices
2	p-n junction, space-charge region and potential barrier, p-n junction biasing, temperature effects on the p-n junction currents, current components in p-n junction,	1	1	Describe P-N junctions and its working principles
3	diode I-V characteristics, diode circuits : piecewise linear model, DC and AC resistance component	1, 2	1	Explain basic principles behind Diode
4	Solving series and parallel diode network,	1	1	Solve simple diode circuits
	half wave and full wave rectifier			to find current and voltages
5	Clipper and Clamper circuit with analysis and examples	1, 2	1	Solve Clipper and Clamper Circuits
6	Zener diode, principle, construction, characteristics curve, operation as voltage regulator	1, 2	1	Recognize principles behind Zener Diode
7	Review			Lecture 1 – 6
8	Class Test 1			Lecture 1 – 6
9	Introduction to BJTs and FETs – application and advantages of one	1, 2	1, 2	Recall and explain BJT

	over the other, differences between BJTs and FETs and JFETs with MOSFETs BJT types, symbols and Operation			
10	BJT In and parameters, BJT current s config uration s	1, 2	1, 2	Solve BJT circuits
11	BJT modes of operation, BJT input and output I-V Characteristics,	1, 2	1, 2	Solve BJT circuits
12	BJT dc biasing – load line and Q-point and biasing circuits, Stability of biasing circuit, BJT as a switch,	3		Lecture 9 – 12
13	Transistor modeling: r_e model, hybrid model, introduction to the small-signal model (transconductance, input conductance, depletion and diffusion capacitance, Early effect),	3	1	Clarify and compare Transistor modeling
14	Brief introduction to FET	1, 2	1	Recall and explain FET
15	Symbols, types, cross-section, operation, transfer and drain characteristics, important parameters of JFET	1, 2	1	Explain JFET
16	JFET current equation JFET dc biasing : Fixed biasing, Self-biasing, Mid-point biasing, Voltage division biasing, load line and Q-point, Q-point stability	3	1	Solve JFET Circuits
17	Problem Solving			Lecture 13 – 16
18	Review			Lecture 9 – 16
19	Class Test 2			Lecture 9 – 16
20	Symbols, cross-section, operation (enhancement, depletion), transfer and drain characteristics, current equation DC biasing : zero bias of D-MOSFET	1, 2	1, 2	Explain and Solve D-MOSFET circuits

21	Symbols, cross-section, operation, transfer and drain characteristics, current equation, channel length modulation effect DC biasing : voltage divider, drain feedback of E-MOSFET	1, 2	1, 2	Explain and Solve E-MOSFET circuits
22	CMOS, biasing of FETs, FET small signal analysis, Low and high frequency response of FETs;	2	1	Compute circuit analysis
23	Problem Solving			Lecture 20 – 22
24	Operational Amplifiers Basics, Different types of Operational Amplifiers, related Explanations	3	1	Solve and Design op-amp circuits
25	Negative Feedback topologies, op-amp Applications	3	1, 2	Solve and Design op-amp Circuits
26	Problem Solving			Lecture 24 – 25
27	Review			Lecture 20 – 25
28	Class Test 3			Lecture 20 – 25
29	Oscillatory Circuit types, their Circuits and example maths	3	1, 2	Explain Oscillatory Circuits
30	Op-amp Multivibrator, Op-amp Comparator, Op-amp Nonstable	2, 3	1, 2	Explain Oscillatory Circuits
31	Operation, Characteristics and application of SCR, TRIAC	2, 3	1, 2	Explain power electronics devices and solve related circuits
32	Operation, Characteristics and application of DIAC, and UJT	2, 3	1, 2	Explain power electronics devices and solve related Circuits
33	Problem Solving		2	Lecture 29 – 32
34	Problem Solving		2	Lecture 24 – 32
35	Project Assignment and Presentation	4		
36	Introduction to IC fabrication processes	1	2	Recognize and recall IC fabrication processes
37	Introduction to IC fabrication processes	1	2	Recognize and recall IC fabrication processes
38	Introduction to IC fabrication processes	1	2	Recognize and recall IC fabrication processes
39	Problem Solving			Lecture 29 – 38
40	Review			Lecture 29 – 38
41	Class Test			Lecture 29 – 38

42	Review Class, Problem Solutions and Suggestion			Lecture 1 – 41

Reading Reference:

1. Electronic Devices and Circuit Theory -Robert L. Boylestad and Louis Nashelsky
2. Electronic Principles – Albert P. Malvino.

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Electronic Devices and Circuits Sessional
Course Code: EEE 122, Credit: 1.5, Total hours: 42

Rationale:

To learn and familiarize with the basics of electronic circuits as well as utilizing electronic devices for practical purposes.

Objective:

1. To learn about electronic circuits.
2. To know and use the electronic circuits and devices for theoretical and practical purposes.
3. To learn about operational amplifier circuits.
4. To solve complex design problems regarding electronics based on realistic aspects.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Practically understand the basic electronic devices such as diode, BJT, MOSFET, FET and also special electronic devices like operational amplifiers.
2. Apply practical knowledge to analyze the characteristics and solving methods of the electronic devices.
3. Demonstrate the concepts of feedback circuits, their modelling and uses.

Course Content:

Study of Diode Characteristics, Study of Diode Rectifier, Study of N-P-N CB (Common Base) Transistor Characteristics, Study of N-P-N CE (Common Emitter) Transistor Characteristics, Study of BJT Biasing Circuits, Study the Characteristics of JFET, Mathematical Operations Using Op-Amp, SCR Operation Characteristic in DC Circuit.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	**Program Outcome (PO)**											
	1	2	3	4	5	6	7	8	9	10	11	12
Practically understand the basic electronic devices such as diode, BJT, MOSFET, FET and also special electronic devices like operational amplifiers.						√						
Apply practical knowledge to analyze the characteristics and solving methods of the electronic devices.							√					
Demonstrate the concepts of feedback circuits, their modelling and uses.							√					

Lecture Outline:

Week	**Lab**	**Topics**
1	Lab 1	Study of Diode Characteristics
2	Lab 2	Study of Diode Rectifier
3	Lab 3	Study of N-P-N CB (Common Base) Transistor Characteristics
4	Lab 4	Study of N-P-N CE (Common Emitter) Transistor Characteristics
5	Lab 5	Study of BJT Biasing Circuits
6	Lab 6	**Lab Test- 01**
7	Lab 7	Study the Characteristics of JFET
8+9	Lab 8+9	Mathematical Operations Using Op-Amp
10	Lab 10	SCR Operation Characteristic in DC Circuit
11	Lab 11	**Lab Test- 02**
12	Lab 12	**Quiz**
13+14	Lab 13+14	**Viva**

Reading Reference:

1. Electronic Devices and Circuit Theory -Robert L. Boylestad and Louis Nashelsky
2. Electronic Principles – Albert P. Malvino.
3. Micro Electronics Circuits-Adel S. Sedra & Keneth C. Smith-Oxford University Press
4. Operation Amplifiers and Linear Integrated Circuits-Robert F. Coughlin-Prentice Hall of India Private Limited

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Mechanics, Wave, Heat and Thermodynamics
Course Code: PHY 121, Credit: 3.0, Total hours: 42

Course Description:

Mechanics: Linear momentum of a particle, Linear momentum of a system of particles, Conservation of linear momentum, Some applications of the momentum principle; Angular momentum of a particle, Angular momentum of a system of particles, Kepler's Law of planetary motion, The Law of universal gravitation, The motion of planets and satellites, Introductory quantum mechanics; Wave function, Uncertainty principle, Postulates, Schrodinger time independent equation, Expectation value, Probability, Particle in a zero potential, Calculation of energy.

Wave: Differential equation of simple harmonic oscillator and its solution, Total energy and average energy, Combination of simple harmonic oscillations, Spring mass system, Torsional pendulum; Two body oscillation, Reduced mass, Damped oscillation, Forced oscillation, Resonance; Progressive wave, Power and intensity of wave, Stationary wave, Group and phase velocities.

Heat and Thermodynamics: Heat and work- the first Law of Thermodynamics and its applications; Kinetic theory of gases - kinetic interpretation of temperature, Specific heats of ideal gases, Equipartition of energy, Mean free path, Maxwell's distribution of molecular speeds, Reversible and irreversible processes, Carnot's cycle, Second Law of Thermodynamics, Carnot's theorem, Entropy, Thermodynamic functions, Maxwell relations, Clausius and Clapeyron equation.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

Course Outcomes (CO) of the Course		Level of Domain				PO
		C	P	A	S	
CO 1	1. **Comprehend** knowledge about fundamentals of engineering principles related to mechanics.	C2				PO 1
CO 2	2. **Analyze** simple systems undergoing simple harmonic motion and be able to derive equations describing the motion and expressions for the oscillation frequency.	C4			CT	PO 2
CO 3	3. **Apply** the concepts and laws of thermodynamics to solve problems in thermodynamic systems such as gases, heat engines and refrigerators etc.	C3				PO 1

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
CO1	S											
CO2		S										
CO3	S											

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Discussion on Course Details		1	Fundamental knowledge about mechanics
2	Linear momentum of a particle and Linear momentum of a system of particles. Applications of the momentum	1	1	Brief knowledge about linear momentum
3	Applications of the momentum	1	1	Applications linear momentum
4	Angular momentum of a particle	1	1	Brief knowledge about angular momentum
5	Kepler's Law of planetary motion.	1	1	Applications angular momentum
6	The motion of planets and satellites (I).	1	1	Applications angular momentum

7	The motion of planets and satellites (II).	1	1	Applications angular momentum
8	Introductory quantum mechanics	1	3	Basic knowledge about quantum mechanics
9	Wave function and fundamental postulates of quantum mechanics	1	3	Knowing postulates of quantum mechanics
10	Class test-1		Mechanics	Lecture 1 – 10
11	Schrodinger Equation.	1	2	Application of quantum mechanics
12	Operators and expectation values	1	3	Define Operators and expectation values
13	Uncertainty Principle.	1	3	Different use of Uncertainty Principle in quantum mechanics
14	Particle in a zero potential and calculation of energy	1	2	Application of quantum mechanics
15	Motions	2	4, 5	Basic knowledge about wave
16	Differential equation of simple harmonic oscillator	2	4, 5	Solve different wave related problems
17	Characteristics of oscillations	2	4, 5	Different applications of waves and oscillation
18	Combinations of simple harmonic oscillations	2	4, 5	Different applications of waves and oscillation
19	Class Test-2		Waves and Oscillation	Lecture 11 – 19
20	Spring mass and Torsional Pendulum	2	4, 5	Solve different wave related problems
21	Two body oscillation, Reduced mass and types of oscillation	2	4, 5	Different applications of waves and oscillation
22	Spring mass system and Torsional pendulum	2	4, 5	Different applications of waves and oscillation
23	Progressive wave	2	4, 5	Different applications of waves and oscillation
24	Power and intensity of wave	2	4, 5	Different applications of waves and oscillation
25	Stationary wave, Group and phase velocities	2	4, 5	Solve different wave related problems
26	Heat and work	3	4	Basic thermodynamics
27	First Law of Thermodynamics	3	4	Solve different thermodynamics problems

28	Specific heats of ideal gases	3	4	Solve different thermodynamics problems
29	Applications of first law of thermodynamics	3	4	Solve different thermodynamics problems
30	Kinetic theory of gases	3	4	Different applications of thermal physics
31	Kinetic interpretation of temperature	3	4	Different applications of thermal physics
32	Class Test-3		Up to Reversible and irreversible processes	Lecture 20 – 32
33	Mean free path	3	4	Different applications of thermal physics
34	Reversible and irreversible processes	3	4	Different applications of thermal physics
35	Second law of thermodynamics and Entropy	3	4	Solve different thermodynamics problems
36	Carnot's cycle	3	4	Different applications of thermal physics
37	Thermodynamic functions and Maxwell relations-1	3	4	Different applications of thermal physics
38	Maxwell's relations in thermodynamics-2	3	4	Different applications of thermal physics
39	Clausius and Clapeyron equation	3	4	Different applications of thermal physics
40	Maxwell's law of distribution of molecular velocities	3	4	Different applications of thermal physics
41	Class Test-3		Up to Maxwell's law of distribution of molecular velocities	Lecture 33 – 41
42	Review Class			

Reading Reference:

1. B.Sc. Physics (Volume I) - C.L. Arora
2. Concepts of Modern Physics – A. Beiser
3. Physics for Engineers (Part 2) - Dr. Giasuddin Ahmed
4. Physics for Engineers (Part 1) - Dr. Giasuddin Ahmed
5. B.Sc. Physics (Volume II) - C.L. Arora

Assessment Methods:

Category	**Marks %**
Category	**Marks %**
Lab Test – 1	25
Lab Test – 2	25
Viva	10
Class Participation	10
Class Assessment	30
Total	**100**

Course Title: Basic Physics Sessional
Course Code: PHY 124, Credit: 1.0, Total hours: 28

Rationale:

To perform experiments to verify practically the theories and concepts learned and understand the difference between theory and application.

Objective:

1. To help learning how to be patient and careful while taking observation.
2. To provide knowledge on how to develop the ability of the students to conduct, observe, analyze and report an experiment.
3. To help to deal with physical models and formulas mathematically.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Demonstrate an ability to make physical measurements and understand the limits of precision in measurements.
2. Demonstrate the ability to use experimental statistics to determine the precision of a series of measurements.
3. Compare to use various methods of circuit analysis both for DC and AC networks and their solution methods.
4. Appraise the characteristics of different electronic devices.

Course Content:

Determination of specific resistance of a wire using a meter bridge, determination of the resistance per unit length of meter bridge wire, determination of mechanical equivalent (J) of heat by the electrical method, determination of electrochemical equivalent (ECE) of copper by the cooper Voltammeter, determination of resistance of a galvanometer by half deflection method, determination of high resistance by the method of deflection, determination of focal length of a concave lens by auxiliary lens method, determination

of radius of a curvature of a Plano convex lens by Newton's ring method, determination of the refractive index of the material of a prism by spectrometer, determination of the specific rotation of sugar solution by Polarimeter, determination of frequency of a tuning fork by the Melde's experiment, determination of the spring constant and the effective mass of a loaded spring and hence to calculate the rigidity modulus of the spring, determination of the value of g acceleration due to gravity by means of a compound pendulum, determination of Young's modulus of a bar by bending method, determination of the modulus of rigidity of a wire by statistical method.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes (CO) of the Course	**Program Outcome (PO)**											
	1	2	3	4	5	6	7	8	9	10	11	12
Demonstrate an ability to make physical measurements and understand the limits of precision in measurements.						√						
Demonstrate the ability to use experimental statistics to determine the precision of a series of measurements.						√						
Illustrate properties of a variety of electrical and optical systems.						√						

Lecture Plan:

Week	**Lecture**	**Topics**
1+ 2	Lab 1+ 2	Determination of specific resistance of a wire using a meter bridge, determination of the resistance per unit length of meter bridge wire, determination of mechanical equivalent (J) of heat by the electrical method
3+ 4	Lab 3+ 4	Determination of electrochemical equivalent (ECE) of copper by the cooper Voltammeter, determination of resistance of a galvanometer by half deflection method,
		determination of high resistance by the method of
		deflection
5+ 6	Lab 5+ 6	Determination of focal length of a concave lens by method auxiliary lens , determination of radius of a curvature of a Plano convex lens by Newton's ring method, determination of the refractive index of the material of a prism by spectrometer
7+ 8	Lab 7+ 8	Determination of the specific rotation of sugar solution by Polarimeter, determination of frequency of a tuning fork by the Melde's experiment, determination of the

		spring constant and the effective mass of a loaded spring and hence to calculate the rigidity modulus of the spring
9+ 10	Lab 9+ 10	Determination of the value of g acceleration due to gravity by means of a compound pendulum, determination of Young's modulus of a bar by bending method, determination of the modulus of rigidity of a wire by statistical method
11+ 12	Lab 11+ 12	**Lab** **Test**
13	Lab 13	**Quiz**
14	Lab 14	**Viva**

Reading Reference:

1. Physics for Engineers, Part-1 & Part-2 (4th ed)- Dr. Giasuddin Ahmad, Hafiz Book Centre (2000).
2. Waves and oscillations (2nd ed)- Brijlal and Subramannyam, Vikas Publishing House Pvt Ltd (1994).
3. A Text Book of Optics - Brijlal and N. Subrahmanyam, S. Chand (2006).
4. Fundamental of Optics (3rd ed)- Jenkine and White, McGraw Hill (1957).
5. Fundamentals of Physics (10th ed)- Halliday, Resnick and Walker, Wiley (2013).
6. Physics part-I & II (2nd ed) - Resnick and Halliday, John Wiley & Sons, Inc. (1968).

Assessment Methods:

Category	**Marks %**
Lab Test	40
Quiz	20
Viva	10
Class Participation	10
Home Assignment/Report	10
Class performance/observation	10
Total	**100**

LEVEL-2 TERM-I

Course Title: Data Structures
Course Code: CSE 211, Credit: 3.0, Total hours: 42

Course Description:

1. Internal data representation, Abstract data types.
2. Algorithm performance and elementary asymptotic analysis (Introduction to Big-O notation).
3. Elementary data structures: array, linked list, stack, queue, tree and tree traversal, graphs and graph representation, heap, binary search tree.
4. Sorting algorithms, Searching: linear search and binary search;
5. Advanced data Structures: balanced binary search trees, skip list, advanced heaps; hashing.

Course Outcome (CO) Matrix:
At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Express** basic concepts of stacks, queues, lists, trees and graphs	C2				PO1
CO 2	**Explain** concepts about searching and sorting techniques	C2				PO1
CO 3	**Apply** data structures to solve problem in an efficient way to increase performance	C3			CT	PO3
CO 4	To **utilize** data structures in various applications including the ability to implement algorithms for the creation, insertion, deletion, searching, and sorting of each data structure.	C3			CT	PO2

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2		S										
CO3	S											
CO4	S											

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Abstract data types	1	1	Name basic Data representation

2	Algorithm performance and elementary asymptotic analysis	1	2	Explain Asymptotic notations
3	Searching: linear search and binary search	2,3	1	Explain and Differentiate linear search and binary search
4	Searching: linear search and binary search	2,3	1	Explain and Differentiate linear search and binary search
5	Sorting algorithms	2,3	1	Explain Selection sort and Bubble sort
6	Sorting algorithms	2,3	1	Explain Insertion sort
7	Sorting algorithms	2,3	1	Explain Merge sort
8	Sorting algorithms	2, 3	1	Explain Quick sort
9	Sorting algorithms	2, 3	1	Explain Bucket sort
10	Problem Solving			Lecture 1– 9
11	Class Test 1			Lecture 1-7
12	Elementary data structures: array	1	1	Explain Array
13	Elementary data structures: stack	1	1	Explain push and pop operation of Stack
14	Elementary data structures: stack	1	1	Explain polish notations and evaluate postfix notation
15	Elementary data structures: stack	1	1	Convert Infix notation into postfix notation
16	Elementary data structures: Recursion	1,4	1	Solve problem using recursion
17	Elementary data structures: queue	1	1	Explain insert and delete operations of Queue
18	Elementary data structures: queue	1	1	Explain deque operations
19	Problem Solving			Lecture 12– 18
20	Class Test 2			Lecture 8 – 15
21	Elementary data structures: linked list	1,3	1	Explain and Differentiate Array and Lined List
22	Elementary data structures: linked list	1	1	Insertion in Linked list
23	Elementary data structures: linked list	1	1	Deletion in Linked list
24	Elementary data structures: Queue	1	1	Explain priority queue operations
25	Review			Lecture 1 – 24
26	Assignment			Based on Lecture 1 – 24
27	Assignment			Based on Lecture 1 – 24
28	Graphs and graph representation	1	1	Explain DFS and BFS
29	Graphs and graph representation	1	1	Explain shortest path algorithms
30	Graphs and graph representation	1	1	Differentiate shortest path algorithms
31	Tree and tree traversal	1	1	Explain Huffman Coding
32	Class Test 3			Lecture 16– 24
33	Heap	3,4	1	Construct heaps and explain Heapsort

34	Sorting algorithms	2,3	1	Explain topological sort
35	balanced binary search trees	2,4	1	Explain balanced binary search trees
36	balanced binary search trees	2,4	1	Analyze balanced binary search trees
37	Class Test 4			Lecture 32– 36
38	Hashing	3,4	1	Explain Hashing
39	Advanced heaps	3,4	1	Explain Advanced heaps
40	Review Class and Problem Solutions			Lecture 28 – 30 and Lecturer 32 – 33
41	Review Class and Problem Solutions			Lecture 34 – 39
42	Suggestion			Lecture 1 – 41

Reading Reference:

1. Theory and Problems of Data Structures - Seymour Lipschutz
2. Introduction to algorithms- Thomas H. Corman, Ronald L. Rivest, Clifford Stein and Charles E. Leiserson

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Data Structures Sessional
Course Code: CSE 212, Credit: 1.5, Total hours: 42

Rationale:

The study of data structure is an essential part of computer science. Data structure is a logical and mathematical model of storing & organizing data in a particular way in a computer. In system programming application programming the methods & techniques of data structures are widely used. The study of data structure helps the students in developing logic & structured programs.

Objective:

1. General Understanding of Basic Data Structures
2. Developing Programming Skills for Advanced Data Structures such as Trees, Hash Tables, and Priority Queues etc.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Practically demonstrate advantages and disadvantages of specific algorithms and data structures.

2. Select basic data structures and algorithms for autonomous realization of simple programs or program parts.
3. Initiate practical knowledge to determine and demonstrate bugs in programs.
4. Formulate new solutions for problems or improve existing code using learned algorithms and data structures.

Course Content:

Implementation of Abstract data types; Introduction to algorithms; Asymptotic analysis: growth of functions, O, Ω and Θ notations; Correctness proof and techniques for analysis of algorithms; Master Theorem; Elementary data structures: arrays, linked lists, stacks, queues, trees and tree traversals, graphs and graph representations, heaps, binary search trees; Graph Traversals: DFS, BFS, Applications of DFS and BFS; Sorting: heapsort, merge sort, quicksort; Data structures for set operations; Methods for the design of efficient algorithms: divide and conquer, greedy methods, dynamic programming.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
Practically demonstrate advantages and disadvantages of specific algorithms and data structures.									√			
Select basic data structures and algorithms for autonomous realization of simple programs or program parts.									√			
Initiate practical knowledge to determine and demonstrate bugs in programs.									√			
Formulate new solutions for problems or improve existing code using learned algorithms and data structures.						√						

Lecture Outline:

Week	Lecture	Topics	Remarks
1	Lec 1 Lec 2 Lec 3	Array List (Implementing List using Array)	
2	Lec 4 Lec 5 Lec 6	Binary Search	
3	Lec 7 Lec 8 Lec 9	Implementing Stack using Array	
4	Lec 10	Implementation of Queue using Array,	

	Lec 11 Lec 12	Circular Queue	
5	Lec 13 Lec 14 Lec 15	Single Linked List Implementation	
6	Lec 16 Lec 17 Lec 18	Practice on Single Linked List	
7	Lec 19 Lec 20 Lec 21	**Lab Test-1**	
8	Lec 22 Lec 23 Lec 24	Double Linked List Implementation	
9	Lec 25 Lec 26 Lec 27	Operation on Binary Search Tree (Insertion, Deletion, Traversal, Merging)	
10	Lec 31 Lec 32 Lec 33	Operation on Heap Trees/Priority Queue (MAX and MIN Heap)	
11	Lec 28 Lec 29 Lec 30	Implementation of Graph Algorithms: BFS	
12	Lec 34 Lec 35 Lec 36	Implementation of Graph Algorithms: DFS	
13	Lec 37 Lec 38 Lec 39	**Lab Test-02**	
14	Lec 40 Lec 41 Lec 42	**Quiz**	

Reading Reference:

1. Classic Data Structures 2nd Ed – Debasis Samanta
2. Data Structures using C 2nd Ed – Aaron M. Tenenbaum, Yedidyah Langsam, Moshe J. Augenstein

Assessment Methods:

Category	**Marks %**
Lab Test – 1	25
Lab Test – 2	25
Viva	10
Class Participation	10

Class Assessment	30
Total	**100**

Course Title: Object Oriented Programming Language
Course Code: CSE 213, Credit: 3.0, Total hours: 42

Course Description:

1. Philosophy of Object Oriented Programming (OOP); Advantages of OOP over structured programming; Encapsulation Classes and objects, Access specifiers, Static and Non-static members;
2. Constructors, destructors and copy constructors;
3. Array of objects, object pointers, and object references, In-line Functions, friend functions, reference;
4. Inheritance: single and multiple inheritance; Polymorphism: overloading, abstract classes, virtual functions and overriding;
5. Exceptions; Object Oriented I/O, inserter, extractor;
6. Template functions and classes;
7. Namespaces, overview of Standard Template Library;
8. Multi-threaded Programming.

Course Outcome (CO) Matrix:
At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	Students should be able to **classify** and **apply** major object-oriented concepts like data abstraction, encapsulation, polymorphism and inheritance	C3			CT	PO1
CO 2	Students should be able to **solve** various programming problems implementing C++ syntax, templates, exceptions, forms of casting, conversions objects, operator overloads, dynamic memory allocation, control structures, modularity, I/O, file I/O and so on.		P4		CT	PO2
CO 3	Students will **take part** in group works like group presentation and group assignment to develop idea of team work.			A1	TW	PO9
CO 4	Students should be able to **replicate** ideas to solve a real life problem focusing on idea of object-oriented software engineering.		P1			PO2

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2		S										
CO3									M			
CO4		M										

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1.	General Review of Structure Programming Language	1	3	Will **use** idea of Logical Programming
2.	General Review of Structure Programming Language	1	3	Will **use** idea of Logical Programming
3.	General Review of Structure Programming Language	1	3	Will **use** idea of Logical Programming
4.	Introduction to OOP	1	1	Students can **identify** the characteristics of OOP
5.	Introduction to C++ Language	1	1	Students can **compare** between C and C++
6.	Introduction to Class and Object	1	1	Students can **identify** basic concept of Object and Class
7.	Brief Discussion on Class and Object	1	1,2	Students will **use** idea of Class and Object
8.	Discussion on Access Specifier and Scope Resolution	1	1	Students will be able to **solve** problems related to class member access.
9.	Discussion on Friend Function and Friend Class with problem solving	2	1,2	Will **adapt** idea of various member methods.
10.	Discussion on Constructor and Destructor	1	1	Can **identify** basic concept of initialization and deletion of object
11.	Discussion on Parameterized Constructor and Inline Function with real life problem solving	2	1,2	Students will be able to **solve** problems related to memory allocation and de-allocation of object.
12	CT1	1, 2	Lecture 4 to 11	
13.	Discussion on Static Class Members	1	1	Will **adapt** idea of member method
14.	Constant Class Member	1	1	
15	Discussion on Function Overloading with problem solving	2	1,2	Will **follow** examples to create several methods using same name
16	Discussion on Pointer and Reference	1	1	Will be able to **explain** usage of pointer variable
17.	Discussion on this pointer with problem solving	CO 2	1,2	Can **illustrate** concept of object address.

18.	Discussion on Dynamic Memory Allocation with problem solving	CO 1	1,2	Can **classify** static memory allocation and dynamic memory allocation.
19	**Copy Constructor and Assignment Operator** with problem solving	CO 2	1,2	Can **specify** how to replicate object
20	CT2	CO 2	Lecture 14 to 18	
21	Review			
Mid Term Break				
22	Default Argument and Passing, Returning Object with problem solving	CO 2	1,2	Will be able to **combine** method overloading and default argument along with object transfer
23.	Inheritance	CO 1	1	Will **define** inheritance
24	Inheritance with problem solving	CO 2	1,2	Can **copy** idea of class hierarchy
25.	Inheritance with problem solving	CO 2	1,2	Can **plan** solutions implementing inheritance concept.
26	Exception Handling with problem solving	CO 2	1,2	Can **implement** idea of exception handler
27	Exception Handling in Derived Class with problem solving	CO 2	1,2	Will **manage** various errors
28	CT3	CO 2	Lecture 23 to 27	
29	Method Overloading Vs Method Overriding	CO 2	1	Will be able to **compare** method overloading and method overriding
30	Virtual Function and Abstract Class with problem solving	CO 2	1,2	Will **solve** problems of more than one copy of parent class
31	Introduction to Operator Overloading	CO 1	1	
32	Operator Overloading with real life problem solving	CO 2	1,2	Will **use** operator for more than one purpose
33	Thread with real life problem solving	CO 4	4	Students will **control** parallel processing
34	Thread with real life problem solving	CO 4	4	Students will **control** parallel processing
35	Thread with real life problem solving	CO 4	4	Students will **control** parallel processing
36	File I/O with real life problem solving	CO 4	1	Students can **adapt** to read and to write files
37	Namespace with real life problem solving	CO 4	1	Students can **adapt** to usage of specific memory region
38	STL with real life problem solving	CO 4	1	Will **implement** templates for easy solutions
39	CT4	CO 2, CO 4	Lecture 31,32, 36 and 38	
40	Presentation	CO 3, CO4		Will **copy** and **integrate** OOP concepts to run a real life software and will **focus** on mutual help and understanding
41	Presentation	CO 3, CO 4		Will **copy** and **integrate** OOP concepts to run a real

				life software and will **focus** on mutual help and understanding
42	Review			

Reading Reference:

1. The Complete Reference C++ by Herbert Schildt , 4th Edition
2. Teach Yourself by Herbert Schildt, 3rd Edition
3. Programming in Ansi C by E. Balagurusamy
4. Online Materials provided by Course Instructor

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Object Oriented Programming Language Sessional
Course Code: CSE 214, Credit: 1.5, Total hours: 42

Rationale:

Object-oriented programming is a programming paradigm that includes or relies on the concept of objects, encapsulated data structures that have properties and functions and which interact with other objects

Objective:

1. To achieve a basic idea on Object Oriented Programming Language
2. Present object-oriented aspects of C++
3. To learn programming with C++

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Practice object oriented Programming and design object oriented solutions for small systems/ problems, involving multiple objects.
2. Demonstrate good programming style and discuss the impact of style on developing and maintaining programs.
3. Identify the relative merits of different algorithmic designs, programming constructs and data structures.

Course Content:
Encapsulation, classes and objects, access specifiers, static and non-static members; Constructors, Destructors and Copy Constructors; Array of objects, object pointers, and object references; In-line functions, friend functions, static functions; Inheritance: single and multiple inheritance; Polymorphism: overloading, abstract classes, virtual functions and overriding; Exception Handling; Object Oriented I/O ; Template functions and classes; Concept of Namespaces, Overview of Standard Template Library (Vectors & Iterators); Multi-threaded Programming, Abstract Data Type

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	**Program Outcome (PO)**											
	1	2	3	4	5	6	7	8	9	10	11	12
Practice object oriented Programming and design object oriented solutions for small systems/ problems, involving multiple objects.									√			
Demonstrate good programming style and discuss the impact of style on developing and maintaining programs.								√				
Identify the relative merits of different algorithmic designs, programming constructs and data structures.									√			
Able to write code, test, document and prepare a professional looking package for specified systems / problems.											√	

Lecture Outline:

Week	**Lecture**	**Topics**	**Remarks**
1	Lec 1 Lec 2 Lec 3	Introductory session on OOP	
2	Lec 4 Lec 5 Lec 6	Structure and Classes with namespace	
3	Lec 7 Lec 8 Lec 9	Class and objects with access specifier	
4	Lec 10 Lec 11 Lec 12	Member Functions, In-line functions, Friend functions, Function Overloading	
5	Lec 13 Lec 14 Lec 15	Introduction to the concept of Constructors and Destructors	
6	Lec 16 Lec 17	Copy Constructors	

	Lec 18		
7	Lec 19 Lec 20 Lec 21	**Lab Test -1**	
8	Lec 22 Lec 23 Lec 24	Inheritance: Introduction, derived and base classes, accessing base class members, access specified for 'protected'	
9	Lec 25 Lec 26 Lec 27	Multiple inheritance, Constructor and destructor in Inheritance	
10	Lec 31 Lec 32 Lec 33	Virtual functions, runtime polymorphism and overriding Abstract class	
11	Lec 28 Lec 29 Lec 30	Operator overloading: Introduction, overloading of unary operators	
12	Lec 34 Lec 35 Lec 36	Operator overloading: Overloading of binary operators, multiple overloading	
13	Lec 37 Lec 38 Lec 39	**Lab Test-2**	
14	Lec 40 Lec 41 Lec 42	**Quiz and Viva**	

Reading Reference:

1. Teach Yourself C++ by Herbert Schildt
2. Object Oriented Programming with C++ by E Balagurusamy
3. Complete Reference C++ by Herbert Schildt
4. Programming with C++ by Schaums Outline Series

Assessment Methods:

Category	**Marks %**
Lab Test – 1	25
Lab Test – 2	25
Viva	10
Class Participation	10
Class Assessment	30
Total	**100**

Course Title: Linear Algebra and Complex Analysis
Course Code: MATH 211, Credit: 3.0, Total hours: 42

Course Description:

Matrices:
Definition of matrix; Algebra of matrices; Multiplication of matrices, Transpose of a matrix, Inverse of matrix; Rank and elementary transformations of matrices; Solution of linear equations, Null space and Nullity of matrix. Characteristic subspace of matrix. Matrix polynomials, Determination of characteristic root and vectors.

Vector Analysis:
Differentiation and integration of vectors together with elementary applications; Definition of line, surface and volume integrals; Gradient, divergence and curl of point functions, various formulae, Gauss's theorem, Stoke's theorem, Green's theorem.

Fourier Analysis:
Real and complex form of Fourier series; Finite transform; Fourier Integral; Fourier transforms and their uses in solving boundary value problems of wave equations.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Explain** how to find different types of matrices, the rank and nullity of a matrix, and **Execute** the relationship of these concepts to associated systems of linear equations.	C2				PO1
CO 2	**Calculate** velocity and acceleration vectors for moving particles and **Recognize** Greens, Gauss & Stocks theorem and their applications.	C1		A4	CT	PO2
CO 3	**Evaluate** Fourier series, Fourier Transformation and its applications.	C5			CT	PO1

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO 12
CO1	S											
CO2		M										
CO3	S											

Lecture Outline:

Lec. No	**Topics/ Assignment**	**COs**	**Reading**	**Lecture Outcomes/Activities**

			Reference	
1	Discuss on Course Appraisal. Discuss on vector and scalars, algebra of vectors.	2	2	Apply vector algebra techniques to analyze problems involving two and three dimensional entities lines, curves, planes and surfaces
2	Multiplication of vectors by scalars.	2	2	Apply vector algebra techniques to analyze problems involving two and three dimensional entities lines, curves, planes and surfaces
3	Scalars and vectors, equality of vectors; Addition and subtraction of vectors;	2	2	Apply vector algebra techniques to analyze problems involving two and three dimensional entities lines, curves, planes and surfaces
4	Sine trigonometric law, Scalar and vector product of two vectors and their geometrical interpretation, Triple products and multiple products	2	2	Understand how to analyze and synthesize given data to solve problems in vector multiplication.
5	Linear equation, Linear dependence and independence of vectors	2	2, 6	Solve Linear dependence and independence of vectors
6	Linear equation, Linear dependence and independence of vectors	2	5	Solve Linear dependence and independence of vectors
7	Differentiation of vectors together with elementary applications	2	2	Determine and apply, the important quantities associated with scalar fields,
8	Gradient, divergence and Curl	2	2	Determine and apply, the important quantities associated with scalar fields, such as partial derivatives of all orders, the gradient vector and directional derivative
9	Gradient, divergence and Curl	2	2	Compute the curl and the divergence of vector fields
10	**CT-01**			algebra of vectors, Linear equation, dot product & cross product, Gradient, divergence and Curl
11	Integration of vectors together with elementary applications.	2	2	Acquire the basic knowledge of vector differentiation
12	Integration of vectors together with elementary applications	2	2	Calculate line integrals along piecewise smooth paths; interpret such quantities as work done by a force
13	Definition of line, surface and volume integrals;	2	2	Define of line, surface and volume integrals;

14	Definition of line, surface and volume integrals;	2	2	Solve a problem of line, surface and volume integrals;
15	Guass,Green's and Stoke's theorem	2	2, 6	Evaluate line, surface, double and triple integrals and use these integrals to verify the seminal integral theorems (Green's theorem in the plane, Gauss' divergence theorem and Stokes' theorem)
16	Guass,Green's and Stoke's theorem	2	2, 6	Apply Stokes' theorem to compute line integrals along the boundary of a surface
17	Guass,Green's and Stoke's theorem	2	2, 6	Use Green's theorem to evaluate line integrals along simple closed contours on the plane
18	Basic concept of matrices and its application.	1	7	Acquire the knowledge of a matrix, basic operations,
19	Definition of different types of matrices and related mathematical problems.	1	1, 7	Differentiate different types of matrices
20	Definition of different types of matrix.	1	1, 7	Understand the various applications of the theory of matrices to a wide variety of problems
21	Definition of different types of matrices and related mathematical problems.	1	1, 7	Understand the various applications of the theory of matrices to a wide variety of problems
22	Inverse of matrix.	1	1	Understand how to analyze given data to solve problems in Inverse matrix
23	**CT-02**			Definition of line, surface and volume integrals, Guass, Green's and Stoke's theorem, Definition of different types of matrices and related mathematical problems..
24	Inverse matrix and Adjoint matrix	1	1	Acquire knowledge of invertible matrices and their properties
25	System of Linear equations	1	7	Solve a System of Linear equations using the inverse of a matrix
26	System of Linear equations	1	7	Solve a System of Linear equations using the inverse of a matrix
27	Rank of a matrix	1	1, 7	Understand various methods for determining rank of a matrix
28	Rank of a matrix	1	1	Understand various methods for determining rank of a matrix
29	Rank of a matrix	1	7	Apply various methods for determining rank of a matrix

30	Determination of characteristic root and vectors	2	1	Solve eigenvalues and eigenvectors
31	**Assingment-01** Determination of characteristic root and vectors	2	1, 7	Apply eigenvalues and eigenvectors
32	**CT-03**			Inverse matrix and Adjoint matrix, System of Linear equations, Rank of a matrix
33	Determination of characteristic root and vectors. Cayley-Hamilton theorem, Using Cayley-Hamilton theorem find the inverse matrix, Verify Cayley-Hamilton theorem for a matrix.	2	1	Apply Cayley- Hamilton Theorem to problems for finding the inverse of a matrix and higher powers of matrices without using routine methods
34	Real and complex form of Fourier series	3	3	Solve the Fourier series expansion of a functions analytically and numerically
35	Fourier series	3	3	Analyse about Fourier Series, its representation and Fourier expansion of odd and even functions, half range series, Parseval‘s formula and practical harmonic analysis
36	Real and complex form of Fourier series	3	3	Analyse about Fourier Series, its representation and Fourier expansion of odd and even functions, half range series, Parseval‘s formula and practical harmonic analysis
37	Fourier Integral	3	3	Understand about Fourier Integral, properties and applications of Fourier Integral
38	Fourier transforms	3	3	Solve the Continuous model problems using Fourier transforms
39	Fourier transforms and their uses in solving boundary value problems of wave equations	3	3	Solve the Continuous model problems using Fourier transforms
40	Fourier transforms and their uses in solving boundary value problems of wave equations	3	3	Use appropriate numerical methods to solve Boundary Value Problems in Partial differential equations
41	**CT-04**			Real and complex form of Fourier series, Fourier transforms and their uses in solving boundary value problems of wave equations

42	Review			----------

Reading Reference:

1. Linear Algebra-Md. Abdur Rahman
2. Vector Analysis- Schaum Outline's Series
3. Mathematical Methods (Vol-2)-Md. Abdur Rahman
4. Analytic and Vector Geometry-Dr. M. F Rahman & Dr. Md. Elias.
5. Linear Algebra – Prof. Dewan Abdul Kuddus
6. Integral calculus and Vector Analysis- Md. Abu Yousuf
7. Matrices-ML. Khanna.

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Electromagnetism, Optics and Modern Physics
Course Code: PHY 211, Credit: 3.0, Total hours: 42

Course Description:

Electricity and magnetism: Electric charge and Coulomb's Law, Electric field and electric field intensity, Concept of electric flux and the gauss's Law - some applications of gauss's Law, Electric potential, Relation between electric field and electric potential, Capacitor and Capacitance, Gradient, Laplace's and Poisson's equations, Electric Current, Drift velocity and Current density, Resistivity. Magnetic field and magnetic force, Ampere's Law, Biot-savart Law and their applications, Electromagnetic induction, Faraday's law of electromagnetic induction, Lenz law and conservation of energy, Maxwell's equations

Modern Physics: Galilean relativity and Einstein's special theory of relativity; Lorentz transformation equations, Length contraction, Time dilation and mass-energy relation. Photoelectric effect, Compton effect; de Broglie matter waves and its success in explaining Bohr's theory, Pauli's exclusion principle, Constituent of atomic nucleus, Nuclear binding energy, Different types of radioactivity, Radioactive decay Law; Nuclear reactions, Nuclear fission, Nuclear fusion, Atomic power plant.

Mechanics: Linear momentum of a particle, Linear momentum of a system of particles, Conservation of linear momentum, Some applications of the momentum principle; Angular momentum of a particle, Angular momentum of a system of particles, Kepler's Law of planetary motion, The Law of universal gravitation, The motion of planets and satelites, Introductory quantum mechanics; Wave function, Uncertainity principle, Postulates, Probability. Schrodinger time independent equation, Expectation value, Particle in a zero potential, Calculation of energy.

Objective:
To extend the student's knowledge and understanding of some of the laws, principles and theories of electricity and magnetism, special theory of relativity, nuclear physics and Newtonian and quantum mechanics. To develop an appreciation of how these laws, principles and theories operate in experiment and applied engineering problems.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

Course Outcomes (CO) of the Course			
Course Outcome	Bloom's Taxonomy	Domain	Program Outcome
CO1: **Comprehends** the basics of electrical circuits, magnetic forces, capacitors and resistors and analyze circuits using Kirchhoff's rules.	Level -2 (Understand)	C	PO 1: Engineering Knowledge K1 (Natural Sciences)
CO2: **Analyze** the fundamental postulates and principles related to modern physics as well as relativity.	Level -4 (Analyze)	C	PO 2: Problem analysis K1 (Natural Sciences)P2(Range of conflicting requirements)
CO3: **Apply** the knowledge about fundamentals of engineering principles related to mechanics.	Level -3 (Apply)	C	PO 1: Engineering Knowledge K1 (Natural Sciences)

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork) **K4: knowledge profile, P2, P3, P4: complex engineering problem**

Mapping of CO and Program Outcomes:

Cos	Program Outcomes S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2		S										
CO3	S											

Lecture Outline:

Week 1		**CTs**	**COs**
Class 1	Electric Charge and Coulomb's Law	CT-01	1

Class 2	Electric Field and Electric Field Intensity;		
Class 3	Gauss' Law and its application		
Week 2			1
Class 4	Poisson's Equation and Laplace's Equation		
Class 5	Electric Potential and Potential Difference		
Class 6	Capacitor and Capacitance		
Week 3			1
Class 7	Ohm's law and its application		
Class 8	Electric Current and Current Density		
Class 9	Laws of Resistance and Resistivity		
Week 4			1
Class 10	Magnetic Force and Ampere's Law		
Class 11	Biot-Savart Law and its applications		
Class 12	Laws of Electromagnetic Induction		
Week 5		CT-02	2
Class 13	Class Test-1		
Class 14	Pauli's exclusion principle, Constituent of atomic nucleus and Nuclear binding energy		
Class 15	Different types of radioactivity and Radioactive decay Law		
Week 6			2
Class 16	Nuclear reactions and Nuclear fission		
Class 17	Nuclear fusion and Atomic power plant		
Class 18	Compton effect & Photoelectric effect		
Week 7			2
Class 19	De Broglie matter waves and its success in explaining Bohr's Theory		
Class 20	General discussion on relativity		
Class 21	Class Test-2		
Week 8		CT-03	2
Class 22	Einstein's special theory of relativity		
Class 23	Galilean relativity		
Class 24	Lorentz transformation equations		
Week 9			2
Class 25	Length Contraction (Relativity of length)		
Class 26	Time dilation (Relativity of time)		
Class 27	Mass-Energy Relation		

Week 10			3
Class 28	Linear momentum of a particle and Linear momentum of a system of particles. Applications of the momentum		
Class 29	Applications of the momentum		
Class 30	Angular momentum of a particle		
Week 11			3
Class 31	Kepler's Law of planetary motion.		
Class 32	The motion of planets and satellites (I).		
Class 33	Class Test-3		
Week 12		CT-04	3
Class 34	The motion of planets and satellites (II).		
Class 35	Introductory quantum mechanics		
Class 36	Wave function and fundamental postulates of quantum mechanics		
Week 13			3
Class 37	Schrodinger Equation.		
Class 38	Operators and expectation values		
Class 39	Uncertainty Principle.		
Week 14			3
Class 40	Particle in a zero potential and calculation of energy		
Class 41	Class Test-4		
Class 42	Review full course		

Reading Reference:

1. B.Sc. Physics (Volume I) - C.L. Arora
2. Concepts of Modern Physics – A. Beiser
3. Physics for Engineers (Part 1) - Dr. Giasuddin Ahmed
4. Physics for Engineers (Part 2) - Dr. Giasuddin Ahmed
5. B.Sc. Physics (Volume II) - C.L. Arora

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Bangla Language
Course Code: HUM 211, Credit: 2.0, Total hours: 28

Course Description:

প্রথমপাঠ- ভাষা

১। বাংলা ধ্বনি/ বাগধ্বনি (Phone/Speech Sound), বর্ণ (Letter), অক্ষর (Syllable) ২। বাংলা ধ্বনির উচ্চারণ স্থান ও রীতি (Point of Articulation and Manner of Articulation) ৩।বাংলা উচ্চারণ- প্রমিত (Standard), আঞ্চলিক (Dialectal), বৈচিত্র (Variation) ৪। অপিনিহিতি, অভিশ্রুতি, স্বরসঙ্গতি, শ্বাসাঘাত (Stress accent), স্বরভঙ্গি / স্বরতরঙ্গ (Intonation)। ৫। বাংলা ও ইংরেজির তুলনা ৬। বাংলা লিখন দক্ষতা- সাধু / চলিত রীতি, বিরামচিহ্ন প্রয়োগ। প্রমিত বাংলা বানানের নিয়ম (বাংলা একাডেমী) ৭। ব্যাবহারিক বাংলাঃ(সংক্ষিপ্ত আলোচনা) ২১ শে ফেব্রয়ারী, মুক্তিযুদ্ধ, বাংলা ভাষা, বিশ্বায়ন, বাংলার উৎসব, ষড়ঋতু, বাংলা নববর্ষ, আধুনিক তথ্য-প্রযুক্তি, বাংলার লোকসংস্কৃতি, মানবতা ও নৈতিকতা।

দ্বিতীয়পাঠ- সাহিত্য

কবিতাঃ ১। আবদুল হাকিম- নূরনামা ২। মাইকেল মধুসূদন দত্ত- বঙ্গভাষা ৩। লালনসাঁই- খাঁচার ভেতর অচিন পাখি ৪। রবীন্দ্রনাথ ঠাকুর- নির্ঝরের স্বপ্নভঙ্গ ৫। কাজী নজরল ইসলাম- আজ সৃষ্টি সুখের উল্লাসে ৬। জীবনানন্দ দাশ- রূপসী বাংলা ৭। হাসান হাফিজুর রহমান- অমর একুশে ৮। আলাউদ্দিন আল আজাদ- স্মৃতি স্তম্ভ ৯। শামসুর রাহমান- তোমাকে পাওয়ার জন্য হে স্বাধীনতা ১০। সৈয়দ শামসুল হক- পরিচয়

প্রবন্ধঃ ১।বঙ্কিম চন্দ্র চট্টোপাধ্যায়- বাঙ্গালা ভাষা ২। রবীন্দ্রনাথ ঠাকুর- সভ্যতার সংকট ৩। হর প্রসাদ শাস্ত্রী- তৈল ৪। প্রমথ চৌধুরী- যৌবনে দাও রাজটিকা ৫। কাজী নজরল ইসলাম- বর্তমান বিশ্ব সাহিত্য ৬। মুহাম্মদ আবদুল হাই- আমাদের বাংলা উচ্চারণ ৭। কবীর চৌধুরী- আমাদের আত্মপরিচয়

ছোটগল্প ও অনন্য রচনাঃ ১। রবীন্দ্রনাথ ঠাকুর- পোস্ট মাস্টার ২। রোকেয়া সাখাওয়াত হোসেন- অবরোধ বাসিনী ৩। বিভূতিভূষণ বন্দোপাধ্যায়- পুঁইমাচা ৪। সৈয়দ ওয়ালিউল্লাহ- নয়নচারা ৫। জাহানারা ইমাম- একাত্তরের দিনগুলি ৬। হাসান আজিজুল হক- ঘরগেরস্থি ৭। আখতারুজ্জামান ইলিয়াস- অপঘাত

নাটকঃ ১। মুনির চৌধুরী- কবর

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Bloom's Taxonomy Level

After completion of the course, the students will be expected to:

Sl. No.	COs	POs	Bloom's Taxonomy		
			C	A	P
1	Develop proper understanding about Bengali grammar	6	2		
2	Gather knowledge about strategy to write accurate Bengali	6	2		
3	Develop interest and curiosity about Bengali language and literature	7	3		

Mapping of COs with Knowledge Profiles, Complex Engineering Problem Solving and Complex Engineering Activities

Course Outcome	Knowledge Profile	Complex Problem Solving	Complex Engineering Activities
CO1	***K7***		

CO2	***K7***		
CO3	***K7***		

Week wise distribution of contents and assessment methods

Week	Topics	Assessment Method(s)
1	১। বাংলা ধ্বনি/ বাগধ্বনি (Phone/Speech Sound), বর্ণ (Letter), অক্ষর (Syllable) ২। বাংলা ধ্বনির উচ্চারণ স্থান ও রীতি (Point of Articulation and Manner of Articulation) ৩।বাংলা উচ্চারণ- প্রমিত (Standard), আঞ্চলিক (Dialectal), বৈচিত্র (Variation)	
2	৪। অপিনিহিতি, অভিশ্রুতি, স্বরসঙ্গতি, শ্বাসাঘাত (Stress accent), স্বরভঙ্গি / স্বরতরঙ্গ (Intonation) ৫। বাংলা ও ইংরেজির তুলনা ৬। বাংলা লিখন দক্ষতা- সাধু / চলিত রীতি, বিরামচিহ্ন প্রয়োগ। প্রমিত বাংলা বানানের নিয়ম (বাংলা একাডেমী)	
3	৭।ব্যাবহারিক বাংলাঃ(সংক্ষিপ্ত আলোচনা) ২১ শে ফেব্রয়ারী, মুক্তিযুদ্ধ, বাংলা ভাষা, বিশ্বায়ন, বাংলার উৎসব, ষড়ঋতু, বাংলা নববর্ষ, আধুনিক তথ্য-প্রযুক্তি, বাংলার লোকসংস্কৃতি, মানবতা ও নৈতিকতা।	Quiz-1
4	১। আবদুল হাকিম- নূরনামা ২। মাইকেল মধুসূদন দত্ত- বঙ্গভাষা ৩। লালনসাঁই- খাঁচার ভেতর অচিন পাখি ৪। রবীন্দ্রনাথ ঠাকুর- নির্ঝরের স্বপ্নভঙ্গ	
5	৫। কাজী নজরল ইসলাম- আজ সৃষ্টি সুখের উল্লাসে ৬। জীবনানন্দ দাশ- রূপসী বাংলা ৭। হাসান হাফিজুর রহমান- অমর একুশে	
6	৮। আলাউদ্দিন আল আজাদ- স্মৃতি স্তম্ভ ৯। শামসুর রহমান- তোমাকে পাওয়ার জন্য হে স্বাধীনতা ১০। সৈয়দ শামসুল হক- পরিচয়	Quiz-2
7	১।বঙ্কিম চন্দ্র চট্টোপাধ্যায়- বাঙ্গালা ভাষা ২। রবীন্দ্রনাথ ঠাকুর- সভ্যতার সংকট ৩। হর প্রসাদ শাস্ত্রী- তৈল	
8	৪।প্রমথ চৌধুরী- যৌবনে দাও রাজটিকা ৫। কাজী নজরল ইসলাম- বর্তমান বিশ্ব সাহিত্য	
9	৬।মুহাম্মদ আবদুল হাই- আমাদের বাংলা উচ্চারণ ৭। কবীর চৌধুরী- আমাদের আত্মপরিচয়	Quiz-3
10	১।রবীন্দ্রনাথ ঠাকুর- পোস্ট মাস্টার ২। রোকেয়া সাখাওয়াত হোসেন- অবরোধ বাসিনী ৩।বিভূতিভূষণ বন্দোপাধ্যায়- পুঁইমাচা	
11	৪।সৈয়দ ওয়ালিউল্লাহ- নয়নচারা ৫। জাহানারা ইমাম- একাত্তরের দিনগুলি	
12	৬।হাসান আজিজুল হক- ঘরগেরস্থি ৭। আখতারুজ্জামান ইলিয়াস- অপঘাত	Quiz-4
13	১।মুনির চৌধুরী- কবর	
14	পর্যালোচনা	

Reading References

১। বাংলা ভাষা ও সাহিত্য – রফিকুল ইসলাম ও সৌমিত্র শেখর

Recommended

১। ভাষাবিজ্ঞানের ভূমিকা ও বাংলা ভাষা – সৌরভ সিকদার

২। পাঠ্য বইয়ের বানান – আনিসুজ্জামান (সম্পা)

৩। Bengali Language and Culture by Afia Dil

Assessment Methods:

Category	Marks %
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5

Final Examination (3 hours)	70
Total	**100**

Course Title: Finance and Investment

Course Code: GED 211, Credit: 3.0, Total hours: 42

Course Description:

1. **Introduction:** Definition of finance, Principles of Finance, Functions of finance, Goal of a firm, Profit Maximization vs. Value creation, Management vs. Shareholders.
2. **Financial Environments:** Sole proprietorship , Partnership , Corporation , Income tax, Corporate tax , Capital gain tax ,Sales tax , VAT , Import duties , Purpose of financial market, Money and capital market, Primary and secondary market, Financial intermediaries and brokers.
3. **Time Value of Money:** Simple and compound interest, Future value (FVn) and present value (PV0), Discount rate, Annuities, Ordinary Annuity and Annuity Due, Compounding more than once a year.
4. **Risk and Return:** Defining risk and return, types of risk, Probability distribution to measure risk , Attributes towards risk , Diversification , Capital-asset pricing model (CAPM), CAPM Assumptions, Deriving the Capital Market Line, CAPM-System and Unsystematic Risk, SML, Test of CAPM, APT, Efficient financial market.
5. **Capital Budgeting:** Capital budgeting and its process , Generating investment , Project proposals , Estimating project after tax incremental operating cash flow , Sunk cost and opportunity cost , Alternative methods for evaluation and selection of project, Payback period (PBP) , Internal Rate of Return (IRR) under interpolation , Net present value (NPV), Profitability Index (PI).
6. **Cost of Capital:** Value creation, Cost of debt , Preferred stock and equity , CAPM approach, Beta , Risk free rate & Market Return,WACC.
7. **Overview of the Portfolio Formation Process: Investment Environment:** Investment, Financial Assets; Securities Market and its Characteristics; Measures of Return and Risk, Investment Management Process; The Asset Allocation Decision: Background and Investment Policy Statements; Investment Style Allocation; Analysis of Exchange Markets: Types of order, Exchange Membership; Margin: Margin on long purchase and margin on short sale.
8. **Security market indicator series:** Time Weighted and value Weighted Index- construction; DGEN; Free Float Index.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				
		C	P	A	S	PO
CO 1	1. **Understand** key ideas and concepts of Finance and Investment.	C2				PO11
CO 2	2. **Evaluate** the financing and investment alternatives.	C2			CT	PO11
CO 3	3. **Execute** analysis through different financing and investment tools and techniques.	C3			CT	PO11

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1											S	
CO2											S	
CO3											S	

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	**Chapter -1:** Definition of finance, Principles of Finance, Functions of finance	1	1,2	Explain Finance
2	Goal of a firm, Profit Maximization vs. Value creation, Management vs. Shareholders	1	1,2	Explain ultimate goal of a firm and management and owners.
3	**Chapter -2:** Sole proprietorship, Partnership , Corporation , Income tax, Corporate tax , Capital gain tax ,Sales tax , VAT , Import duties	1	1,2	Explain basic Business structure and different taxes and duties.
4	Purpose of financial market, Money and capital market, Primary and secondary market, Financial intermediaries and brokers	1	1,2	Explain Financial Market and the Participants.
5	**Chapter -3:** Time Value of Money, Simple and compound interest, Future value (FVn) and present value (PV0)	1,2,3	1,2	Understand interest and time value of money
6	Annuities, Ordinary Annuity and Annuity Due, Compounding more than once a year.	1,2,3	1,2	Understand annuities and compounding interest more than once a year.
7	Mathematics on time value of money	2,3	1,2	Analyze the financing and investment proposals
8	**Class Test 01-** From Lecture 1-7			Lecture 1 - 7
9	**Chapter- 04:** Defining risk and return, types of risk, Probability distribution to measure risk , Attributes towards risk	1,2	1, 2	Explain & Measure Risk and Return
10	Diversification, Capital-asset pricing model (CAPM), CAPM Assumptions, Deriving the Capital Market Line	1,2	1, 2	Understand CAPM and CML
11	CAPM-System and Unsystematic Risk, SML, Test of CAPM, APT, Efficient financial market.	1,2	1, 2	Explain fundamental concept CAPM, APT, Efficient Financial Market.
12	**Chapter- 05:** Capital budgeting and its process , Generating investment , Project proposals	1,2,3	1,2	Understand Capital Budgeting, Investment and projects
13	Estimating project after tax incremental operating cash flow, Sunk cost and opportunity cost	1,2,3	1, 2	Estimate the project cash flow and opportunity cost
14	Payback period (PBP) , Net present value (NPV)	1,2,3	1, 2	Explain different techniques of capital budgeting
15	Internal Rate of Return (IRR), Profitability Index (PI).	1,2,3	1, 2	Explain different techniques of capital budgeting
16	**Class Test 02-** From Lecture 9-15			Lecture 9-15

17	**Chapter- 06:** Value creation, Cost of debt , Preferred stock and equity	1,2,3	1,2,3	Explain cost of debt preferred stock and equity
18	CAPM approach, Beta	1,2,3	1,2,3	Explain CAPM and Beta
19	Risk free rate & Market Return,WACC	1,2,3	1,2	Explain Risk free rate & Market Return,WACC
20	**Chapter- 07:** Investment, Financial Assets; Securities Market and its Characteristics	1,2,3	2,3	Explain Investment and Different Financial Assets
21	Measures of Return and Risk, Investment Management Process; The Asset Allocation Decision: Background and Investment Policy Statements;	1,2,3	2,3	Explain Asset Allocation Decision and measurement of Risk and return
22	Investment Style Allocation; Analysis of Exchange Markets: Types of order	1,2,3	2,3	Explain Exchange Markets, Types of Order
23	Exchange Membership; Margin: Margin on long purchase and margin on short sale.	1,2,3	2,3	Explain Exchange Membership Margin on long purchase and margin on short sale.
24	**Chapter- 08:** Time Weighted and value Weighted Index- construction.	2,3	3	Explain Time Weighted and value Weighted Index.
25	Time Weighted and value Weighted Index-construction; DGEN; Free Float Index.	2,3	3	Construct Time Weighted and value Weighted Index, Free Float Index.
26	Class Test 3: Lecture 17-25			Lecture 17 – 25
27	Review Class			
28	Review Class			

Reading Reference:

1. Principles of Managerial Finance – Lawrence J. Gitman, Chad J. Zutter.
2. Fundamentals of Financial Management– Brigham & Houstom.
3. Investment Analysis &Portfolio Management–FRANK K. REILLY & KEITH C. BROWN

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

LEVEL-2 TERM- II

Course Title: Computer Algorithms and Complexity
Course Code: CSE 221, Credit: 3.0, Total hours: 42

Course Description:

1. Introduction to algorithms, correctness proof and techniques for analysis of algorithms. 2. Methods for the design of efficient algorithms: divide and conquer, greedy method and dynamic programming.
3. Graph algorithms: Basic search and traversal techniques, topological sorting, connected components, spanning trees and shortest paths.
4. Flow algorithms; Lower bound theory; NP-completeness, NP-hard and NP-complete problems.
5. Coping with Hardness: backtracking, branch and bound, approximation algorithms.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Analyze** the asymptotic performance and write rigorous correctness proofs for algorithms	C2				PO1,PO2, PO4, PO12
CO 2	**Estimate** the performance of various algorithms and data structures	C2				PO1,PO2, PO3, PO12
CO 3	**Apply** important algorithmic design paradigms and methods of analysis for problem solving	C3			CT	PO1,PO2, PO4, PO12
CO 4	**Device** efficient algorithms in common engineering design situations	C3			CT	PO1,PO2, PO3,PO4, PO12

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	P O 10	PO11	PO 12
CO1	S	S		S								S
CO2	S	S	S									S
CO3	S	S		S								S
CO4	S	S	S	S								S

Lecture Outline:

Class	Topics/Assignment	COs		Lecture Outcomes/ Activities
1	Introduction to algorithms	1	1	Explain Algorithm definition characteristics and complexity,
				Asymptotic notations
2	Correctness proof and techniques for analysis of algorithms	1,2	1,2	Complexity analysis using Frequency Count method
3	Correctness proof and techniques for analysis of algorithms	1,2	1,2	Complexity analysis using Frequency Count method
4	Correctness proof and techniques for analysis of algorithms	1,2	1,2	Complexity analysis using substitution method for solving recurrences
5	Methods for the design of efficient algorithms: divide and conquer	1,2	2	Complexity analysis using recursion-tree

				method for solving recurrences
6	Methods for the design of efficient algorithms: divide and conquer	1,2	2	Complexity analysis using Master Theorem
7	Methods for the design of efficient algorithms: divide and conquer	2,3	1	Explain Merge Sort and Quick Sort
8	Methods for the design of efficient algorithms: divide and conquer	2, 3	1	Explain finding the k-th element and finding the maximum and minimum
9	Methods for the design of efficient algorithms: divide and conquer	2, 3	1	Explain Strassen's Mstrix multiplication
10	Methods for the design of efficient algorithms: greedy method	2, 3		Explain Knapsack problem
11	Class Test 1			Lecture 1– 8
12	Methods for the design of efficient algorithms: greedy method	2, 3	1	Explain Tree Vertex Splitting
13	Methods for the design of efficient algorithms: greedy method	2, 3	1	Explain Job Sequencing with deadlines
14	Methods for the design of efficient algorithms: greedy method	2, 3	1	Explain Minimum Cost Spanning Tree
15	Methods for the design of efficient algorithms: greedy method	2, 3	1	Explain Single Source Shortest Path
16	Methods for the design of efficient algorithms: greedy method	2, 3	1	Explain Huffman Coding
17	Class Test 2			Lecture 9– 10, Lecture 12– 16
18	Methods for the design of efficient algorithms: dynamic programming	2, 3	1	Explain Single Source Shortest Path

19	Methods for the design of efficient algorithms: dynamic programming	2, 3		Explain All Pair Shortest Path
20	Methods for the design of efficient algorithms: dynamic programming	2, 3		Explain Longest Common Subsequence
21	Assignment			Based on Lecture 1– 20
22	Review class			Lecture 1– 21
23	Methods for the design of efficient algorithms: dynamic programming	2, 3,4	1	Explain Travelling Salesman Problem
24	Methods for the design of efficient algorithms: dynamic programming	2, 3,4	1	Explain 0/1 Knapsack
25	Methods for the design of efficient algorithms: dynamic programming	2, 3,4		Explain Rod Cutting
26	Basic search and traversal techniques	2, 3		Explain BFS and DFS
27	Connected components	2, 3		Explain Connected components
28	Backtracking	2, 3	1	Explain N-queens Problem
29	Class Test 3			Lecture 18– 20, Lecture 23– 26
30	Backtracking	2, 3	1	Explain Graph Coloring
31	Backtracking	2, 3,4	1	Explain Hamiltonian Cycles
32	Backtracking	2, 3,4		Explain Sum of Subsets
33	Flow algorithms	3,4	1	Explain Flow Network
34	Flow algorithms	3,4	1	Explain Ford-Fulkerson Method
35	Class Test 4			Lecture 28, Lecture 29-34

36	NP-hard and NP-complete problems	1,3, 4	1	Explain Basic Concepts
37	NP-hard and NP-complete problems	1,3, 4		Explain Cook's Theorem
38	Approximation algorithms	3,4	1	Explain Vertex Cover Problem
39	Approximation algorithms	3,4	1	Explain Set Cover Problem
40	Review Class and Problem Solutions			Lecture 22-39
41	Review Class and Problem Solutions			Lecture 1 – 39
42	Suggestion			Lecture 1 – 41

Reading Reference:

1. Fundamentals of Computer Algorithms – Ellis Horowitz, Sartaj Sahni and Sanguthevar Rajasekaran
2. Introduction to Algorithms - Thomas H. Corman, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Computer Algorithms and Complexity Sessional
Course Code: CSE 222, Credit: 1.5, Total hours: 42

Rationale:
The study of data structure is an essential part of computer science. Data structure is a logical and mathematical model of storing & organizing data in a particular way in a computer. In system programming application programming the methods & techniques of data structures are widely used. The study of data structure helps the students in developing logic & structured programs.

Objective:

1. General Understanding of Basic Data Structures
2. Developing Programming Skills for Advanced Data Structures such as Trees, Hash Tables, and Priority Queues etc.

Course Outcomes (CO):
Upon completion of the course, the students will be able to:

1. Practically demonstrate advantages and disadvantages of specific algorithms and data structures.
2. Select basic data structures and algorithms for autonomous realization of simple programs or program parts.
3. Initiate practical knowledge to determine and demonstrate bugs in programs.
4. Formulate new solutions for problems or improve existing code using learned algorithms and data structures

Course Content:

Implementation of Abstract data types; Introduction to algorithms; Asymptotic analysis: growth of functions, O, Ω and Θ notations; Correctness proof and techniques for analysis of algorithms; Master Theorem; Elementary data structures: arrays, linked lists, stacks, queues, trees and tree traversals, graphs and graph representations, heaps, binary search trees; Graph Traversals: DFS, BFS, Applications of DFS and BFS; Sorting: heapsort, merge sort, quicksort; Data structures for set operations; Methods for the design of efficient algorithms: divide and conquer, greedy methods, dynamic programming.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	**Program Outcome (PO)**											
	1	2	3	4	5	6	7	8	9	10	11	12
Practically demonstrate advantages and disadvantages of specific algorithms and data structures.									√			
Select basic data structures and algorithms for autonomous realization of simple programs or program parts.									√			
Initiate practical knowledge to determine and												

demonstrate bugs in programs.								√				
Formulate new solutions for problems or improve Existing code using learned algorithms and data structures.					√							

Lecture Outline:

Week	Lecture	Topics	Remarks
1	Lec 1 Lec 2 Lec 3	Array List (Implementing List using Array)	
2	Lec 4 Lec 5 Lec 6	Binary Search	
3	Lec 7 Lec 8 Lec 9	Implementing Stack using Array	
4	Lec 10 Lec 11 Lec 12	Implementation of Queue using Array, Circular Queue	
5	Lec 13 Lec 14 Lec 15	Single Linked List Implementation	
6	Lec 16 Lec 17 Lec 18	Practice on Single Linked List	
7	Lec 19 Lec 20 Lec 21	**Lab Test-1**	
8	Lec 22 Lec 23 Lec 24	Double Linked List Implementation	
9	Lec 25 Lec 26 Lec 27	Operation on Binary Search Tree (Insertion, Deletion, Traversal, Merging)	
10	Lec 31 Lec 32 Lec 33	Operation on Heap Trees/Priority Queue (MAX and MIN Heap)	
11	Lec 28 Lec 29 Lec 30	Implementation of Graph Algorithms: BFS	
12	Lec 34 Lec 35	Implementation of Graph Algorithms: DFS	

	Lec 36		
13	Lec 37 Lec 38 Lec 39	**Lab Test-02**	
14	Lec 40 Lec 41 Lec 42	**Quiz**	

Reading Reference:

1. Classic Data Structures 2nd Ed – Debasis Samanta
2. Data Structures using C 2nd Ed – Aaron M. Tenenbaum, Yedidyah Langsam, Moshe J. Augenstein

Assessment Methods:

Category	**Marks %**
Lab Test – 1	25
Lab Test – 2	25
Viva	10
Class Participation	10
Class Assessment	30
Total	**100**

Course Title: Digital Logic Design
Course Code: CSE 223, Credit: 3.0, Total hours: 42

Course Description:

Number systems and codes; Digital logic: Boolean algebra, De-Morgan`s theorems, logic gates and their truth tables, canonical forms, combinational logic circuits, minimization techniques; Arithmetic and data handling logic circuits, decoders and encoders, multiplexers and de-multiplexers; Flip-flops, race around problems; Counters; asynchronous counters, synchronous counters and their applications; Registers and basic memory unit; Synchronous and asynchronous logic design; Design of sequential circuit: State diagram; State minimizations and assignments; Pulse mode logic; Fundamental mode design; PLA design.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				
		C	P	A	S	PO
CO 1	**Apply** principles of Boolean algebra to logic functions.	C3				PO1
CO 2	**Use** K-maps to realize two-level minimal/optimal combinational circuits with up to 4-5 variables.	C3				PO1
CO 3	**Construct** gate-level implementation of a combinational logic function using fundamental logic gates, multiplexers, decoders and programmable logic devices.	C3				PO3
CO 4	**Analyze** and **Design** sequential circuits built with various flip-flops, registers, counters.	C4, C6			CT	PO3

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking)

Mapping of CO and Program Outcomes:

Cos	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO 9	PO10	PO11	PO12
CO1	S											
CO2	S											
CO3			S									
CO4			S									

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Overall idea of Digital Logic Design (DLD), lecture plans, examples of DLD in Computer Science and Engineering, review of question pattern, purpose and implementation of DLD etc.	N/A		Importance of DLD in Computer Science and Engineering and real world, Overall idea about course details.
2	Number systems, Converting from one number system to another, Fractional Numbers.	1	1	To be familiar with different number systems and their conversions.
3	Computer arithmetic, addition, subtraction, etc.	1	1	How ALU and CU operate the arithmetic operations.
4	Different Complementary methods, Multiplication, division etc.	1	1	To be familiar with other arithmetic operations.
5	Binary codes, Error detection codes, The Reflected code.	1	1	Definition of digital and analog signal, their difference and signal representation in different domains.
6	Binary storage and Registers	1,4	1	Here, students will learn about how the binary data is stored in registers and their data flow.

7	Binary Logic and Integrated Circuits.	1	1	Description of different logic operation of binary number and ICs.
8	Boolean algebra, basic theorem and properties.	1	1	Description of different theorems of Boolean functions and algebra.
9	Venn diagram, Boolean functions.	1	1	Representation of Venn diagram and Boolean functions.
10	Algebraic manipulation, Canonical and standard forms.	1	1	Here, students will learn about Algebraic manipulation, Canonical and standard forms.
11	Sum of minterms and product of maxterms.	1	1	Representation of sum of minterms and product of maxterms.
12	Conversion between canonical forms.	1	1	Technique of different conversion between canonical forms.
13	Simplification of Boolean functions, The Map method, 2 and 3 variable maps.	2	1,2	Method of how to simplify the Boolean functions, the K-Map method for 2 and 3 variables.
14	K-Map: Four-variable map, five.	2	1,2	Simplify the Boolean function using K-Map for 4 and 5 variables.
15	K-Map for Six variable maps, Product of sums simplification.	2	1,2	Simplify the Boolean function using K-Map for 6 variables and POS.
16	Don't care conditions, Exercises.	1,3	1,2	Simplify the Boolean function using K-Map for Don't care conditions.
17	Combinational Logic, Introduction, Design procedure, Adders: Half and Full adders.	1,3	1,2	Description of combinational logic and it's design procedure.
18	Half and Full sub tractors, Code Conversion.	3	1,2	Here, students will learn about functions of the sub tractors and code conversion techniques.
19	Analysis procedure, Universal gate.	3	1	Analysis procedure of combinational logic and discussion of universal gate.
20	Combinational logic, Introduction, Binary parallel adder.	3	1,2	Here, students will know about different kinds of combinational logic circuits.
21	Design a BCD to ex-3/2421/84-2-1 code converter.	3	1,2	Code conversion and simplification using K-Map.
22	Carry propagation, Look-ahead carry generator.	3	1,2	How to propagate carry and carry generator.
23	4-bit full adders with look-ahead carry.	3	1,2	Propagation of full adders with look-ahead carry.
24	Decimal adder: BCD adder.	3	1,2	Students will learn about different adders.
25	Decoders.	3	1,2	Students will learn about decoders.
26	De-multiplexers.	3	1,2	Students will learn about job functions of De-multiplexers.
27	Encoders, Multiplexers.	3	1,2	Job functions and simplification of encoders and multiplexers.

28	Boolean function implementation with multiplexers.	3	1	Technique of implementation of Boolean function by using multiplexers.
29	ROM, Types of ROMs.	3	1	Will know about different types of ROM and their work functions.
30	Programmable Logic Array, PLA table, Exercises.	3	1	Process of PLA and will solve various exercises of this chapter.
31	Sequential Logic: Flip-Flops, Basic Flip-Flops Circuit.	4	1,2	Basic of Sequential circuits/logic and Flip-Flops.
32	Clocked RS Flip-Flop, D Flip-Flop,	4	1,2	Students will learn how the data flow in RS and D Flip-Flop.
33	JK Flip-Flop, T Flip-Flops.	4	1,2	Students will learn how the data flow in JK and T Flip-Flop.
34	Master Slave Flip-Flops, Edge-Triggered Flip-Flops.	4	1,2	Description of Master Slave and Edge-Triggered Flip-Flops.
35	Analysis of Clocked sequential circuits, State table, state diagrams, State Equations.	4	1,2	Process of clocked sequential circuits and their generation of state table, state diagrams and state equations.
36	Counters: Asynchronous Counters,	4	1,2	Job functions of Counters and its basic idea.
37	Ripple up counter, Ripple down counter.	4	1,2	Details and technique of Ripple up and down counter.
38	Synchronous Counters, Shift Register counter.	4	1,2	Description of Synchronous and Shift Register counter.
39	Ring counter, Johnson counter.	4	1,2	Description of Ring counter and Johnson counter.
40	Register Buffer register.	4	1,2	Different types of Register.
41	Shift right register, Shift left register.	4	1,2	Details and technique of Shift right and Shift left register.
42	Review class for all above topics.	N/A		Review class and Problem solving

Reading Reference:

1. Digital Logic and Computer Design-M. Morris Manno, Latest Edition.
2. Digital Computer Electronics - Albert P. Malvino, Jerald A Brown, Latest Edition.

Assessment Methods:

Category	Marks %
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Digital Logic Design Sessional
Course Code: CSE 224, Credit: 1.5, Total hours: 42

Rationale:
This course aims to provide students with a knowledge of problem solving with digital logic circuits & systems. The basic building blocks of combinational and sequential circuits are introduced to enable students to develop circuit solutions to problems and to understand the design and operation of hardware models of digital systems

Course Outcomes (CO):
Upon completion of the course, the students will be able to:

1. Demonstrate the basic tools for designing with combinational and sequential digital logic and state machines.
2. Interpret practical knowledge to state simple digital circuits in preparation for computer engineering.
3. Describe the different switching algebra theorems and apply them for logic functions.
4. Practically correlate a given problem, design with different ICs and equipment.
5. Practically predict or syntheses simple projects and design those projects with different equipment.

Course Content:

Number systems, complements and codes; Digital logic: Boolean algebra, De-Morgan`s theorems, logic gates and their truth tables, canonical forms, combinational logic circuits, minimization techniques; Arithmetic and data handling logic circuits, decoders and encoders, multiplexers and de-multiplexers; Flip-flops, race around problems; Counters; asynchronous counters, synchronous counters and their applications; Registers and basic memory unit; Synchronous and asynchronous logic design; Design of sequential circuit: State diagram; Mealy and Moor machines; State minimizations and assignments; Pulse mode logic; Fundamental mode design; PLA design using MSI and LSI components.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
Demonstrate the basic tools for designing with combinational and sequential digital logic and state machines.									√			
Interpret practical knowledge to state simple digital circuits in preparation for computer engineering.									√			
Describe the different switching algebra theorems and apply them for logic functions.							√					
Practically correlate a given problem, design with different ICs and equipment.						√						
Practically predict or syntheses simple projects and design those projects with different equipment.									√			

Lecture Outline:

Week	Lecture	Topics	Remarks
1	Lec 1 Lec 2 Lec 3	Verify Basic Logic Gates and Truth table of the Logic Gates.	
2	Lec 4 Lec 5 Lec 6	Combionational Circuit (Light Your Lamp)	
3	Lec 7 Lec 8 Lec 9	Experiments Based on Truth tables and Boolean functions	
4	Lec 10 Lec 11 Lec 12	Experiments Based on Truth tables and K-maps	
5	Lec 13 Lec 14 Lec 15	Design and implementation of the Logic Circuits using K-maps (7 Segment Display)	
6	Lec 16 Lec 17 Lec 18	Experiments Based on Adder/Subtractor	
7	Lec 19 Lec 20 Lec 21	**Lab Test-1**	
8	Lec 22 Lec 23 Lec 24	Experiments Based on Comparator	
9	Lec 25		

	Lec 26 Lec 27	Design and Implementation of Combinational Circuits using Multiplexer	
10	Lec 31 Lec 32 Lec 33	Design and Implementation of encoder and decoder	
11	Lec 28 Lec 29 Lec 30	Design and implement Flip Flop using basic gates	
12	Lec 34 Lec 35 Lec 36	Design and Implements Counters using Flip-Flops	
13	Lec 37 Lec 38 Lec 39	Lab Test- 2	
14	Lec 40 Lec 41 Lec 42	**Quiz & Viva**	

Reading Reference:

1. Digital Logic and Computer Design-M. Morris Manno
2. Digital Computer Electronics - Albert P. Malvino, Jerald A Brown

Assessment Methods:

Category	**Marks %**
Lab Test – 1	25
LabTest – 2	25
Viva	10
Class Participation	10
Class Assessment	30
Total	**100**

Course Title: Numerical Methods
Course Code: CSE 225, Credit: 3.0, Total hours: 42

Course Description:

1. Introduction; Solution of algebraic and transcendental equations: method of iteration, False Position method, Newton-Rhapson method; Least squares approximation of functions: linear and polynomial regression, fitting exponential and trigonometric functions.

2. Solution of simultaneous linear equations: Cramer's rule, iteration method, Gauss-Jordan Elimination method, Choleski's process;

3. Interpolation: diagonal and horizontal difference, differences of a polynomial, Newton's formula for forward and backward interpolation, Spline interpolation;

4. Integration: Gauss quadrature formula, Trapezoidal rule, Simpson's rule, Weddle's rule;

5. Solution of ordinary differential equations: Euler's method, Picard's method, Milne's method, Taylor's series method, Runge-Kutta method;

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Recall** basics of accuracy and finite precision, round of errors, truncation errors, and stability of numerical algorithms.	C1				PO1
CO 2	**Apply** appropriate techniques for numerical methods and calculate the computational cost in the field of Science and some fields of Engineering.	C3			CT	PO2
CO3	**Formulate** Numerical techniques for root finding, curve fitting, differentiation, and integration.		P2		CT	PO2

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2		S										
CO3		S										

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Introduction of Numerical Methods Uses and Applications	1	1	Understand basic information about numerical methods
2	Accuracy and Errors Truncation, Rounding-off, Algorithmic, Absolute errors	1	1	Relate different error of numerical methods
3	Solution of algebraic and transcendental equations: Theory of Bisection Method, Algorithm, Uses & Applications, Advantages & Disadvantages	2,3	1	Overview of method and understand uses and applications.

4	Implementation of Bisection Method	2,3	1	Solve equations to find the real roots using bisection method.
5	Method of False Position Theorem, Algorithm, Uses & Applications, Advantages & Disadvantages	2,3	1	Differentiate false position and bisection method, and overview of this method
6	Implementation of False position	2,3	1	Solve equations to find the real roots using false position method.
7	Method of Iteration, Successive Approximation Theorem, Algorithm , Uses & Applications, Advantages & Disadvantages	2,3	1	Overview of Successive methods.
8	Newton-Raphson Method Convergence of N-R Method, Error Estimate, Algorithm , Uses & Applications, Advantages & Disadvantages	2,3	1	Estimate algorithm and theorem of newton -Raphson method.
9	Implementation of Iteration and Newton-Raphson Method	2,3	1, 3	Solve equations to find the real roots using iteration and newton-Raphson method.
10	**Solution of Simultaneous Linear Equations:** Gauss Elimination Method, Problem Solving	2, 3	1, 3	Solve a system with equations using gauss-elimination method.
11	Gauss Jordan Method, Problem Solving	2, 3	1, 3	Solve a system with equations using gauss Jordan method.
12	Review			Lecture 1– 11
13	Class test 1			Lecture 1– 11
14	Gauss Jacobi's method, problem solving	2, 3	1, 3	Solve a system with equations using Gauss Jacobi method.
15	Problem Solving of gauss seidel method	2, 3	1, 3	Solve a system of linear equations using Gauss Seidel method.
16	Problem solving			Lecture 14-15
17	Finite Differences Operators Basic explanation	2	3	Introduction of finite difference
18	Forward Differences	2	3	Overview of forward difference
19	Backward Differences, Central Differences	2	3	Overview of backward and central difference
20	Review			Lecture 17 – 19
21	Class test 2			Lecture 14-19
22	Newton's Formula for Forward Interpolation	3	2	Obtain value from given data using Newton's Forward Interpolation
23	Newton's Formula for Backward Interpolation	3	2	Obtain value from given data using Newton's Forward Interpolation

24	Problem Solving			Lecture 22-23
25	Gauss's forward interpolation	2,3	2	Apply Gauss's forward interpolation to find value from given data.
26	Gauss's backward interpolation	2,3	2	Apply Gauss's backward interpolation to find value from given data.
27	Review			Lecture 25 – 26
28	Class Test 3			Lecture 25 – 26
29	Numerical Integration , Quadrature Formula Theory, Equation Derivation,	2	2	Overview of Numerical Integration.
30	Proof of Trapezoidal Rule and Simpson's Rule	3	2	Evaluate integral equations using Trapezoidal Rule and Simpson's Rule
31	Problem solving			Evaluate integral equations using Trapezoidal Rule and Simpson's Rule
32	Method of Group of Average Fitting Straight Line (2 variables)	2,3	2	Obtain equation of straight-line using group of averages.
33	Method of Group of Average Fitting Parabola (3 variables)	2,3	2	Fit a parabola using group of averages.
34	Principle of Least Squares Method Fitting Straight Line (2 variables)	2,3	2	Obtain equation of straight-line using least-square method.
35	Principle of Least Squares Method Fitting Parabola (3 variables)	2,3	2	Fit a parabola using least-square method.
36	**Class test 4**			Lecture 32-35
37	**Solution of Ordinary Differential Equations:** Taylor's series Method	2,3	2	Apply Taylor's series method.
38	Theorem of Euler's Method and problem solving	2,3	2	Apply Euler's series method.
39	Theorem of Improved Euler's Method and problem solving	2,3	2	Apply Improved Euler's series method
40	Theorem of Modified Euler's Method and problem solving	2,3	2	Apply Modified Euler's series method
41	Review			Lecture 36 – 39
42	Review Class, Problem Solutions and Suggestion			Lecture 1 – 41

Reading Reference:

1. Numerical Methods for Engineers – Steven C Chapra, Raymokand P Canale
2. Numerical Methods – E Balagurusamy
3. Numerical Methods -- S. Balachandra Rao, C.K.Shantha
4. Numerical Methods for Scientific and Engineering Computation -- M.K. Jain, Iyengar, R.K. Jain

Assessment Methods:

Category	Marks %
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Numerical Methods Sessional
Course Code: CSE 226, Credit: 1.0, Total hours: 28

Rationale:

In numerical analysis, a numerical method is a mathematical tool designed to solve numerical problems. The implementation of a numerical method with an appropriate convergence check in a programming language is called a numerical algorithm

Objective:

1. To calculate the solutions of nonlinear equations in one variable
2. To analyse the interpolation and approximation
3. To analyse numerical differentiation and integration
4. To understand direct methods for solving linear systems, numerical solution of ordinary differential equations.

Course Outcomes (CO):
Upon completion of the course, the students will be able to:

1. Demonstrate a familiarity with major algorithms and data structures.
2. Write rigorous correctness proof and analyze the running time of the advanced algorithms and data structures for some classic problems;
3. Apply important algorithmic design paradigms and methods of analysis.
4. Synthesize efficient algorithms and data structures in common engineering design situations.

Course Content:

Introduction; Mathematical Model. Sources of errors, their propagation and the idea of conditioning. Solution of algebraic and transcendental equations: Iterative methods for root finding. Rates of convergence. False Position method, Newton-Rhapson method; Solution of simultaneous linear equations: Gauss-Jordan Elimination method, Choleski's process; Dolittle and Crout factorization; Interpolation: diagonal and horizontal difference, differences of a polynomial, Newton's formula for forward and backward interpolation, Spline interpolation; Integration: Gauss quadrature formula, Trapezoidal rule, Simpson's

rules, Weddle's rule; Solution of ordinary differential equations: Euler's method, Picard's method, Milne's method, Taylor's series method, Runge-Kutta method; Least squares.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
Demonstrate a familiarity with major algorithms and data structures.									√			
Write rigorous correctness proof and analyze the running time of the advanced algorithms and data structures for some classic problems						√						
Apply important algorithmic design paradigms and methods of analysis.									√			
Synthesize efficient algorithms and data structures in common engineering design situations.											√	

Lecture Outline:

Week	Lecture	Topics
1	Lec 1 Lec 2	Introduction; Mathematical Model. Sources of errors, their propagation and the idea of conditioning
2	Lec 3 Lec 4	Solution of algebraic and transcendental equations: Iterative methods for root finding
3	Lec 5 Lec 6	Rates of convergence. False Position method, Newton-Rhapson Method
4	Lec 7 Lec 8	Solution of simultaneous linear equations: Gauss-Jordan Elimination method
5	Lec 9 Lec 10	Choleski's process; Dolittle and Crout factorization
6	Lec 11 Lec 12	Interpolation: diagonal and horizontal difference, differences of a Polynomial
	Lec 13	

7	Lec 14	**Online 1**
8	Lec 15 Lec 16	Newton's formula for forward and backward interpolation, Spline Interpolation
9	Lec 17 Lec 18	Integration: Gauss quadrature formula
10	Lec 19 Lec 20	Trapezoidal rule, Simpson's rules, Weddle's rule
11	Lec 21 Lec 22	Solution of ordinary differential equations: Euler's method, Picard's method, Milne's method
12	Lec 23 Lec 24	Taylor's series method, Runge-Kutta method; Least squares
13	Lec 25 Lec 26	**Online 2**
14	Lec 27 Lec 28	**Quiz / Viva**

Reading Reference:

1. Data Structures - Reingold and Hansen
2. Data structures and algorithm – Hopcroft, Ullman
3. Data Structures and Algorithms in Java - Michael T. Goodrich, Roberto Tamassia
4. Introduction to Algorithms - T. H. Cormen, C. E. Leiserson, R. L. Rivest and C. Stein, MIT Press

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Probability and Statistics
Course Code: MATH 221, Credit: 3.0, Total hours: 42

Course Contents:

Probability and Statistics:
Introduction. Sets and probability. Random variable (Discrete and Continuous variable) and its probability distributions' (PMF, PDF, JPD, MD, CD, Independent variable), Mathematical expectation. Some discrete probability distributions. Normal distribution, Sampling theory (Treatment of grouped sampled data). Estimation theory. Test of hypothesis and Significance. Curve Fitting, Regression, and correlation. Analysis of variance.

Objective:
Targets of the course are the students to be able to:

1. understand the basic probability axioms and rules and the moments of discrete and continuous random variables as well as be familiar with common named discrete and continuous random variables.
2. understand several well-known distributions.
3. develop the concept of the central limit theorem
4. perform hypotheses testing.

Course Outcomes (CO):

Course Outcomes (CO) of the Course			
Course Outcome	Bloom's Taxonomy	Domain	Program Outcome
CO1: **Understand** elementary probability and statistical theory and practical data analysis techniques and students can **implement** a small report summarizing, presenting and analyzing an appropriate data set and distributions.	Level -2 (**Understand**)	C	PO 1: Engineering Knowledge K2(Conceptually based mathematics)
CO2: **Explain** hypothesis testing using statistical methods and estimation and **compute** and interpret the results of correlation and regression.	Level -3 (**Apply**)	C	PO 2: Problem analysis K2(Conceptually based mathematics)

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork (These domains are cognitive (thinking), affective (emotion/feeling), and psychomotor (physical/kinesthetic**))**
K4: knowledge profile, P2, P3, P4: complex engineering problem

Mapping of Course Outcomes and Program Outcomes:

Cos	Program Outcomes S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2		M										

Lecture Schedule:

Week 1		**CTs**	**COs**
Class 1	Basic concept of Probability.		CO 1
Class 2	Probability		
Class 3	Sets and probability		
Week 2			
Class 4	Basic probability (Conditional Probability)		CO 1
Class 5	Basic probability (Conditional Probability)		
Class 6	Random variable (Discrete and Continuous variable)		
Week 3			CO 2
Class 7	Random variable (Discrete and Continuous variable)		
Class 8	Random variable (Discrete and Continuous variable)		
Class 9	Probability distributions (PMF, PDF)		
Week 4			CO 2
Class 10	Probability distributions' (JPD ,MD)	CT-1	
Class 11	Class Test on Probability Introduction and conditional probability		
Class 12	Probability distributions' (JPD ,MD)		
Week 5			CO 2
Class 13	Mathematical expectation	Assignment-1	
	Assignment based on Random variable (Discrete and Continuous variable)		
Class 14	Mathematical expectation		
Class 15	Mathematical expectation		
Week 6			CO 2
Class 16	Some discrete probability distributions		
Class 17	Some discrete probability distributions		
Class 18	Some discrete probability distributions		
Week 7			CO 2
Class 19	Normal distribution		
Class 20	Normal distribution	CT-2	
Class 21	Class Test on Probability distributions' (JPD ,MD), Probability distributions' (CD, Independent variable)		
Week 8			CO 1
Class 22	Sampling Theory		
Class 23	Sampling Theory		

Class 24	Sampling Theory		
Week 9			CO 1
Class 25	Estimation Theory		
Class 26	Estimation Theory		
Class 27	Estimation Theory		
Week 10		CT-3	CO 2
Class 28	Curve Fitting.		
Class 29	Regression.		
Class 30	Class Test on Sampling Theory, Estimation Theory		
Week 11			CO 2
Class 31	Regression.		
Class 32	Regression.		
Class 33	Correlation.		
Week 12			CO 2
Class 34	Correlation.		
Class 35	Correlation.		
Class 36	Analysis of Variance		
Week 13		Assignment -2	CO 2
Class 37	Analysis of Variance		
Class 38	Analysis of Variance		
Class 39	Analysis of Variance		
	Assignment based on Curve Fitting., Regression and Correlation		
Week 14		CT- 4	CO 2
Class 40	Analysis of Variance		
Class 41	Class Test on Test of hypothesis and Analysis of Variance		
Class 42	Review		

Text and Reference Books:

1. Probability and Statistics-M Nurul Islam
2. Probability and Statistics-Murray R Spiegel
3. Fundamentals of Probability & Probability Distributions – Manindra Kumar Roy

Linkage of Course Outcome with Assessment Methods and their Weights:

COs	Assessment Method	(100%)
	Class Assessment	
	Attendance	5%
	Class Performance	5%
CO 1-2	Assignment	5%
CO 1-2	Class Test	15%
	Exam	
CO 1-2	Final Exam	70%

Course Title: Engineering Economics
Course Code: GED 221, Credit: 3.0, Total hours: 42

Course Description:

- Introduction to economics.
- Supply and demand
- Various forms of market
- Analysis of costs.
- Analysis of national income.
- Banking- commercial bank and central Bank.
- Government Budgets and Fiscal Policy
- Economic development and growth.

Course Learning Outcomes (CLO):

Upon successful completion of this course, students will be able to:

1. develop preliminary knowledge of economic concepts for understanding of national and international economy.
2. understand how markets work, when markets fail and how the government can affect market outcomes.
3. evaluate macroeconomic policies based on their effects on the economic variables of interest.
4. apply different techniques of investment decisions.

Mapping of Course Outcome to Program Outcome

	PLO1	PLO2	PLO3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1										m		
CLO2												
CLO3												
CLO4											s	

Class	**Topics/Assignment**	COs	Reading Reference
1	The Fundamentals of Economics	1,3	1,2
2	Central Problems of an Economy	1	1,2
3	Different Economic systems	1	1,2
4	Production Possibility Curve (or Transformation Curve)	1	1,2
5	Utility, Diminishing marginal utility	1	1,2
6	Theory of Demand	1	1,2
7	Elasticity of demand- price, income and cross elasticity	1	1,2
8	Theory of Supply.	1	1,2
9	Equilibrium	1	1,2
10	Class test -1		
11	Consumer surplus and Producer surplus	1	1,2
12	Indifference curve technique	1	1,2
13	Theory of Production	1	1,2
14	Analysis of cost.	1	1,2
15	Various forms of market- monopoly, perfect competition, monopolistic competition, oligopoly, duopoly.	2	1,2
16	National Income.	1,3	1,2
17	Consumptions, Savings and Investment.	1,3	1,2
18	Inflation and effect of inflation on production and distribution of wealth.	1,3	1,2
19	Types of inflation	1,3	1,2
20	Unemployment	1,3	1,2
21	Class test-2		
22	Banking- Commercial Banking and Central Bank	1	1,2
23	Government Budgets and Fiscal Policy	1,3	1,2
24	Poverty and Economic Inequality	1,3	1,2
25	Economic Growth	1,3	1,2
26	Theories of developments	3	1,2
27	Development partners (World Bank, Asian Development Bank, World Trade Organization, International Monetary Fund)	3	1,2
28	Cost benefit analysis, NPV	4	1,2
29	IRR, Payback period	4	1,2

Reference Book(s):

1. Dewett, K K. and Chand Adarsh, Modern Economic Theory, Shyamlal Charitable Trust, New Delhi, 2003.
2. Samuelson, Paul & Nordhaus, William D., Economics, McGraw-Hill Inc., New York, 1995
3. Parkin, Michael, "Microeconomics", Addison – Wesley Publishing Company.

Assessment Methods:

Category	Marks %
Class Participation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

LEVEL- 3 TERM-I

Course Title: Operating System
Course Code: CSE 311, Credit: 3.0, Total hours: 42

Course Description:

1. Introduction of Operating System, Types of OS; Process: process managements, process states, job and process scheduling, CPU scheduling algorithms, process coordination, critical section problems, semaphores, Inter-Process Communication (IPC), classical IPC problems, multiprocessing and timesharing, message and mailbox etc

2. Memory management: swapping, memory allocation schemes, Paging and segmentation, virtual memory, page replacement strategies, working sets, demand paging; Input/output: hardware/software, disk, disk scheduling algorithms, Secondary storage management, terminals, clocks

3. Deadlock: resource allocation, detection, prevention, avoidance and recovery.

4. Main Features of Windows, UNIX, LINUX; Introduction to distributed operating systems.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Illustrate** the fundamentals concepts of computer system organization and the structure of operating systems.	C2				PO1

CO 2	**Demonstrate** different CPU and disk scheduling algorithms and their respective importance.	C3			CT	PO3
CO 3	**Analyze** the working procedure and structure of main and virtual memory.	C4			CT	PO2
CO 4	**Outline** the various aspects of process management mechanisms and deadlock.	C4			CT	PO2

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	M											
CO2			S									
CO3		S										
CO4		S										

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Introduction of OS (Operating System), Relationship between OS and Hardware, The importance of OS, Major Issues related to OS, Three important views of OS.	1	1,2	Overview of OS and its importance in computer system.
2	Types of OS with their benefits and shortcomings.	1	1,2	Classification of OS.
3	Types of OS with their benefits and shortcomings.	1	1,2	Classification of OS.
4	Computer System Structure, Components of Computer System, Start up process of Computer, Interrupts in Computer System,	1	1,2	Overview of Computer Structure and OS operations.
5	Dual Mode Operation of OS, Activities of OS, OS Services.	1	1.2	OS operations and activities.
6	System Call, Types of System Call, System Programs, Different types of OS structures	1	1,2	OS operations, activities, structure.
7	Review Class			Lecture 1-7
8	Class Test – 01			Lecture 1-7
9	Process concept, Process states, PCB, Process Scheduling.	4	1,2	Process Management
10	Context Switch, Process creation and termination, Inter Process Communication (IPC).	4	1,2	Process Management
11	Inter Process Communication (IPC).	4	1,2	Process Management
12	Process Synchronization, Basic Concepts, The Critical-Section Problem, Peterson's Solution,	4	1,2	Process Management
13	Synchronization Hardware, Semaphores and Classic problems of synchronization	4	1,2	Process Management
14	Synchronization Hardware, Semaphores and Classic problems of synchronization	4	1,2	Process Management
15	Concepts of deadlock, resource allocation graph and wait for graph,	4	1,2	Introduction to Deadlock
16	Deadlock detection Algorithm (Banker's Algorithm and Resource Request Algorithm	4	1,2	Deadlock Detection
17	Deadlock detection Algorithm (Banker's Algorithm and Resource Request Algorithm	4	1,2	Deadlock Detection
18	Deadlock prevention, avoidance and recovery.	4	1,2	Methods of deadlock handling

19	Review Class			Lecture 9-18
20	Class Test – 02			Lecture 15-18
21	Basic concepts of CPU scheduling, scheduling criteria, scheduling algorithms and simulations: FCFS.	2	1	Overview of CPU Scheduling
22	Scheduling algorithms and simulations: SJF, PS.	2	1	CPU Scheduling Algorithms
23	Scheduling algorithms and simulations: RR, SRTF.	2	1	CPU Scheduling Algorithms
24	Scheduling algorithms and simulations: Multilevel queue and Multilevel feedback queue scheduling.	2	1	CPU Scheduling Algorithms
25	**Review Class**			Lecture 21-24
26	Class Test – 03			Lecture 21-24
27	Memory management Background, Swapping, Memory Allocation (MFT)	3	1,2	Main Memory Management
28	Memory Allocation (MVT, Segmentation).	3	1,2	Main Memory Management
29	Memory Allocation (Paging).	3	1,2	Main Memory Management
30	Introduction of virtual memory, demand paging, performance of demand paging.	3	1,2	Virtual Memory Management
31	Page fault handling, page replacement algorithms (FIFO, LRU, Optimal)	3	1,2	Virtual Memory Management
32	LRU Implementation, Frame Allocation, EAT and thrashing	3	1,2	Virtual Memory Management
33	**Review Class**			Lecture 27-32
34	Class Test – 04			Lecture 27-29
35	Overview of Mass storage structure, Disk scheduling (FCFS, SSTF)	2	1,2	Mass Storage Management
36	Disk scheduling (SCAN, C-SCAN, C-LOOK)	2	1,2	Mass Storage Management
37	Review Class			Lecture 35-36
38	Main Features of Windows, UNIX, LINUX; Introduction to distributed operating systems	1	1	Case Study of some mostly used OS
39	Main Features of Windows, UNIX, LINUX; Introduction to distributed operating systems	1	1	Case Study of some mostly used OS
40	Review			Lecture 38-39
41	Review Class, Problem Solutions and Suggestion Lecture 1 – 41			Lecture 1-40
42	Review Class, Problem Solutions and Suggestion Lecture 1 – 41			Lecture 1-40

Reading Reference:

1. Modern Operating System- Andrew S. Tanenbaum
2. Operating Systems – William Stallings

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Operating System Sessional
Course Code: CSE 312, Credit: 1.0, Total hours: 28

Rationale:
Understand the basic components of a computer operating system, and the interactions among the various components on the policies for scheduling, deadlocks, memory management, synchronization, system calls, and file systems.

Objective:
1. To learn basic OS concepts and to be familiar with the design principles of Operating System
2. Developing Internal and design principles of Operating System.
3. Be Familiar with the boot loader, kernel and how they works

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Evaluate major operating systems like Windows, Linux.
2. Discuss and apply internal design principles of Operating System.
3. Able to develop and design algorithms for process, thread and memory management.
4. Able to configure system software to enhance system capacity and security

Course Content:

Introduction of Linux Operating System, Installation of Linux in various modes, Installation of windows application programs on Linux, Installation of Linux application programs on Windows, Basic Command Line commands, Linux Kernels and Office Environments, Orientation with Shell Programing, Making own kernel, Harding Windows, Harding Linux.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	**Program Outcome (PO)**											
	1	2	3	4	5	6	7	8	9	10	11	12
Evaluate major operating systems like Windows, Linux.											√	
Discuss and apply internal design principles of Operating System.										√		
Able to develop and design algorithms for deadlock and memory management.							√					
Able to configure system software to enhance									√			

system capacity and security.												

Lecture Outline:

Week	Topics
1-2	Introduction of Linux Operating System Installation of Linux in various modes Installation of windows application programs on Linux Installation of Linux application programs on Windows Basic Command Line commands
3-4	Playing with Linux Kernels and Office Environments Assigning related Project
5-6	Orientation with Shell Programing
7-8	Introduction with a lightweight open source OS (For e.g., NACHOS) Implementation of Process & Thread Synchronization
9-10	Evaluation on Synchronization
	Implementation of Memory management
11-12	Evaluation on Memory management
13-14	Quiz & Final Report Submission

Reading Reference:

1. Modern Operating Systems (4th ed) - Andrew S. Tanenbaum; Prentice Hall (2014)
2. **"UNIX Shell Programming** " - Kanetkar

Assessment Methods:

Category	Marks %
Lab Test	40
Quiz	20
Viva	10
Class Participation	10
Home Assignment /Report	10
Class performance/observation	10
Total	**100**

Course Title: Database Management System
Course Code: CSE 313, Credit: 3.0, Total hours: 42

Course Description:

1. Introduction to concepts and methods for storing and manipulating data in stored form.
2. File retrieval and organization.
3. Database models and designing of database systems.

4. The principles of database management systems.
5. Relational database management systems.
6. Query formulation and language.
7. Database administration.
8. Methods used for the storage, selection and presentation of Data.
9. Database integrity and security.
10. Students will work with database languages and popular application packages. Common database management systems. Structure of SQL and principals behind the design of SQL. Students must complete four SQL assignments in the lab. The course includes a compulsory 3 hour laboratory work each week.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Describe** the theoretical concepts of database systems such as, database architecture, database security, data models, relational algebra, query processing and transactions.	C4	P1	A1		PO1
CO 2	**Design** database systems using ER/EER diagrams from given data requirements	C6			CT	PO1
CO 3	**Apply** relational schema algorithm for mapping ER/EER models.	C2			CT	PO2
CO 4	**Prepare** standard queries using Structured Query Languages (SQL) for the storage and manipulation of data.	C2		A4		PO3
CO 5	**Develop** a complete database application using SQL and other effective programming languages and tools as a group project.		P6		TW	PO9
CO 6	**Apply** functional dependencies, normalization and integrity constraints for superior database design.		P1		CT	PO3

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2	S											
CO3		S										
CO4			S									
CO5									M			
CO6			S									

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Over view on Syllabus, orientation of lectures and topics of the subject, Database-system Applications, Purpose of Database system	1	1	

2	Introduction to data storage and data types (e.g. structured, unstructured), database vs file-based approach	1	1	
3	Types of database users, View of Data, popularly used data models	1	1	
4	Structure of relational Database, Database Schema and Keys	1,3	1	
5	Schema Diagrams, Relational query languages	1,3	1	
6	Relational operations, constructing relational query using relational operations.	4,5	1	
7	Review			Lecture 1 - 6
8	Class Test 1			Lecture 1 - 6
9	Over view of the SQL Query Language,SQL Data Definition	4,5	1	
10	Basic Structure of SQL Quries,Set Operations	4,5	1	
11	Null Values, Aggregate Functions	4,5	1	
12	Nested Subqueries, Modification of Database	5		
13	Join Expressions, Views,	5	1	
14	Transactions ,Integrity Constraints,	5	1	
15	SQL Data Types and Schemas Authorization	5	1	
16	Relational Algebra	1	1	
17	Tuple and Domain Relational Calculus	1		
18	Review			Lecture 9-17
19	Class Test 2			Lecture 9-17
20	Overview of the database design process, The Entity-Relationship Model.	3	1	
21	Constraints, Removing Redundant Attributes in Entity sets, Entity-Relationship Diagrams	3	1	
22	Reduction to Relational Schemas, Extended ER Features	3	1	
23	Features of Good Relational Designs, Atomic Domains and First Normal Form	6	1	
24	Decomposition using Functional Dependencies and 2nd Normal form	6	2	
25	Decomposition using multivalued Dependencies and 3rd Normal form	6	2	
26	Other Normal forms and some practices	6	2	
27	Review			20-26
28	Class Test 3			20-26
29	RAID Levels, File organization	1	1	
30	Organization of Records in fiels,Data Dictionary Storage	1	1	
31	Query Processing, Query Optimization	1	1	
32	Transaction Concept, A simple transaction model, Transaction Atomicity and Durability	1	1	
33	Transaction isolation, Serializability, Transaction Isolation and Atomicity	1	1	
34	Lock –Based Protocols,Deadlock Handling	1	1	
35	Failure Classification, Recovery and Atomicity, Recovery Algorithm	1	1	
36	Centralized and Client-server Architectures, Server System Architectures, parallel and distributed systems	1	1	
37	Homogenous and Heterogeneous Databases,Distributed data storage	1	1	
38	Decision Support Systems, Data warehousing, Datamining	1	1	
39	Classification, Association rules, Clustering	1	1	

40	Review		1	
41	Class Test			Lecture 29-40
42	Review Class, Problem Solutions and Suggestion			Lecture 1 – 40

Reading Reference:

1. Database System Concepts- Abraham Silbeschatz, Henry F. Korth, S. Sundarshan
2. Files and Databases- An Introduction, Peter D. Smith and G.M. Barnes, Addison-Wesley

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Database Management System Sessional
Course Code: CSE 314, Credit: 1.5, Total hours: 42

Rationale:
This course is designed to introduce the basic concepts of database, learn how to design database and gain first-hand experience through developing a real world e- commerce database application in a term project. Also, to learn the design of a database starting from the conceptual design to the implementation of database schemas and user interfaces to a database.

Objective:

1. To introduce the basic concepts of database.
2. Developing a real world database application.
3. To learn the design of a database starting from the conceptual design to the implementation of database schemas and user interfaces to a database.

Course Outcomes (CO):
Upon completion of the course, the students will be able to:

1. Demonstrate the knowledge in projects with a commercial relational database system (Oracle) and design a team-based project.
2. Utilize the database design principles, SQL and PL SQL.
3. Demonstrate the relational database theory and be able to write relational algebra expressions for queries.

Course Content:
Introduction, Oracle Installation, Authentication, Security, Table Creation, SQL, Simple Query, Data Expressions, Join, Constraints, Advanced Query (GROUP Function etc.), Sub-queries, Single-row function, Numeric function, Manipulation function, Conversion

function, Nesting of function, Abstract data type, Database Trigger/ Procedure, PL/SQL Packages, Indexing, View, Introduction to PL/SQL

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
Demonstrate the knowledge in projects with a commercial relational database system (Oracle) and design a team-based project.									√			
Utilize the database design principles, SQL and PL SQL.									√			
Demonstrate the relational database theory and be able to write relational algebra expressions for queries.									√			

Lecture Schedule:

Week	Lecture	Topics	Project
1	Lec 1	Introduction, Oracle Installation, Authentication, Security	Project group distribution Project assignment
2	Lec 2	Table Creation, SQL	Presentation on the project proposal with submission of a report
3	Lec 3	Simple Query, Data Expressions etc.	Submission of the E- R diagram, Schema diagram
4	Lec 4	JOIN Operations	
5	Lec 5		Project Update - 01
6	Lec 6	Online Exam – 1	
7	Lec 7	Constraints	
8	Lec 8	Advanced Query (GROUP Function etc.), Sub-queries	
9	Lec 9	Single-row function, Numeric function, Manipulation function, Conversion function, Nesting of function, Abstract data type etc.	Project Update - 02
10	Lec 10	Online Exam – 2	
11	Lec 11	Database Trigger/ Procedure	
12	Lec 12	PL/SQL Packages, Indexing, View	
13	Lec 13	Introduction to PL/SQL	
14	Lec 14	Additional Online Exam	

Reading Reference:

1. Database System Concept, Abraham Silberschatz, Henry F. Korth, S. Sudarshan, Fourth edition
2. The Complete Reference JAVA2, Herbert Schildt
3. Microsoft C# Professional Projects,Geetanjali Arora, B. Aiaswamy, Nitin Pandey
4. The Complete Reference PHP 5.2 Steven Holzner

Assessment Methods:

Category	**Marks %**
Online Test – 1	25
Online Test – 2	25
Viva	10
Class Participation	10
Class Assessment	30
Total	**100**

Course Title: Computer Architecture and Design
Course Code: CSE 315, Credit: 3.0, Total hours: 42

Course Description:

1. Fundamentals of computer Design: Processor Design; Data paths Design: single cycle and multi cycle implementations; Control Unit design: hardware and micro programmed, Arithmetic Logic Unit (ALU) Design, RISC and CISC processor.

2. System organization, Parallel Processing, Pipeline: pipelined data path and control, Hazards, Exceptions, superscalar and dynamic pipelining, Pipeline structure vector supercomputers.

3. Fixed Point Arithmetic: Addition, Subtraction, Multiplication and Division, Floating point Representation and Arithmetic.

4. Memory organization: Register, Cache, RAM, ROM

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Recall** the fundamental concepts of the structure of a Computer.	C1				PO1
CO 2	**Illustrate** different types of memory organization along with their significance in a computer system.	C2				PO1
CO 3	**Compute** different performance measurements and arithmetic operations of a Computer System.	C3			CT	PO2
CO 4	**Design** the ALU Unit, Control unit: hardware, micro programmed, overall data-path for specific operation like load, store, arithmetic and branch control operations and parallelism	C6			CT	PO3

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	M											
CO2	S											
CO3		S										
CO4			S									

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Introduction to computer architecture, The significance of computer architecture, The Function view of a computer system.	1	1	Importance of Computer Architecture and functions of a computer.
2	Structural specification of a computer system, Moore's Law and its consequences.	1	1	The structure of a computer system and the importance of Moore's law in case of Computer Design.
3	Performance measurement of a computer system, Performance metrices: Execution time, CPU time, Clock rate.	3	1	Measuring Performance of a Computer System
4	Problem solving using CPU execution time as a performance measure	3	1	Measuring Performance of a Computer System
5	MIPS, MFLOPS problem solving using MIPS and MFLOPS as a performance measure	3	1,2	Measuring Performance of a Computer System
6	Amadahl's Law, Performance Improvement Problem Solving using Amadahl's Law	3	2	Measuring Performance of a Computer System
7	Spec & Benchmark, Characteristics of a Benchmark, Benchmark metrices.	3	2	Measuring Performance of a Computer System
8	Problem Solving Related to Benchmark	3	2	Measuring Performance of a Computer System
9	Review			Lecture 3-8
10	Class Test 1			Lecture 3-8
11	Instructions: Language of the Computer, Introduction to MIPS assembly Language, Instruction sets.	4	1, 2	Computer Instructions
12	Conversion from High level code to MIPS equivalent form. (Arithmetic and Memory Reference) and MIPS instruction to Machine Code.	4	1, 2	Computer Instructions
13	Conversion from High level code to MIPS equivalent form. (Conditional and Loop) and MIPS instruction to Machine Code.	4	1, 2	Computer Instructions
14	MIPS addressing Modes	4	1, 2	Computer Instructions
15	Review			Lecture 11-14
16	Class Test 2			Lecture 11-14
17	Signed and Unsigned number representation, addition and Subtraction and Overflow of addition and Subtraction	3	1	Fixed Point Arithmetic

18	**1 Bit ALU Design for AND, OR, NOT, ADD instructions**	4	1	ALU Design
19	**1 Bit ALU Design for AND, OR, ADD, SUB, SLT instructions**	4	1	ALU Design
20	4/8/16/32 bit ALU Design with overflow and zero detection.	4	1	ALU Design
21	Multiplication Algorithm, hardware design and problem solving (1st and 2nd).	3,4	1,2	Fixed point arithmetic with ALU
22	Multiplication Algorithm, hardware design and problem solving (Final and Booth).	3,4	1,2	Fixed point arithmetic with ALU
23	Division Algorithm, hardware design and problem solving (1st and 2nd).	3,4	1,2	Fixed point arithmetic with ALU
24	Division Algorithm, hardware design and problem solving (Restoration).	3,4	1,2	Fixed point arithmetic with ALU
25	Floating point representation and Arithmetic operations	3	1,2	Floating point Arithmetic
26	Review			Lecture 17 – 25
27	Class Test 03			Lecture 17 – 25
28	MIPS non-pipelined data path/control path: how single cycle data path works for Arithmetic and load operations	4	1	Processor unit: Data path design
29	MIPS non-pipelined data path/control path: how single cycle data path works forr store and branch operations	4	1	Processor unit: Data path design
30	MIPS non-pipelined data path/control path: Single cycle and Multicycle data path design and advantages and difference	4	1	Processor unit: Data path design
31	Control Unit design: hardware and micro programmed	4	2	Control Unit design
32	Parallel Processing, Pipeline: Hazards and Solution for Structural and Data	4	1	Parallel processing unit Design
33	Pipeline: Hazards and Solution for control pipelined data path and control, superscalar and dynamic pipelining,	4	1	Parallel processing unit Design
34	Exceptions and Pipeline structure vector supercomputers, RISC and CISC processor and their differences	4	1,2	Parallel processing unit Design
35	Review			Lecture 28-34
36	Memory Hierarchy, Memory Technology, Cache Introduction and organizations.	2	1, 2	Memory Organization
37	Cache Examples, Handling Cache miss for data and Instruction	2	1, 2	Memory Organization
38	Memory Device Characteristics: Registers, Access method, Volatile and Non-Volatile.	2	1, 2	Memory Organization
39	Memory Device Characteristics: SRAM, DRAM, ROM: MROM, PROM, EPROM and EEPROM.	2	1, 2	Memory Organization
40	Review class			Lecture 36 – 39
41	Class Test 04			Lecture 36 – 39
42	Review Class, Problem Solutions and Suggestion			Lecture 1 – 41

Reading Reference:

1. Computer Organization and Design by David A Patterson
2. Computer Organization and Architecture by William Stalling

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Software Engineering and Design Patterns
Course Code: CSE 317, Credit: 3.0, Total hours: 42

Course Description:

1. Introduction to the fundamental concept of software engineering;
2. Understand basic software development process model;
3. Introduction to agile software development;
4. Understand different types of project requirement;
5. Preparing project document and prepared architectural design of project.
6. Test software and assess the software quality.
7. Estimate software cost and make a project management plan.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	6. **Explain** the fundamental concept of software engineering.	C2				PO1
CO 2	7. **Identify** the project requirement, project risk, and make a mitigation plan.	C1			CT	PO2
CO 3	8. **Design** the software architecture**.**	C6				PO3
CO 4	9. **Assess** the software quality.	C5				PO4
CO 5	10. **Estimate** the project cost and time scheduling	C5				PO11

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											

CO2		S										
CO3			M									
CO4				S								
CO5											S	

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Introduction to Software Engineering	1	1	Software Engineering, Attributes of good software, Fundamental Activities of software process.
2	Introduction to Software Engineering	1	1	Importance of software engineering, Relation among computer science, system engineering and software engineering, software engineering diversity.
3	Process Models	1,3	1	Fundamental activities of software engineering, Waterfall Model
4	Process Models	1,3	1	Incremental Model, Re-use oriented software engineering model.
5	Process Models	1,3	1	Evolutionary process model (Spiral Model).
6	Agile Method	1,3	1	Agile Method, Where agile is successful, Principles of Agile Method, Difficulties of principles of Agile Method to understand.
7	Agile Method	1,3	1	Relation between plan driven and agile specification, Extreme Programming with figure of Extreme Programming, Pair Programming, and The scrum Process for Agile Project Management.
8	Class Test 1			Lecture 1 - 5
9	Requirement Engineering	2	1	User requirement and system requirement, functional requirement non-functional requirement
10	Requirement Engineering	2	1	Types of non-functional requirement with detailed figure, Problems of non-functional requirement, Metrics of Specifying non-functional requirement, Definition of

				requirement documents with users of requirement documents
11	Requirement Engineering	2	1	Basic structure of requirement documents, requirement specification with ways of writing requirement specification, Spiral Model of Requirement Engineering Process, The Requirement elicitation and analysis process.
12	Requirement Engineering	2	1	Interviewing, Requirement Validation and Requirement Document, Requirement Validation Technique, Decisions on Requirement Management Stage, Principles of Change Management Process.
13	System Modeling	3	1	Context Model, sequence diagram, use-case diagram.
14	System Modeling	3	1	Structural Model, class diagram.
15	System Modeling	3	1	Generalization and Aggregation of Behavioral Model.
16	System Modeling	3	1	Data-driven modeling, Model Driven Engineering.
17	Design Concept	3,4	2	Pyramid structure of Design Model, Characteristic of good design, Quality Guidelines of Design.
18	Design Concept	3,4	2	Quality Attributes of Design.
19	Class Test 2			Lecture 9 – 16
20	Design Concept	3,5	2	of Procedural Abstraction and data abstraction, Pattern and Modularity
21	Design Concept	4	2	Functional Impedance, Refinement, Aspect, Refactoring.
22	Software Quality Concept	4	2	Software quality, Quality of design and quality of conformance. Quality of Design and quality of conformance
23	Software Quality Concept	4	2	McCall's quality factors, ISO 9126 quality factors, Targeted quality factors, Quality dilemma.
24	Software Quality Concept	4	2	Prevention, Appraisal, Failure cost, Quality and security.

25	Software Quality Concept	4	2	Elements of software quality assurance, SQA tasks, Software quality goals, Attribute and Metrics measures of software reliability and availability
26	Software Testing Strategies	4	2	Software testing, Verification and Validation, Spiral testing strategy.
27	Software Testing Strategies	4	2	Software testing steps, Strategic issues, Smoke testing.
28	Class Test 3			Lecture 22– 25
29	Software Testing Strategies	4	2	System testing, Software debugging. Characteristics of test, Alpha testing and Beta testing.
30	Software Testing Strategies	4	2	Attributes of good test, Black box and White box testing and comparison between them.
31	Product Matrics	5	2	Measures, Metrics and Indicators, Characteristics of measurement process, Principles of metrics characterization and validation.
32	Product Matrics	5	2	Attributes of effective software metrics, Function point. Length, Volume ratio, Program volume, Program level, Halstead effort, Percentage of testing report.
33	Estimation for Software Project	5	2	Software estimation attributes, Types of resources, Types of reusable resources
34	Estimation for Software Project	5	2,3	Software cost estimation, COCOMO model
35	Risk Management	5	2	Software risk, Reactive versus Proactive risk strategies, Types of software risk.
36	Risk Management	2,5	2	Generic risk and Product-specific risk, Subset of known and predictable risk, Risk components, Risk projection steps.
37	Risk Management	2,5	2	Impact assessment, Risk management concern, Calculation of risk exposure.
38	Risk Management	2,5	2	Create risk table, RMMM plan.

39	Review Class, Problem Solutions and Suggestion			Lecture 31 – 35
40	Class Test 4			Class Test 4
41	Review Class, Problem Solutions and Suggestion			Lecture 15 – 28
42	Review Class, Problem Solutions and Suggestion			Lecture 29 – 38

Reading Reference:

1. Software Engineering – Ian Sommerville.
2. Software Engineering A Practitioner's Approach – Roger S.Pressman.
3. COCOMO Model – Version 2 CSE IIT, kharagpur.

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Software Engineering and Design Patterns Sessional
Course Code: CSE 318, Credit: 1.0, Total hours: 28

Course Description:

Learn software design pattern for software development.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

Course Outcome Description		Level of Domain				PO
		C	P	A	S	
CO 1	**Demonstrate** the software design concepts with UML diagram.			A3		PO1+PO2
CO 2	**Analyze** well-known design patterns (such as Iterator, Observer, and Visitor).	C4				PO1+PO2
CO 3	**Apply** design pattern to software development.	C3				PO2 + PO3

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S	M										
CO2	S	M										

CO3		S	S									

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Design Pattern	2,3	1	Introduction to design pattern, design pattern in Smalltalk MVC
2	The Catalog of Design Patterns	2,3	1	The Catalog of Design Patterns
3	Solve Design Problem	2,3	1	Solve design problem, select design problem.
4	Solve Design Problem	2,3	1	use design pattern
5	Unified Modeling Language	1	2	Introduction to UML diagram.
6	Unified Modeling Language	1	2	Design different UML diagram.
7	Case Study	2,3	1	Design Problems, Design Problems,
8	Case Study	3,3	1	Document Structure, Formatting, Embellishing the User Interface, Supporting Multiple Look-and-Feel Standards
9	Structural Patterns	2	1	Adapter, Bridge.
10	Structural Patterns	2	1	Composite, Decorator
11	Behavioral Patterns	2	1	Chain of Responsibility, Command.
12	Behavioral Patterns	2	1	Interpreter, Iterator Observer,
13	Presentation - 1			Assignment
14	Viva + Quiz			Sessional 1 - 12

Reading Reference:

1. Design Patterns: Elements of Reusable Object-Oriented Software by Kevin Zhang.

2. A Uml Pattern Language (The Mtp Software Engineering Series) by paul Evitts.

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Theory of Computation
Course Code: CSE 319, Credit: 3.0, Total hours: 42

Course Description:

Finite automata: deterministic finite automata, nondeterministic finite automata, equivalence and conversion of deterministic and nondeterministic finite automata, pushdown automata; Context free languages; Context free grammars; Pushdown automata;
Regular languages: regular expressions, non-regular languages, the pumping lemma;
Turing Machines: basic machines, configuration, computing with Turing machines, combining Turing machines; Un-decidability.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

CO 1	**Understand** the fundamental concepts of computation, automation, and compilation system and apply logical reasoning to solve a variety of problems
CO 2	**Differentiate** among DFA, NFA, e-NFA and sketch DFA, NFA, e-NFA diagrams of some problems
CO 3	**Demonstrate** the representation and design of Context Free Grammar (CFG) and construct parse trees of any string according to some CFG.

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO 12
CO1	S	M										
CO2		S	S									
CO3		S		S								

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1.	Introduction the Theory of Computation.			
2.	Introduction to Automata Theory			
3.	Introduction to Finite Automata			
4.	Deterministic Finite Automata			
5.	Various design of Deterministic Finite Automata			
6.	Non-Deterministic Finite Automata			

7.	Various design of Non-Deterministic Finite Automata			
8.	Non-Deterministic Finite Automata with €-Transaction			
9.	**Class Test-1**			
10.	Equivalence of Deterministic Finite Automata			
11.	Equivalence of Non Deterministic Finite Automata			
12.	Conversion of Deterministic Finite Automata			
13.	Conversion of Non-Deterministic Finite Automata			
14.	€-NFA			
15.	€- Closures			
16.	Introduction to Regular Expressions			
17.	Various Proof of RE			
18.	RE to NFA conversion			
19.	**Class Test-2**			
20.	Pumping Lemma Proof			
21.	Pumping Lemma Application			
22.	Context Free Languages			
23.	Context Free Grammars			
24.	Application of CFG			
25.	Various Example of CFG			
26.	Various Example of CFG			
27.	**Class Test-3**			
28.	Introduction to Pushdown Automata			
29.	Designing of PDA			
30.	Designing of PDA			
31.	Language of PDA			
32.	Uses and Application of PDA			
33.	Equivalence of PDA and CFG			
34.	Overview of PDA			
35.	Turing Machines			
36.	Basic Machines, Configuration			
37.	Computing With Turing Machines			
38.	Combining Turing Machines			

39.	Undecidability			
40.	**Class Test-4**			
41.	Review Class			
42.	Review Class			

Reading Reference:

1. Introduction to the Theory of Computation. CENGAGE Learning, 3rd ed., 2012 - M. Sipser.
2. Introduction to Automata Theory, Languages, and Computation. Addison-Wesley Longman Publishing Co., Inc., 3rd ed., 2006 - J. E. Hopcroft, R. Motwani, and J. D. Ullman.
3. Elements of the Theory of Computation. Upper Saddle River, NJ, USA: Prentice Hall PTR, 2nd ed., 1997 - H. R. Lewis and C. H. Papadimitriou.

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Engineering Drawing
Course Code: CE 311, Credit: 1.0, Total hours: 14

Rationale:
It is a drawing course where students can learn drawing different linear and curved geometric figures e.g pentagon, hexagon, octagon, ellipse, parabola, hyperbola; solid geometry. Concept of isometric objects and orthographic views are discussed for clear understanding of students.

Course Content:
Lines and lettering; plane geometry: drawing of linear and curved geometric figures, e.g. pentagon, hexagon, octagon, ellipse, parabola, hyperbola; solid geometry: concept of isometric view and oblique view, theory of projections; drawing of isometric view of 3D objects such as cube, prism, pyramid, cone and cylinder; projections of cube, prism, cone, cylinder; developments of cube, pyramid, cone, cylinder.

Objective:

- To get familiar with different drawing instruments and technical standards.
- To develop a deep understanding of different geometric figures
- To gain knowledge about drawing isometric and orthographic views.

Course Outcomes and Skill Mapping:

No.	COURSE OUTCOMES (COs)	PROGRAMME OUTCOMES (POs)											
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
1	**Recognize** different drawing equipment and technical standards.	√											
2	**Understand** 2D and 3D views of simple objects.	√											

Course Outcomes and Generic Skills:

No.	Course Outcomes	Corresponding Pos	Bloom's Taxonomy	CP(WP)	CA(EA)	KP(WK)	Assessment Methods
CO1	**Recognize** different drawing equipment and technical standards.	1	C1	4	-	3	Class Assessment
CO2	**Understand** 2D and 3D views of simple objects.	1	C2	2	-	4	Quiz
WP= Washington Accord Complex Problem Solving/ CP= Complex Problem Solving; EA= Engineering Activities/ CA= Complex Activities; WK= Washington Accord Knowledge Profile/ KP= Knowledge Profile							

Lecture Schedule:

Week	Topic	Assignments
1	Introduction	Drawings
2	Use of Instruments	
3	Lines and Dimensioning	
4	Concepts of Isometric view, orthographic and 3D objects	
5	Acquaintance with sheet layout and title block for each day submission	
6	Class Test-1	
7	Plane Geometry: Pentagon, Hexagon, Octagon etc.	
8	Plane Geometry:	
9	Practice on Isometric Views from 3D view	
10	Practice on Isometric Views & Orthographic views of 3D Object	
11	Sectional views of 3D Object	
12	Visualization of 3D view from Isometric view	
13	Class Test-2	
14	Review	

Reference Books:

1. Prathomik Engineering Drawing by - Hamonto Kumar Bhottacharjo

Assessment Methods:

Components	Grading	CO	Blooms Taxonomy
Class Test	20%	CO1	C1, C2
Assignment	20%	CO2	C1, C2
Final Exam	60%	CO1, CO2	C1, C2
Total Marks	100%		

Course Title: Engineering Drawing & CAD Sessional
Course Code: CE 312, Credit: 1.0, Total hours: 28

Rationale:

To provide the idea of basic electronic devices and amplifiers, analyze the concepts of circuits, their modeling and uses.

Objective:

1. Understand the Engineering drawing which helps to communicates ideas and information.
2. Communicate all needed information from the engineer to workers or manufacturer who constructs any kind of structural element.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Explain the Engineering drawing which helps to communicate ideas and information.
2. Communicate all needed information from the engineer to workers or manufacturer who constructs any kind of structural element.
3. Design solutions for complex structural engineering problems for different environmental context.

Course Content:

Engineering Drawing & CAD Sessional Introduction, Lettering, numbering and heading, Instrument and their use, Sectional views and isometric views of solid geometrical figure, Plan, Elevation and Section of one-story building, Building service drawing (Electrical layout and Plumbing), Detailed drawing of lattice towers, Use of AutoCAD software.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	1	2	3	4	5	6	7	8	9	10	11	12
Explain the Engineering drawing which helps to communicate ideas and information.										√		
Communicate all needed information from the engineer to workers or manufacturer who constructs any kind of structural element.									√			
Design solutions for complex structural engineering problems for different environmental context.						√						

Lecture Outline:

Week	Lab	Topics
1	Lab 1	Engineering Drawing & CAD Sessional Introduction
2	Lab 2	Lettering, numbering and heading
3	Lab 3	Instrument and their use
4	Lab 4	Sectional views of solid geometrical figure
5	Lab 5	Isometric views of solid geometrical figure
6	Lab 6	Plan, Elevation and Section of one-story building
7	Lab 7	**Lab Test- 01**
8	Lab 8	Building service drawing (Electrical layout and Plumbing)
9	Lab 9	Detailed drawing of lattice towers
10	Lab 10	Use of AutoCAD software
11	Lab 11	**Lab Test- 02**
12	Lab 12	**Quiz**
13+ 14	Lab 13+ 14	**Viva**

Reading Reference:

1. Prathomik Engineering Drawing - Hemanta Kumar Bhattacharyya

Assessment Methods:

Category	Marks %
Lab Test	40
Quiz	20
Viva	10
Class Participation	10
Home Assignment /Report	10
Class performance/observation	10
Total	**100**

LEVEL-3 TERM-II

Course Title: Data Communication
Course Code: CSE 321, Credit: 3.0, Total hours: 42

Course Description:

Introduction: Communication models, data communication tasks, data communication network standards and organization introduction to TCP/IP models. Data Transmission basics: Analog

and digital data, spectrum and bandwidth, Transmission impairments, data rate, and channel capacity. Data Encoding, NRZI Manchester and Differential Manchester encoding, ASK, FSK, PSK, QPSK, QAM encoding, speared spectrum technique, Sampling theorem and pulse code modulation techniques and speech digitization. Data Transmission: Asynchronous and synchronous data transmission techniques, error, detection techniques, parity cheeks and CRC error correction and hamming code interfacing and EIA 232D or RS 232c. DSL technology: Data Link Control: Line configurations. Multiplexing: Frequency division multiplexing, international FDM carrier standards, Synchronous time division multiplexing, international TDM carrier standards, statistical time division multiplexing, SONET and SDH.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Describe** the various types of signals and their features.	C2				PO1
CO 2	**Identify** key considerations in selecting various transmission media in networks.	C1				PO1
CO 3	**Explain** the role of line codes and digital communications devices in a data communications network	C2				PO1
CO 4	**Analyze** practical problems and to apply data communication theory in practice	C4			CT	PO2

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2	S											
CO3	S											
CO4		S										

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Overall idea of Data Communication (DC), lecture plans, examples of DC in Computer Science and Engineering, review of question pattern, purpose and implementation of DC in real world, etc.	N/A		Importance of DC in Computer Science and Engineering and real world, Overall idea about course details.

2	A Communication model, A Data Communication model.	1	1	Structure of Communication model, Data Communication model and working principle of those models.
3	The transmission of information, Addressing in the TCP/IP protocol suite.	1,2	1	Details of transmission method of information and TCP/IP protocol structure, work flow, advantage and application
4	OSI versus TCP/IP, Introduction to Physical layer, What is Data and Signal, their differences.	1	1	OSI model, difference between OSI and TCP/IP, layers of OSI model. Basic description about Data, Signal and their difference.
5	Distinguish between analog and digital signal, difference between time and frequency domain representation of signal.	1	1	Definition of digital and analog signal, their difference and signal representation in different domain.
6	Bandwidth of a signal, Bit interval and bit rate of a signal, signals and communication, composite and periodic signal, Few problems regarding the above topics.	1,4	1	Basic definition and description of a signal property, Different type of signals and their pros and cons.
7	Baseband transmission, Modulation of a signal, few examples regarding the above topics.	1,2	1	Description of baseband transmission, Modulation of a signal and problem solving regarding these topics.
8	Transmission impairments: attenuation, distortions and noise.	1,2	1	Description of different transmission impairment and their effect on signal.
9	Signal to Noise ratio, few problems regarding the topics. Data rate limits, Noiseless channel, noisy channel.	1	1	Different techniques regarding to solving transmission impairment problem in different types of channels.
10	Performance, bandwidth, Throughput, latency, bandwidth delay product, Queuing time, jitter, few problems regarding the topics.	1,4	1	Different properties in transmitting a signal and problem solving for these topics.

11	Transmission media, Quality of transmission	2	1	Description of transmission media, types and their quality.
12	Guided media: twisted pair cable, Coaxial cable, Fiber Optic cable.	2	1	Definition of guided media, types of guided media and their pros and cons.
13	Propagation mode: multimode: step index and Graded index and single mode,	2	1	Description of Different types of propagation mode .
14	Unguided media: wireless- Radio waves	2	1	Definition of guided media, types of guided media and details of Radio waves.
15	Microwaves and Infrared.	1,2	1	Details of Microwaves and Infrared.
16	Antennas, Satellite communication, line of sight transmission	1,2	1	Details of Antennas, Satellite communication, line of sight transmission
17	Free space loss, Interference, few problems regarding the topics.	1,4	1	Description of Free space loss, Interference, few problems regarding the topics.
18	Various conversion technique, Digital to Digital conversion: Line coding, block coding, Scrambling.	3	1	Types of conversion technique and Digital to digital conversion technique.
19	Line coding: Unipolar,	3	1	Process of line coding (Unipolar), advantage, properties.
20	Polar	3	1	Process of line coding (polar), advantage, properties.
21	Bipolar	3	1	Process of line coding (Bipolar), advantage, properties.
22	Line coding: Multilevel	3	1	Process of line coding (Multilevel), advantage, properties.
23	Multi-transition,	3	1	Process of line coding (Multi-transition), advantage, properties.
24	Scrambling schemes: B8ZS, HBD3.	3	1	Process of line coding (Scrambling schemes: B8ZS, HBD3), advantage, properties.
25	Block coding: mB/nB, redundancy.	3	1	Process of Block coding :(mB/nB) ,advantage, properties and redundancy.

26	Analog to Digital Transmission: Pulse code modulation (PCM), steps of PCM,	1,2	1	Process of Analog to Digital Transmission: Pulse code modulation (PCM), steps of PCM,
27	Delta Modulation, few problems regarding the topics.	1,4	1	Process and details of Delta Modulation, few problems regarding the topics.
28	Transmission modes: parallel transmission, serial transmission.	1,4	1	Description of Transmission modes: parallel transmission, serial transmission.
29	Digital to analog conversion: ASK, FSK	1	1	Process of Digital to analog conversion: ASK, FSK
30	PSK, QPSK, Constellation diagrams	1	1	Process of PSK, QPSK, Constellation diagrams
31	Analog Data to analog signal conversion: AM, FM and PM modulation,	1	1	Process of Analog Data to analog signal conversion: AM, FM and PM modulation,
32	Bandwidth utilization: Multiplexing and Spreading.	1	1	Details of Bandwidth utilization: Multiplexing and Spreading.
33	Multiplexing: FDM, WDM, STDM, STDM	1	1	Description of Multiplexing: FDM, WDM, STDM, STDM
34	Spread Spectrum: Frequency Hopping Spread spectrum (FHSS), Direct Sequence Spread Spectrum (DSSS)	1	1	Description of Spread Spectrum: Frequency Hopping Spread spectrum (FHSS), Direct Sequence Spread Spectrum (DSSS)
35	Error detection and Correction techniques, types of errors,	1	1,2	Process of Error detection and Correction techniques, types of errors,
36	Block coding: Error detection, correction, Hamming Distance.	1	1,2	Process of Block coding: Error detection, correction, Hamming Distance.
37	Parity checks, SONET and SDH.	1	1,2	Details and technique of Parity checks, SONET and SDH.
38	Cellular Wireless Networks, Cellular network organization, cell organization,	1	1,2	Description of Cellular Wireless Networks, Cellular network organization, cell organization,

39	Frequency concept, reuse, capacity, channels, Mobile radio propagation effects,	1	1,2	Description of Frequency concept, reuse, capacity, channels, Mobile radio propagation effects,
40	Different types of Fading, fading effects, Mobility, handoff, location management,	1	1,2	Different types of Fading, fading effects, Mobility, handoff, location management,
41	Digital-AMPS, 3rd and 4th generation network.	1	1,2	Digital-AMPS, 3rd and 4th generation network
42	Review class for all above topics.	N/A		Review class and Problem solving

Reading Reference:

1. Data Communication and Networking
   Author: Behrouz A Forouzan.
   Latest Edition.
2. Data and Computer Communication
   Author: William Stallings.
   Pearson, ISBN-978-93-325-1886-5, Ninth Edition.

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Data Communications Sessional
Course Code: CSE 322, Credit: 1.0, Total hours: 28

Rationale:

The main course is to learn the working knowledge of data transmission concepts, line control and line sharing and also is to understand the operation of compression optimizing data transfer algorithms.

Objective:

1. Understand and explain Data Communications System and its components.
2. Introduce students to the digital and analogue representations and channels.

3. Describe the mechanism and techniques of encoding.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Apply amplitude, frequency and time division multiplexing techniques to share network bandwidth among multiple users.
2. Compare each data transmission methods using both signal processing devices and lab software.
3. Describe the knowledge of data link layer fundamental in error detection, correction and flow control techniques.

Course Content:

Introduction to MATLAB / Delta Modulation, , AM/FM / AM Modulator & Demodulator, Digital to digital Conversion; Line Coding / DSB-SC and SSB Demodulators, FM Modulator and Demodulator, ASK/PSK/FSK, CDMA, Checksum.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	**Program Outcome (PO)**											
	1	2	3	4	5	6	7	8	9	10	11	12
Apply amplitude, frequency and time division multiplexing techniques to share network bandwidth among multiple users.						√						
Compare each data transmission methods using both signal processing devices and lab software.									√			
Describe the knowledge of data link layer fundamental in error detection, correction and flow control techniques.						√						

Lecture Outline:

Week	**Topics**
1 & 2	Introduction to MATLAB / Delta Modulation
3 & 4	Evaluation On Lab-1, AM/FM / AM Modulator & Demodulator
5 & 6	Digital to digital Conversion; Line Coding / DSB-SC and SSB Demodulators
7	**Lab Test-1**
8 & 9	Digital to digital Conversion; Line Coding / FM Modulator and Demodulator
10 & 11	ASK/PSK/FSK

12 & 13	CDMA, Checksum
14	**Viva/Quiz + Lab Test-2**

Reading Reference:

1. Data Communication and Networking(4th ed) - Behrouz A Forouzan (2017)
2. Data and Computer Communication - William Stallings
3. Data Communication & Networks – R L Brewster

Assessment Methods:

Category	Marks %
Lab Test-1	25
Lab Test-2	25
Viva	10
Class Participation	10
Class Assignment/Report	30
Total	**100**

Course Title: Web Technologies
Course Code: CSE 223, Credit: 2.0, Total hours: 28
Course Description:

1. Introduction To Web: The Internet model, Web browsers, Useful tools, Layers of the Internet World Wide Web, Domain Name Service, Uniform Resource Locator, Use and necessity of Internet, Extranet, Intranet. Server Client Architecture, User Client Relationship

2. HTML: History of Markup Language, HTML Basics, Tags, Formatting Text, Creating Links, Adding Images, Lists, Tables, Frames, Forms, and etc., Introduction to CSS: Border, Margin, Padding, Image, Graphics, Class, Id, Align, Float, Fonts and etc.

3. Introduction to Bootstrap: Responsive user interface, Responsive design Button, Textbox, Radio Button, Grid, and etc., Introduction to JavaScript: Variables, Loops, Array, Input, Output, String, Functions, Dates and etc.

4. Introduction to PHP web programming language: Generating HTML Dynamically, Processing Forms, Maintaining State in Web Applications, Session and Cookies, Cookies to Security, Data Tier, Back-end Database Support, SQL Primer, Database Interface in PHP, Searching in Web Applications, Regular Expressions and Matching, Multimedia and Interactivity, Audio on the Web, Video on the Web

5. Introduction to Framework: MVC architecture, Laravel Framework: Installation, Artisan CLI, HTML Template to Laravel Blade Template, Migrations, Creating a registration & user login form, Admin Panel Setup, Using Controllers and Routes for URLs and APIs, Eloquent ORM (Storing and Using Data), Creating and Using Composer Packages,

6. Introduction to JSON, MongoDB and AJAX technologies.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

Course Outcome Description		Level of Domain				PO
		C	P	A	S	
CO 1	**Evaluate** the concepts of basic web programming	C5				PO1
CO 2	**Conceptualize** different practical situations for using different web based technologies and frameworks			A4		PO3
CO 3	**Design** large scale websites by modelling front end and back end web based programming.				CT	PO9 + PO12

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2			S									
CO3									S			M

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Introduction To Web: The Internet model, Web browsers, Useful tools, Layers of the Internet World Wide Web	1	1	Idea about Web basics
2	Domain Name Service, Uniform Resource Locator, Use and necessity of Internet, Extranet, Intranet. Server Client Architecture, User Client Relationship	1	2, Internet	Knowledge about internet and client server architecture
3	HTML: History of Markup Language, HTML Basics, Tags, Formatting Text	1	2, Internet	Knowing HTML language basics
4	HTML: Creating Links, Adding Images, Lists, Tables, Frames, Forms, and etc.	2	2, Internet	Analyze HTML codes
5	Introduction to CSS: Border, Margin, Padding, Image, Graphics, Class, Id, Align, Float, Fonts and etc.	2	2, Internet	Knowledge about web design with CSS
6	Review			Lecture 1 - 5
7	Class Test 1			Lecture 1 - 5

8	Introduction to Bootstrap: Responsive user interface, Responsive design Button	1	1,2	Applying Bootstrap for responsive website creation
9	Bootstrap: Textbox, Radio Button, Grid, and etc.	2	1,2	Applying knowledge about buttons in web forms
10	Introduction to JavaScript: Variables, Loops	1	4, Internet	Understand Front-end activities
11	Javascript: Array, Input, Output	1	4, Internet	Knowing different JS syntax
12	Javascript: String, Functions, Dates and etc.	1	4, Internet	Applying JS in front end design
13	Review			Lecture 8 - 12
14	Introduction to PHP web programming language: Generating HTML Dynamically, Processing Forms, Maintaining State in Web Applications	1	3, Internet	Idea about basic web programming language
15	Session and Cookies, Cookies to Security	1,2	3, Internet	Security issue analysis in login
16	PHP: Data Tier, Back-end Database Support, SQL Primer, Database Interface in PHP	2	3, Internet	Understanding web database and creation of database
17	PHP: Searching in Web Applications, Regular Expressions and Matching, Multimedia and Interactivity, Audio on the Web, Video on the Web	2	3, Internet	Creating web based applications using php
18	Review			Lecture 14 - 17
19	Class Test 2			Lecture 14 - 17
20	Introduction to Framework: MVC architecture, Laravel Framework: Installation, Artisan CLI	1, 3	Internet, Tutorials	Knowledge about PHP framework
21	Laravel: HTML Template to Laravel Blade Template, Migrations, Creating a registration & user login form	1, 3	Internet, Tutorials	Differentiate between HTML and Blade
22	Laravel: Admin Panel Setup, Using Controllers and Routes for URLs and APIs	2, 3	Internet, Tutorials	Designing admin panel, controller and routing with laravel
23	Laravel: Eloquent ORM (Storing and Using Data)	1, 3	Internet, Tutorials	Creating ORM
24	Laravel: Creating and Using Composer Packages	1, 3	Internet, Tutorials	Creating Composer packages
25	Review			Lecture 20 - 24
26	Class Test 3			Lecture 20 - 24
27	Introduction to JSON, MongoDB and AJAX technologies	1, 2	Internet, Tutorials	Understanding and Applying Ajax and JSON
28	Overall Syllabus Review			Lecture 1 - 27

Reading Reference:

1. Head First Web Design by Ethan Watrall and Jeff Siarto.

2. Learning Web Design: A Beginner's Guide to (X)HTML, StyleSheets, and Web Graphics Author: Aaron Gustafson.
3. PHP and MySQL Web Development., Author: Laura Thomson.
4. Learning JavaScript, Author: Shelley Powers.

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Web Technologies Sessional
Course Code: CSE 324, Credit: 1.5, Total hours: 42

Course Description:

Hands-on website design based on CSE-323 contents.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

Course Outcome Description		Level of Domain				PO
		C	P	A	S	
CO 1	**Evaluate** the concepts of basic web programming	C5				PO1
CO 2	**Conceptualize** different practical situations for using different web based technologies and frameworks			A4		PO3
CO 3	**Design** large scale websites by modelling front end and back end web based programming.				CT	PO9 + PO12

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2			S									
CO3									S			M

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	HTML demonstration	1	2, Internet	Idea about HTML
2	CSS demonstration	1	2, Internet	Knowledge about web design with CSS
3	Building personal websites using HTML and CSS	2	1, 2	Applying HTML and CSS
4	Decorate website with bootstrap and Javascript	2	1, 2, Internet	Adding extra feature to HTML and CSS
5	PHP basic coding in building a website	1	3, Internet	PHP installation and application
6	Session and Cookies, Cookies to Security	2	3, Internet	Give security to login credentials
7	Database Addition and Front-end, Back-end integration using PHP, JS, Bootstrap, HTML and CSS	2	1,2,3	Creating a web database
8	External feature addition in website	3	1,2,3	Design of a php based website
9	Presentation - 1	3		Sessional 1 - 8
10	Laravel Installation and CLI introduction	1, 2	Internet	Creating Environment for PHP framework
11	Admin Panel Setup	2	Internet	Creating distinct panels
12	Eloquent ORM and Composer	1	Internet	Creating ORM and Composer packages in the environment
13	Ajax and MongoDB in websites	2	Internet	Knowing about Ajax for improving searching
14	Presentation - 2			Full Stack Web Development
15	Viva + Quiz			Sessional 1 - 14

Reading Reference:

1. Head First Web Design by Ethan Watrall and Jeff Siarto.
2. Learning Web Design: A Beginner's Guide to (X)HTML, StyleSheets, and Web Graphics Author: Aaron Gustafson.
3. PHP and MySQL Web Development., Author: Laura Thomson.
4. Learning JavaScript, Author: Shelley Powers.

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15

Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Microprocessor, Assembly Language and Computer Interfacing
Course Code: CSE 325, Credit: 3.0, Total hours: 42

Course Description:

1. Architectural overview of Intel Family, Microprocessor and its operation, CMP, Multi-core processor, Many core processor , CISC, RISC;

2. Intel 8086 Microprocessor : Internal architecture, register structure, programming model, addressing modes, instruction set; I/O Pin diagram and Control signals; I/O port organization and accessing; Cache Memory, TLB Structure; Memory Management in Intel 80X86 Family, DMA controller, Co-Processor;

3. Interrupts and Exception in Intel 80X86 families of processors, type of Interrupts, Interrupts in real mode and protected mode, Interrupt descriptor tables, Interrupts Priorities;

4. Introduction to Microcontroller, Atmel AVR microcontroller family: features, Layout, peripherals, Atmega32 Programmer Model: Memory, registers, pin out and descriptions, basic embedded system programming;

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Explain** different elements of microprocessor and microcontroller and their characteristics	C2				PO1
CO 2	**Demonstrate** the connections between microprocessor and it's peripherals	C3			CT	PO1
CO 3	**Analyze** the functionality of Atmega32 microcontroller using basic embedded C language.	C4				PO2

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M= Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2	S											
CO3		M										

Lecture Outline:

Class	Topics/Assignment	COs	Reading Ref.	Lecture Outcomes/ Activities
1	Architectural overview of Intel Family	1	1	Describe about different Processors of Intel Family
2	Microprocessor and its operation	1	1	Name different parts of Microprocessor
3	CMP, Multi-core processor, Many core processor , CISC, RISC	1	1	Identify processor organization
4	Intel 8086 Microprocessor : Internal architecture	1	1	Explain various parts of 8086 and their relations
5	Intel 8086 Microprocessor : register structure	1	1	Explain various registers of 8086
6	Intel 8086 Microprocessor :programming model	1	1	Identify programming model of 8086
7	Review			Lecture 1 - 6
8	Class Test 1			Lecture 1 - 6
9	Intel 8086 Microprocessor : addressing modes	1	1	Differentiate 8086 memory addressing modes
10	Intel 8086 Microprocessor :I/O Pin diagram and Control signals- Part-I	1	1	Describe 8086 pin diagram
11	Intel 8086 Microprocessor :I/O Pin diagram and Control signals- Part-II	1	1	Describe 8086 pin diagram
12	Intel 8086 Microprocessor :I/O Pin diagram and Control signals-Part-III	1	1	Describe 8086 pin diagram
13	Interrupts and Exception in Intel 80X86 families of processors	2	1,2	Identify interrupts and exceptions of

				8086
14	Type of Interrupts, Interrupts in real mode and protected mode, Interrupt descriptor tables	2	1, 2	Identify interrupts type, mode
15	Interrupts Priorities -Part I	2	1, 2	Describe priorities of interrupt and their application
16	Interrupts Priorities- Part-II	2	1, 2	Describe priorities of interrupt and their application
17	Interrupts Priorities- Part-III	2	1 ,2	Describe priorities of interrupt and their application
18	Review			Lecture 9 – 17
19	Class Test 2			Lecture 9 – 17
20	Introduction to Microcontroller	3	1	Explain different basic elements of Microcontroller
21	Atmel AVR microcontroller family: features, Layout, peripherals	3	1	Identify AVR microcontroller family and their characteristics
22	Atmega32 Programmer Model: Memory, registers	3	1	Explain Atmega32 and its functionality
23	Atmega32 Programmer Model: pin out and descriptions Part-I	3	1	Explain Atmega32 and its functionality
24	Atmega32 Programmer Model: pin out and descriptions Part-II	3	1	Explain Atmega32 and its functionality
25	Basic embedded system programming: Introduction to Embedded C language.Part-I	3	3	Analyze functionality of Atmega32 using embedded C language
26	Embedded C language Part-II	3	3	Analyze functionality of Atmega32 using embedded C language

Reading Reference:

1. The Intel Microprocessors-Barry B. Brey.
2. Microprocessors and Interfacing- Douglas V HALL.
3. The AVR microcontroller and embedded system using assembly and C- Muhammad Ali Mazidi

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Microprocessor, Assembly Language and Computer Interfacing Sessional
Course Code: CSE 326, Credit: 1.5, Total hours: 42

Rationale:
This course introduces basics of assembly language, microprocessor architecture, and discusses different interfaces and the design of systems based on microprocessors and microcontrollers.

Objective:

1. To achieve knowledge on the low level language of microprocessor
2. To provide an understanding of microprocessor-based systems and their use in instrumentation, control and communication systems.
3. To Investigate microprocessor-based systems, produce software for a microprocessor-based system, interface microprocessor-based systems and understand usage of programmable logic controllers

Course Outcomes (CO):
Upon completion of the course, the students will be able to:

1. Understand how the high level language is converted to low level languages and how a processor executes a program line by line.
2. Work with basic microprocessors using assembly language.
3. Understand the basic method of how a microcomputer works with its associate components.
4. Design 8051 microcontroller based system.
5. Know about different family and generation of microprocessors and microcontrollers.

Course Content:

Basic of Assembly Language - Compilation, input, output, variables, basic instructions, memory model, data segment, stack segment, code segment, Input Output Instruction; **Flow Control Instruction** - Conditional and unconditional jump instructions, If-then-else, case, for loop, while loop, repeat loop; **Logic, Shift and Rotate Instructions** - AND, OR, XOR, complement, shift left, shift right, rotate left, rotate right, rotate carry left, rotate carry right, Binary , Hexa Input Output; **Stack and Procedure** - Push, Pushf, Pop, Popf; **Multiplication**

and Division – Mul, IMul, Div, IDiv; **Array and Addressing moods –** 1D Array, DUP operator, Addressing mood, register indirect mood **String Instructions** - Moving string, load string, scan string, compare string; **File Operations** – File errors, opening and closing a file, reading a file, writing a file. Basic Idea of MDA 8086; LED, Seven Segment display, LCD, Keyboard, Motor, Dot matrix Interface with 8086; Basic idea of ATMEGA 16 microcontroller and simulation.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	**Program Outcome (PO)**											
	1	2	3	4	5	6	7	8	9	10	11	12
Describe how the high level language is converted to low level languages and how a processor executes a program line by line and solve problems using to low level languages.										√		
Work with basic microprocessors using assembly language.						√						
Work on Group Project based on basic micro-controllers with presentation.								√				

Lecture Outline:

Week	**Lecture**	**Topics**
1	Lec 1	**Basic of Assembly Language** - Compilation, input, output, variables, basic instructions, memory model, data segment, stack segment, code segment, Input Output Instruction
2	Lec 2	**Flow Control Instruction** - Conditional and unconditional jump instructions, If-then-else, case, for loop, while loop, repeat loop
3	Lec 3	**Logic, Shift and Rotate Instructions** - AND, OR, XOR, complement, shift left, shift right, rotate left, rotate right, rotate carry left, rotate carry right, Binary , Hexa Input Output
4	Lec 4	**Stack and Procedure** - Push, Pushf, Pop, Popf **Multiplication and Division** – Mul, IMul, Div, IDiv
5	Lec 5	**Array and Addressing moods –** 1D Array, DUP operator, Addressing mood, register indirect mood
6	Lec 6	**String Instructions** - Moving string, load string, scan string, compare string **File Operations** – File errors, opening and closing a file, reading a file, writing a file
7	Lec 7	**Lab Test-1**

8	Lec 8	Project Idea
9	Lec 9	Basic Idea of MDA 8086 LED and Seven Segment display interface
10	Lec 10	Operation of DOT matrix using 8086 kit LCD interface with 8086
11	Lec 11	Keyboard interface with 8086 Motor interface with 8086
12	Lec 12	Project Update
13	Lec 13	**Lab Test-2**
14	Lec 14	Final Project Submission Quiz

Reading Reference:

1. Assembly Language Programming and Organization of the IBM PC--Ytha Yu, Charles Marut
2. The Intel Microprocessors - Barry B Brey
3. Microprocessors and Interfacing - Douglas V. Hall

Assessment Methods:

Category	**Marks %**
Lab Test – 1	25
Lab Test – 2	25
Viva	10
Class Participation	10
Class Assessment	30
Total	**100**

Course Title: Computer Networks
Course Code: CSE 327, Credit: 3.0, Total hours: 42

Course Description:

1. Protocol hierarchies: Data link control, HLDC, DLL in Internet, DLL of ATM and LAN Protocols: Standards IEEE 802, Hubs, Bridges, and Switches, FDDI, Fast Ethernet.
2. Network layer: Routing Algorithm, Internetworking, WAN, Fragmentation, Firewalls, IPV4, IPV6, ARP, RARP, Mobile IP
3. Transport Protocols: Transmission Control Protocol, Connection Management, Transmission Policy, Congestion Control, Timer Management, UDP, AAL of ATM
4. Domain Name System: Name servers, Email and Its privacy, SNMP, HTTP, World Wide Web
5. Network security: Cryptography, DES, IDEA, public key algorithm, Authentication, Digital signatures, Principles of Reliable Data Transfer, FTP.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Identify** functionality of different layers of the network architecture.	C1				PO1
CO 2	**Apply** mathematical foundations to solve computational problems in computer networks.	C3			CT	PO2
CO 3	**Formulate** a plan to find the best approaches for connection establishment	C6			CT	PO3
CO 4	**Design** a network architecture.	C6			CT	PO3

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2		S										
CO3			M									
CO4			S									

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Introduction and Uses of Computer Networks	1	1	About Computer Network
2	Details about Network hardware, LAN, MAN, WAN, Protocol Hierarchies	1	1	Idea about different types of network
3	OSI Reference Model, TCP/IP Model	1	1,2	Distinguish between OSI & TCP/IP, OSI Layers and TCP/IP layers
4	The Physical layer, Theoretical basis for data communication, Nyquist's theorem, Shannon's theorem	2	1,2	Measuring maximum number of data bits
5	Guided and Unguided transmission medium	3	2,5	Differences between Guided and Unguided medium, speed and capacity of the medium
6	Circuit switching, Packet Switching, Message Switching	3	2,5	Comparing switching techniques
7	Class Test -1			Lecture 4-6
8	Asynchronous Transfer Mode: Virtual Circuit, Reference model, Reference model	3	1,5	Familiarization of ATM model
9	Network Topology: Advantages and disadvantages of Mesh, Star, Bus, Ring topology	3	1,5	Evaluate which topology is needed under a circumstances
10	The Data Link Layer: Four basic Framing methods	2	1,2	Finding the starting and ending point of a Frame

11	Error Detection and Correction: Hamming distance, Cyclic Redundancy check	2	1,2	Calculate Hamming distance, CRC for error detection and correction in a frame
12	Data link protocol: High-Level Data link Control, Point-to-point protocol	1	1,2	Idea about Data link layer protocol
13	Class Test - 2			Lecture 10-11
14	The Network layer: IP datagram header format, IP structure, IPv4, IPv6, Classful IP addressing	4	2,5	Familiarization of IP addressing
15	IP address format, Address range, Special IP address, Network & Broadcast address, Subnet mask, Public & Private IP	4	2,5	Familiarization of IP addressing
16	Subnetting: Concept of Fixed length subnet mask and its application	2,4	2,5	Calculating IP address for hosts using FLSM method
17	Subnetting: Concept of Variable length subnet mask and its application	2,4	2,5	Calculating IP address for hosts using VLSM method
18	Review Subnetting			Calculating IP address for hosts using FLSM & VLSM method
19	Classless IP address, classless InterDomain Routing, Datagram & Virtual circuit	4	2,5	Differentiate between class-full and classless IP addressing
20	Routing Algorithms: Shortest Path Routing, Dijkstra's algorithm	2,4	2,5	Calculate shortest path using Dijkstra's algorithm
21	Distance Vector Routing, Count to infinity problem in Distance Vector Routing	2,4	2,5	Calculate shortest path using Distance Vector Routing algorithm
22	Congestion Control Algorithms: Congestion control in Virtual circuit subnet & Datagram subnet	1	2,5	Mechanism about congestion control in virtual circuit and datagram
23	Warning Bit, Choke packet, Hop-by-hop choke packet	3	2,5	Idea about how datagram declares a warning state
24	Internetworking, Tunneling, Fragmentation	2	1,5	Calculate Frame Fragmentation
25	IPv4 header Datagram format, Internet Control Message protocol: ARP, RARP	3	1,5	Calculating ARP and RARP technique
26	BOOTP, DHCP, Mobile IP	3	1,5	Identification about ICMP protocol
27	IPv6 addressing	2	1,5	Calculating IP address using IP version 6
28	Class Test- 3			Lecture-14-18
29	The Transport Layer: TPDU, TCP segment header, TCP connection establishment	3	3,4	Identify three way handshaking technique in Transport Layer
30	TCP connection Release, Transmission policy, Congestion control	1,3	3,4	Idea about TCP congestion control
31	The Medium Access Control Sublayer: CSMA/CD, 1 persistent CSMA, nonpersistent CSMA	3	3,4	Differentiate between 1 persistent CSMA and nonpersistent
32	Ethernet(IEEE 802.3), OSI 7 layer devices: Repeater, Hub, Bridge	1	1,2	Idea about OSI layer devices, advantages and disadvantages

33	OSI 7 layer devices: Switch, Router, Gateway	1	1	Idea about OSI layer devices, advantages and disadvantages
34	The Application Layer: Domain name system, email, www, URL	1	1,5	Configuring how Domain name works
35	Class Test - 4			Lecture 29-30
36	Network security: Cryptography, DES, IDEA	3	1	Idea about network security
37	Public key algorithm, Authentication	3	1	Idea about network security
38	Digital signatures, Principles of Reliable Data Transfer, FTP	3	1	Idea about network security
39	Review class			Lecture 29-33
40	Review class			Lecture 20-23
41	Mathematical problem solving			Lecture 16-18
42	Mathematical problem solving			Lecture 16-18

Reading Reference:

1. Computer Networks - Andrew S. Tanenbaum
2. Computer Networks: Protocols, Standards, and Interfaces - Uyless Black
3. Internetworking with TCP/IP: Principles, Protocols, Architecture - D. E. Comer
4. TCP/IP Illustrated Vol. I - W. R. Stevens
5. Complete Networking : A Top Down Approach Featuring the Internet – 3rd edition , James F. Kurose, Keith W. Ross

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Computer Network Sessional
Course Code: CSE 328, Credit: 1.5, Total hours: 42

Rationale:

Apply, discuss, analyze, simulate, and present architectures, applications of different types of computer networks to facilitate communication and resource-sharing among a wide range of users.

Objective:

1. Understand, analyze, simulate, and present architectures of different types of computer networks & develop, and present contemporary and new protocols of computer networks.
2. Understand, Design, analyze different types of Network Layers.
3. Identify applications of computer networks with determining suitable alternatives of the networks.
4. Achieve a basic idea about Cisco Packet tracer, WireShark,Ns2

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Discuss, analyze, simulate, and present architectures of different types of computer networks; develop, and present contemporary and new protocols of computer networks.
2. Describe, design, and analyze different types of network layers.
3. Identify applications of computer networks with determining suitable alternatives of the networks.
4. Apply the basic functionalities of Cisco Packet Tracer, WireShark, NS-2.

Course Content:

IP Addressing, Basic Configuration of Cisco Packet Tracer, Socket Programing, Basic Network Configuration (Static) Data, Variable Length Subnet Mask (VLSM), RIP, EIGRP, Dynamic Host Configuration Protocol (DHCP) , Open Shortest Path First (OSPF), Physical Network Interface Connection/ Router & Switch Configuration, Access Control List (ACL), VLAN, InterVLAN, VTP, Information Gathering using Wireshark, Introduction to NS2.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	**Program Outcome (PO)**											
	1	2	3	4	5	6	7	8	9	10	11	12
Discuss, analyze, simulate, and present architectures of different types of computer networks; develop, and present contemporary and new protocols of computer networks.									√			
Describe, design and analyze different types of network layers											√	
Identify applications of computer networks with determining suitable alternatives of the networks.									√			
Apply the basic functionalities of Cisco Packet Tracer, WireShark,Ns2.											√	

Lecture Outline:

Week	**Topics**

1	IP Addressing, Basic Configuration of Cisco Packet Tracer
2	Socket Programing
3	Basic Network Configuration (Static) Data
4	Variable Length Subnet Mask (VLSM)
5	RIP, EIGRP
6	Dynamic Host Configuration Protocol (DHCP) , Open Shortest Path First (OSPF)
7	**Lab Test-1**
8	Physical Network Interface Connection/ Router & Switch Configuration
9	Access Control List (ACL)
10	VLAN, InterVLAN, VTP
11	Information Gathering using Wireshark
12	Introduction to NS2
13	**Lab Test-2**
14	Quiz

Reading Reference:

1. Computer Networks - Andrew S. Tanenbaum
2. Computer Networks: Protocols, Standards, and Interfaces - Uyless Black Internetworking with TCP/IP: Principles, Protocols, Architecture - D. E. Comer
3. TCP/IP Illustrated Vol. I - W. R. Stevens
4. Complete Networking : A Top Down Approach Featuring the Internet – James F. Kurose, Keith W. Ross

Assessment Methods:

Category	**Marks %**
Lab Test-1	25
Lab Test-2	25
Viva	10
Class Participation	10
Class Assignment	30
Total	**100**

Course Title: Integrated Design Project
Course Code: CSE 301, Credit: 2.0, Total hours: 56

Rationale:

Culminating demonstration of skills and knowledge achieved to date to apply and solve real life problems solvable through computer technology.

Objective:

To apply technical knowledge and skills for further research and design of computer system at professional engineering scale.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Develop systems' requirement specification from top-level customer requirements.
2. Analyze and compare design alternatives, at the system and subsystem levels, and use measures of performance or other criteria to rank alternatives.
3. Plan and organize an engineering design project using tools such as Gantt charts to develop a work breakdown structure, develop a schedule including milestones, and estimate effort and costs incorporating the ethical, financial and environmental issues.
4. Develop a design concept and elaborate it through to a detailed design by decomposing a system concept into component subsystems, identifying the subsystem requirements and applicable standards, and defining interfaces between the subsystems.
5. Build prototypes of key subsystems.

Course Content:

Knowledge Acquisition:

Introduction and brief with software, Discussion and submission Project Proposal, Learning Version Control System: Github, Interface design, Database Connectivity: Sqlite, Json, Cloud, Firebase database, Google Api, Sensor.

Implementation:

Idea Submission, Objective, Methodology, Literature Review, High Level Design, Low Level Design, Evaluation and feedback, Design & Partial Implementation (Prototype/Demo).

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
Develop system requirements from top-level customer requirements.	√	√				√						
Analyze and compare design alternatives, at the system and subsystem levels, and use measures of performance or other criteria to rank alternatives.		√	√		√							
Plan and organize an engineering design project using tools such as Gantt charts to develop a work breakdown structure, develop a schedule including milestones, and				√		√						

estimate effort and costs.											
Develop a design concept and elaborate it through to a detailed design by decomposing a system concept into component subsystems, identifying the subsystem requirements and applicable standards, and defining interfaces between the subsystems			√	√							
Build prototypes of key subsystems.					√			√			

Lecture Outline:

Week	Topics	Remarks
1	Introduction and brief with softwares and Tools	
2	Learning Version Control System: Github	
3	Interface Design, activity and navication.	
4	Database Connectivity: Sqlite, Json, Cloud* Firebase database	
5	Google Api	
6	Sensor	
7	Database Integration with the application	
8-9	Topic Selection and Project Plan	
10-11	Objective, Methodology, Literature Review	
12-14	Design & Partial Implementation (Prototype/Demo)	

Reading Reference:

1. Android Programming: The Big Nerd Ranch Guide (3rd Edition) (Big Nerd Ranch Guides) 3rd Edition
2. Professional Android, Reto Meier, Ian Lake; 4th Edition
3. Website: https://www.tutorialspoint.com/android/

Assessment Methods:

Category	Marks %
Project	40
Quiz	20
Viva/Presentation	10
Class Participation	10
Home Assignment /Report	10
Class performance/observation	10
Total	**100**

Course Title: Management Information System
Course Code: CSE 329, Credit: 2.0, Total hours: 28

Course Description:

1. To understand managerial challenges of the changing world, management functions, management involvement to get competitive advantages
2. To find and analyze environmental forces that influences organization and to know how to respond/adapt to the environments.
3. To know all about decision making and to understand how to use tools and techniques to take creative decisions.
4. To understand planning, planning processes and strategic planning and to know how companies can achieve competitive advantages through good planning and good strategy.
5. To know all about organization structure and to construct good organization's structure to execute plan and to achieve organizational goals.
6. To know how to manage human resources to gain competitive advantages.
7. To understand all about leadership and know how a good leader can motivate to achieve organizational goals.
8. To know how to supervise, monitor and work as per plans and to control the execution of the plans.
9. To understand all about team, change management and international business.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Identify** managerial challenges, strategic planning & leadership	C1				PO1
CO 2	**Develop** an idea for forecasting the necessary steps to start-up a new business			A4	CT	PO11

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
CO1	M											
CO2											S	

Lecture Outline:

Time	Topics/Assignment	COs	Reading Reference	Teaching Strategies
Class 1 & 2	Course Overview/Introduction to Management - Course Introduction - Introduction to Management: Definition, Competitive advantages, Managerial function, Level of management, productivity-performances etc.	1	Reading Chapter # 1 of the text # 1 and chapter # 1 of the text # 2 and handout	Lecture Multimedia
Class 3 & 4	Forecasting Fundamentals of forecasting, application of forecasting in business, forecasting models	2	Reading Chapter # 3 of the text # 2 and handout Assignment	Lecture presentation
Class 5 & 6	Capacity Planning Applications of capacity planning in service and manufacturing industries, different capacity planning models	2	Reading Chapter # 5 of the text # 2 and handout Assignment	Lecture presentation Group discussion
	Class Test 01			Lecture 1-4
Class 7 & 8	Inventory Management - Capacity Planning Inventory models and inventory control	1	Reading chapter # 11 & 13 of the text # 2 and handout	Lecture presentation Group discussion

Class 9 & 10	Decision Making Decision and Decision making: managerial decisions and stages of decision making; barriers to decision making; group decision making.	2	Reading Chapter # 3 of the text # 1 and handout Assignment	Lecture presentation Group discussion
Class 11 & 12	Decision Making Tools - Formulation of decision problem - Graphical presentation - Linear Programming - Financial analysis: NPV, IRR, etc.	2	Reading Chapter # 6 of the text # 2 and handout Assignment	Lecture presentation Group discussion
	Class Test 02			Lecture 9-12
Class 13 & 14	Technology & Manufacturing Management - Basic of Technology, components of technology, selection of technology, automation and technology, technology and productivity	1	Reading chapter # 7 of the text # 2 and handout Assignment	Lecture presentation Group discussion
Class 15 & 16	Quality Control and Management - Quality system - Quality Management and Assurance - Quality tools - Inspection Statistical Quality Control	1	Reading chapter # 9 & 10 of the text # 2 and handout Assignment	Lecture presentation Group discussion Industrial Visit
Class 17 & 18	Scheduling & Project Management - Scheduling Operations	1	Reading chapter # 13 & 17 of the	Lecture presentation
	- Project and Scheduling Network diagram: CPM/PERT		text # 2 and handout Assignment	Group discussion
	Class Test 03			Lecture 13-16
Class 19 & 20	Facility Selection - Identification of product/services processing techniques, techniques of plant layout design, facility location techniques - Transportation Model	1	Reading chapter # 6 of the text # 2 and handout Assignment	Lecture presentation Group discussion
Class 21 & 22	Queuing Theory Definition, Characteristics, Discipline	1	Reading chapter # 18 of text # 2	Lecture presentation
Class 23 &	Supply Chain Management - Needs, Benefits and	1	Reading chapter # 16	Lecture presentation

24	Elements of Supply Chain Management - Logistics; Creating an effective Supply Chain		of the text # 2 and handout	Group discussion
Class 25 & 26	Review Class			

Reading Reference:

Text

1. Bateman & Snell, *Management – The New Competitive Landscape,* 6th edition. McGraw-Hill.
2. Stevenson, William J., *Operations Management*, 9th edition, McGraw-Hill.

Reference

1. Kast & Rosenzweig, Organization & Management, 6th edition
2. Koontz, *Management*, 8th edition
3. Adam & Ebert, Production and Operations Management, 6th edition, Prentice Hall

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

LEVEL-4 TERM-I

Course Title: Machine Learning
Course Code: CSE 411, Credit: 2.0, Total hours: 28

Rationale:

Machine learning provides appropriate learning algorithm to best suit the current need and enhance the learning parameters for maximum performance.

Objective:

1. To learn paradigms in different environmental setting and apply the appropriate learning algorithm to best suit the current need.
2. To enhance the learning parameters to achieve maximum performance.

Course Content:

Introduction to Machine Learning; Regression analysis: Logistic Regression, Linear Regression; Supervised and Unsupervised learning; Bayesian Learning; Decision Tree Learning; Rule based learning; Instance based learning; Neural Nets; Support Vector Machine; Genetic Algorithms; Reinforcement learning; Ensemble learning; Hidden Markov Models; Maximum Likelihood Estimates, Parameter Estimation; Computational learning theory.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Discuss the different learning paradigms in different environmental setting
2. Select and apply the appropriate learning algorithm to best suit the current need.
3. Enhance the learning parameters to achieve maximum performance.

Mapping of CO and Program Outcomes:

Course Outcomes(CO) of the Course	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
Discuss the different learning paradigms in different environmental setting.	√											
Select and apply the appropriate learning algorithm to best suit the current need.		√										
Enhance the learning parameters to achieve maximum performance.		√										

Lecture Outline:

Week	Lecture	Topics	Class Test
1	Lec 1 Lec 2	Introduction to Machine Learning	
2	Lec 3 Lec 4	Regression analysis Logistic Regression	
3	Lec 5 Lec 6	Linear Regression	**Class Test 1**
4	Lec 7	Supervised learning	**Class Test 2**
	Lec 8	Unsupervised learning	

	Lec 9		
5	Lec 10 Lec 11	Bayesian Learning Decision Tree Learning	
6	Lec 12 Lec 13	Rule based learning Instance based learning	
7	Lec 14	Neural Nets	
8	Lec 15 Lec 16	Support Vector Machine Genetic Algorithms	
9	Lec 17 Lec 18	Reinforcement learning	**Class Test 3**
10	Lec 19 Lec 20	Ensemble learning	
11	Lec 21 Lec 22	Hidden Markov Models	
12	Lec 23 Lec 24	Maximum Likelihood Estimates	
13	Lec 25 Lec 26	Parameter Estimation	**Class Test 4**
14	Lec 27 Lec 28	Computational learning theory	

Reading Reference:

1. Pattern Recognition and Machine Learning - Christopher M. Bishop; Springer
2. Machine Learning - Tom Mitchell, McGraw Hill
3. Pattern Recognition –SergiosTheodoridis and KonstantinosKoutroumbas; Elsevier Inc.

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Machine Learning Sessional
Course Code: CSE 412, Credit: 1.0, Total hours: 28

Rationale:

To be oriented different algorithm of machine learning practically to best suit the current need and implement the enhanced learning parameters for maximum performance.

Objective:

1. To implement the appropriate learning algorithm to best suit the current need.
2. To use practical knowledge to enhance the learning parameters to achieve maximum performance and enhance the learning parameters to achieve maximum performance.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Use and apply the appropriate learning algorithm to best suit the current need.
2. Use practical knowledge to enhance the learning parameters to achieve maximum performance.
3. Enhance the learning parameters to achieve maximum performance.

Course Content:

Logistic Regression, Linear Regression; Supervised and Unsupervised learning; Bayesian Learning; Decision Tree Learning; Rule based learning; Instance based learning; Neural Nets; Support Vector Machine; Genetic Algorithms;

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	**Program Outcome (PO)**											
	1	2	3	4	5	6	7	8	9	10	11	12
Use and apply the appropriate learning algorithm to best suit the current need.											√	
Use practical knowledge to enhance the learning parameters to achieve maximum performance.						√						
Enhance the learning parameters to achieve maximum performance.										√		

Lecture Schedule:

Week	**Lecture**	**Topics**	**Remarks**
1	Lec 1 Lec 2	Orientation Practical on Logistic Regression, Linear Regression;	

2	Lec 3 Lec 4	Implementing Supervised and Unsupervised learning	Assignment
3	Lec 5 Lec 6	Implementing Bayesian Learning Implementing Decision Tree Learning;	Lab Test 1
4	Lec 7 Lec 8	Implementing Rule based learning and Instance based learning;	Assignment
5	Lec 9 Lec 10	Implementing Neural Net	Lab Test 2
6	Lec 11 Lec 12	Implementing Support Vector Machine Implementing Genetic Algorithms	
7	Lec 13 Lec 14	Final Quiz Viva	

Assessment Methods:

Category	**Marks %**
Lab test	40
Quiz	20
Viva	10
Class Performance	10
Report	10
Class Assessment	10
Total	**100**

Course Title: Compiler Design and Construction
Course Code: CSE 413, Credit: 3.0, Total hours: 42

Course Description:

1. To understand the concepts of System software, Application Software and different hypothetical machine architectures.
2. Familiarize with source file, symbol table creation (pass-1), object file creation (pass-2), loaders and linkers.
3. To understand the fundamental concepts of translators and strategies for parsing techniques.
4. Devise and perform syntax-directed translation schemes for compilers for better optimization.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				**PO**
		C	**P**	**A**	**S**	
CO 1	**Review** the fundamentals concepts of computer language processing system,	C1				PO1

	different phases of a compiler, and analysis-synthesis model of compilation					
CO 2	**Design** regular expressions for recognizing tokens, building symbol table, and transforming the regular expressions to finite state machines for developing lexical analyzer.	C2				PO2
CO 3	**Design** context-free-grammars and parsers for syntax analysis to validate string of tokens of a programming language.	C2				PO2
CO 4	**Construct** syntax-directed definitions, translation schemes, annotated parse trees, dependency graph for semantic analysis and type checking.		P4			PO3
CO 5	**Illustrate** intermediate codes, and code optimization techniques for efficient compiler design.	C4			CT	PO2
CO 6	**Generate** codes for target programming language considering memory, instruction cost, available registers and run-time memory allocation.	C6			CT	PO3

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2		S										
CO4		S										
CO6			M									

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference
1	Introduction to Compilers Review the fundamentals concepts of computer language processing system, different phases of a compiler, and analysis-synthesis model of compilation	1	1
2	Lexical Analysis: Token, Pattern, Lexeme, Regular Expressions, DFA, NFA	1	1
3	Lexical Analysis: RE to NFA, NFA to DFA using subset construction, NFA to CFG	1	1
4	Lexical Analysis: RE directly to DFA by computing followpos	1	1
5	Lexical Analysis: NFA to RE using Subset Construction Algorithm, DFA Minimization using Hopcroft's Algorithm	1	1
6	Practice on L-2, L-3, L-4 and L-5	1	1
7	Review		1
8	Class Test 1		1
9	Syntax Analysis: Introduction to Parsing, Context Free Grammar, Parse tree	2, 3	1

10	Syntax Analysis: Error Recovery, Ambiguous Grammar, Left Recursion Elimination, Left factoring	2, 3	1
11	Syntax Analysis: LL Grammar, Top down parsing, computing first and follow sets, Dealing with errors in Top down parsing, Panic and phrase level error recovery	2, 3	1
12	Practice on L-9, L-10 & L-11		1
13	LL(1) Grammar, Bottom up parsing algorithm	2, 3	1
14	Computing closure and goto, basic SLR parser	2, 3	1
15	Practice on L-12, L-13 & L-14	2, 3	1
16	Review	2, 3	1
17	Class Test 2		1
18	Syntax Directed Translation: Semantic rules, synthesized and inherited attributes	2,3	1
19	Syntax Directed Translation: SDD vs SDT; SDD based on bottom up parsing, Annotated parse tree, Dependency graph, Evaluation order	2,3	1
20	Syntax Directed Translation: SDD based on top down parsing	3	1
21	Practice on L-18, L-19, L-20	3	1
22	Intermediate Code Generation: Linear and Graphical representation		1
23	Three address code, DAG, Implementation of DAG, Value number method, Triple, Quadruple, Indirect triples	3	1
24	Intermediate Code Generation: Translating Expressions, SDT into three address code, short circuit code	3	1
25	Intermediate Code Generation: SDT for flow of control statements, Generating for three address code for Booleans	3	1
26	Practice on L-22, L-23, L-24 L-25		1
27	Review		1
28	Class Test 3		1
29	Type Checking, Type Conversion	3	1
30	Run time environment, Activation Tree Activation Record	3	1
31	Problem Solving	3	1
32	Assembly code generation, Cost calculation of instructions	3	1
33	Basic block construction, Flow graph generate and loop identification from a pseudo code	3	1
34	Practice on L-32, L-33	3	1
35	Code Optimization techniques, Dead Code Elimination	4	1
36	optimization of basic blocks, loop in flow graphs, global data flow analysis, iterative solution of data flow equations	3	1
37	Practice on L-36	3	1
38	Review	3	1
39	Class Test 4		1
40	Problem Solving		1
41	Assignment Evaluation		1
42	Review Class, Problem Solutions and Suggestion		1

Reading Reference:

1. Compilers Principles, Techniques and Tools -- Alfred V. Aho, Jeffrey D. Ullman

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5

Final Examination (3 hours)	70
Total	**100**

Course Title: Compiler Design and Construction Sessional
Course Code: CSE 414, Credit: 1.0, Total hours: 28

Rationale:

To implement tokenizer, arithmetic calculator and to be able to write the code by using Flex and Bison.

Objective:

1. To learn to implement different phases of a compiler.
2. To learn the use of Flex and Bison tools used for designing a compiler.
3. To understand the different types of parsing techniques and to solve the problem.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Use the basic techniques of compiler construction and tools that can be used to perform syntax-directed translation of a high-level programming language into an executable code.
2. Describe the working of lex and yacc compiler for debugging of programs
3. Discuss the new tools and technologies used for designing a compiler

Course Content:

Symbol Table, Tokenizer, Tokenizer Using Flex, Arithmetic Calculator using Bison, Intermediate Code Generator (Flex + Bison).

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes (CO) of the Course	**Program Outcome (PO)**											
	1	2	3	4	5	6	7	8	9	10	11	12
Use the basic techniques of compiler construction and tools that can used to perform syntax-directed translation of a high-level programming language into an executable code.						√						
Describe the working of lex and yacc compiler for debugging of programs.									√			

Discuss the new tools and technologies used for designing a compiler.										√	

Lecture Schedule:

Week	Lecture	Topics
1+ 2	Lab 1+2	Symbol Table
3+ 4	Lab 3+ 4	Tokenizer
5+ 6	Lab 5+6	Tokenizer Using Flex
7+ 8	Lab 7+ 8	Arithmetic Calculator using Bison
9+ 10	Lab 9+ 10	Intermediate Code Generator (Flex + Bison)
11+ 12	Lab 11+ 12	Online
13	Lab 13	Quiz
14	Lab 14	Viva

Reference Books:

1. Compilers: Principles, Techniques & Tools (2nd ed)- Alfred V Aho, Monica S Lam, Ravi Sethi, and Jeffrey D Ullman, Pearson/Addison Wesley (2006).
2. Engineering A Compiler (2nd Ed)- Linda Torczon and Keith Cooper, Morgan Kaufmann Publishers Inc (2011)

Assessment Methods:

Category	Marks %
Online Test – 1	25
Online Test – 2	25
Viva	10
Class Participation	10
Class Assessment	30
Total	**100**

Course Title: Research Methodology and Complex Engineering Activities
Course Code: CSE 415, Credit: 2.0, Total hours: 42

Rationale:
Research Methodology is a hands- on and advanced level course designed to impart education in the foundational methods and techniques of research in the context of social sciences and business. Through this course students will be familiar with the main components of a research framework including problem definition, research design, and data collection, ethical issues in research, report writing, and presentation. Students will also be familiar with the quantitative,

qualitative, and mixed approaches to research. Once equipped with this knowledge, participants would be well- placed to conduct disciplined research in their areas of interest.

Course Objectives:
The overarching objective of this course is to impart the fundamental knowledge of research to students. The specific objectives of this course are:

a. to familiarize participants with basic of research and the research process;
b. to enable the participants in conducting research work and formulating research synopsis and report;
c. to familiarize participants with statistical packages, such as SPSS, EXCEL etc.;to impart knowledge for enabling students to develop data analytics skills and meaningful interpretation to the data sets so as to solve the business/research problem.

Course Learning Outcomes (CLO):
Upon successful completion of this course, students will be able to:

CLO1	Understand and define the basic framework of research process
CLO2	Develop a comprehensive research proposal for a research problem
CLO3	Demonstrate a sound knowledge of statistical terms at an introductory level
CLO4	Apply appropriate sampling techniques and develop hypothesis and test it
CLO5	Analyze qualitative and quantitative data using Excel and SPSS software

Mapping of Course Outcome to Program Outcome:

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1		✓							
CLO2								✓	
CLO3						✓			
CLO4						✓			
CLO5				✓					
CLO6									

SL NO	COURSE CONTENT	Hrs.	CLOs
1.	**Introduction:** Definition of business research, Types of research, The language of research, Steps of research process, Significance of research to the managers, Ethics in business research.	3	CLO1
2.	**Research Design:** Classification of research design, Secondary data searches, Qualitative research, Observation studies, Surveys, Experimentation.	3	CLO4
3.	**Research Process:** The management research question, Research process issues, Designing research proposal.	3	CLO4
4.	**Sources and Collection of Data:** Nature of measurement, Measurement scales, Rating scales, Ranking scales, Selecting	3	CLO2

	measurement scales, Scaling, Validity and Reliability, Questionnaire and instruments.		
5.	**Sampling Design:** Nature, Steps in sampling design, Probability and non-probability sampling, Sample size determination.	3	CLO6
6.	**Analysis and Presentation of Data:** Editing, Coding, Data entry, Exploring, displaying and examining data, Hypothesis testing, Vicariate and multivariate analysis.	3	CLO4
7.	**Presenting Insights and Findings:** The written research report, Research report component, Writing the report, Presentation of statistics, Oral presentation, Bibliography and reference writing.	3	CLO5
8.	**Proposal Writing, Publication Strategies, etc.**	3	CLO3
9.	Measurement and Scaling: nature - Components and Process Scales - nominal ordinal, interval and ratio, Tests - validity and reliability. Testing of Hypotheses, Format and use of graphic aids.	3	CLO6
10.	Data Collection Methods - Types - Primary and Secondary, Interviewing, Questionnaires, Observation.	3	CLO4
11.	Data Analysis: Data editing- Cronbach's Alpha, Descriptive and inferential statistics, Software package- SPSS, Excel, Structural Equation Model (SEM).	3	CLO5

Reading Reference:

Recommended Texts:

1. Zikmund, W. G., Carr, J. C., Babin, B., & Griffin, M. (2013). *Business research methods*. Nelson Education.
2. Kothari, C. R. (2004). *Research methodology: Methods and techniques*. New Age International.

References:

1. Cooper, D. R., Schindler, P. S., & Sun, J. (2006). *Business research methods* (Vol. 9). New York: McGraw-Hill Irwin.
2. Lind, D. A., Marchal, W. G., & Wathen, S. A. (2012). *Statistical techniques in business & economics*. New York, NY: McGraw-Hill/Irwin.
3. Malhotra, N. K., & Dash, S. (2016). *Marketing research: An applied orientation*. Pearson.

Assessment Methods:

CIE-Continuous Internal Evaluation (30 Marks)

Blooms' Category	Tests	Assignments	Case/Viva/ Presentation	External Curricular or Co-curricular Activities
Remember	5			
Understand	5			
Apply			5	
Analyze		5		

Evaluate			5	
Create		5		

MSE- Mid-term and Semester-final Examination (70 Marks)

Blooms' Category	Tests
Remember	10
Understand	15
Apply	15
Analyze	20
Evaluate	10
Create	10

Elective 1

Course Title: Digital Image Processing
Course Code: CSE 443, Credit: 3.0, Total hours: 42

Course Description:

1. Digital image fundamentals: representation, sampling and quantization, image acquisition, basic
   relationships between pixels, imaging geometry;
2. Image transforms: discrete Fourier transform, discrete cosine transform, analyzing digital
   images in frequency domain;
3. Image enhancement: in spatial domain and in frequency domain, image smoothing and sharp-
   ening;
4. Image restoration: degradation models, inverse filter, Wiener filter;
5. Color, color models and pseudo-color image processing;
6. Image segmentation: detection of discontinuities, thresholding, region-oriented segmentation,
   the use of motion analysis in segmentation

Course Outcome (CO) Matrix:
At the end of the course, the students should be able to:

Course Outcome Description		Level of Domain				PO
		C	P	A	S	
CO1	**Discuss** the fundamental concepts and various applications of digital image processing.	C2				PO1, PO3
CO2	**Acknowledge** the theory and algorithms that are widely used in digital image processing			A1		PO1, PO2

CO3	**Evaluate** the techniques for image enhancement and morphological image processing	C5				PO2, PO3, PO4
CO4	**Demonstrate** various edge detection, image segmentation, restoration, compression, object recognition techniques and also **analyze** different color models along with their usage.	C4	P4			PO1, PO2, PO4

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S		M									
CO2	H	H										
CO3	M		S	M								
CO4	H	H		M								

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1.	Introduction to Digital Image Processing and its various applications	1	1	Idea about digital images and its various applications
2.	Fundamental steps in Digital Image Processing, Components of an Image Processing System	1	1	Solve problems using rules of Probability
3.	Structure of the Human Eye, Light and the Electromagnetic Spectrum	1	1	Mechanism of human eye and its perception of images
4.	Image Sensing and Acquisition, A simple Image Formation Model	1, 2	1	Formation of an image in a plane
5.	Review			Lecture 1 - 4
6.	Practice			Lecture 1 - 4
7.	Class Test 1			Lecture 1 - 4
8.	Basic concepts in Sampling and Quantization, Representing Digital Images	2	1	Transforming an Analogue image into a Digital one
9.	Spatial and Intensity Resolution, Image Interpolation	2	1	Measurements of various resolution types,
10.	Basic Relationships between Pixels, Linear versus Nonlinear Operations	2	1	Neighbors of a pixel, Adjacency, Linear-Non Linear methods.
11.	Binary Images, Gray Scale Images, Histogram	1, 2	1, 2	Different forms of digital image, importance of

				histogram in image processing
12.	Review			Lecture 8 - 11
13.	Practice			Lecture 8 - 11
14.	Histogram Equalization	2	2	Enhancing the contrast of digital image
15.	Importance of Morphological Image Processing, Erosion and Dilation	3	2	Study about different shapes, extend or reduce parts of images.
16.	Morphological Image Processing: Opening and Closing , OTSU Method	3	1, 2	Finding out the best threshold for performing histogram
17.	Fourier Series and Fourier Transformation , Analyzing Images in Frequency Domain	2, 3	Inter net	Analyzing images in Frequency domain
18.	Review			Lecture 14 - 17
19.	Practice			Lecture 14 - 17
20.	Class Test 2			Lecture 14 - 17
21.	Kernels, Convoluting Images using Masks	3	2, Inter net	Demonstrate regression using modern tools
22.	Importance of Image Compression, Lossless and Lossy Image Compression Methods	4	1, Inter net	Compressing images to utilize the storage
23.	JPEG and Huffman Compression Techniques	4	Inter net	Compare between the JPEG and Huffman coding method
24.	Introduction to Color Models, Usage and Applications of Different Color Models: RGB, CMY and HSI Model	4	1, Inter net	Importance and necessity of color models in image processing
25.	Review			Lecture 21 - 24
26.	Practice			Lecture 21 - 24
27.	Image Enhancement, Sharpening and Smoothing using Spatial Filters	3	1, 3	Procedure of improving the quality and information content of original data before processing
28.	Image Restoration, Weiner Filter and Inverse Filter	4	1, 2, 3	Recovering the original image from the degraded one
29.	Image Segmentation: Detection of discontinuities, Segmentation by thresholding, Region based segmentation	4	2, 3	Partitioning a digital image into multiple segments
30.	Class Test 3			Lecture 27 - 29
31.	Wavelets and multiresolution processing, Wavelet transforms in one	2	1	Converting a digital image into a set of mathematical

	dimension: Discrete and Continuous Wavelet Transform			expressions that can then be decoded by the receiver
32.	The Fast Wavelet Transform, Wavelet transforms in two dimension	2		Applying Wavelet transformation in 2D
33.	Image Feature Extraction and Representation	4	1	Efficiently represents interesting parts of an image as a compact feature vector
34.	Object Recognition: Patterns and Pattern Classes, Recognition based on Decision-Theoretic Methods	4	1	Identifying objects of interest in digital images
35.	Review			Lecture 31 - 34
36.	Practice			Lecture 31 - 34
37.	Class Test 4			Lecture 31 - 34
38.	Object Recognition: Structural Methods	4	1, 4	Representing objects as strings, trees or graphs and then defining descriptors and recognition rules based on those representations
39.	Introduction to Image Reconstructions from Projections, Principles of Computed Tomography	4	1, 4	Generates tomographic images from X-ray projection data acquired at many different angles
40.	Image Reconstructions from Projections: Projections and Radon Transform, The Fourier Slice Theorem	4	1	Provides a very useful relation between the 2D Fourier transform of the original function and the 1D Fourier transform of its Radon transform along the detector coordinate
41.	Problem Solving			Lecture 1 - 40
42.	Overall Course Discussion			Lecture 1 - 40

Reading Reference:

1. R. C. Gonzalez, R. E. Woods, --- "Digital Image Processing", Prentice-Hall, 3rd Edition
2. Anil K. Jain, --- "Fundamentals of Digital Image Processing", Prentice-Hall, 7th edition
3. S Jayaraman, S Esakkirajan, T Veerakumar --- "Digital Image Processing", TMH Publication 5th Edition
4. Milan Sonka, Vaclav Hlavac, Roger Boyle --- "Image Processing, Analysis and Machine Vision", CENGAGE Learning, 4th Edition

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15

Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Business Communication
Course Code: GED-411, Credit: 2.0, Total hours: 56

Course Description:

1. Introduction to Communication; history of Communication
2. Fundamentals of communication process, business communication skills for solving engineering problems;
3. Principle of Effective Communication; Classification and Methods of Communication, Media of Communication
4. Communication Across Culture; Effective Meeting: Written Communication; Office Memos
5. Technology-Enabled Communication: Short Essay writing on Topics relating to Business World; Fundamentals of Report Writing.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	1. **Understand** the business communication theory to solve workplace communication issues	C2				PO1
CO 2	2. **Develop** the communication skills required in the workplace to solve different business project problem.	C2				PO1
CO 3	3. **Apply** complex business communication ideas accurately in written and spoken formats to accelerate the process of solving business problems.	C3			CT	PO2

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1									M	S		
CO2		M			W				W	S	M	
CO3		M	M							M		

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	**The Basics of Communication**	1,2	1,2	Elements of Communication process; Steps in communication process
2	**The Basics of Communication**	1,2	1,2	Importance and Benefits of Effective Communication; Factors affecting Communication
3	**The Basics of Communication**	1,2	1,2	Potential communication barriers and ways to remove them.
4	**Classification and Methods of Communication**	1,2	1,2	Two Ways, External, Internal, Formal, Informal, Oral, Written Communication, Visual, Audiovisual and Computer based communication, Mass Communication
5	**Classification and Methods of Communication**	1,2	1,2	Communication patterns, Downward Communication: Objectives Media, Limitations, and essentials of effective Downward communication;
6	**Classification and Methods of Communication**	1,2	1,2	Horizontal communication, Grapevine: How the grapevine consensus process, Advantages and Limitations of consensus.
7	**Classification and Methods of Communication**	1,2	1,2	Upward Communication: Importance, methods, Limitations, and Essentials of effective Upward communication
8	Class Test 1			Lecture 1 - 7
9	**Media of Communication**	1,2	1, 2	Written Communication: purpose, Merits and Limitations;
10	**Media of Communication**	1,2	1, 2	Oral Communication: Importance, Uses, Preparation, Delivery, Style of Delivery, Speech, Face-to- face conation, Interviews, meeting, Telephonic Conversation, limitations, Do's and Don'ts in Oral Communication
11	**Media of Communication**	1,2	1, 2	Audio-Visual Communication: Merits and Limitation. Non-verbal Communication: Merits and Limitations Symbols, Gestures, Nods, Body Language, Visual Aids in Communication,

				How to Make Audio-Visual Communication Effective.
12	**Principle of Effective Communication**	1,2	1,2	The Seven C's of Effective Communication: Completeness, Conciseness, Consideration, Concreteness, Clarity, Courtesy, and Correctness.
13	**Effective Meeting**	1,2	1, 2	Preparation, Conduction, Strategy and Follow-up-Agenda and Minutes-Effective versus Ineffective groups, Small Group Communication.
14	**Communication Across Culture**	1,2	1, 2	How Culture Affects Business Communication; Cultural Barriers to Communication
15	**Communication Across Culture**	1,2	1, 2	Guidelines for effective multicultural communication; Intercultural Communication Model; National Cultural Variables; Individual Cultural Variables.
16	Logic Gates and Circuits	1,2	1,2	Design or Simplify complex circuit expressions
17	Universal Gates & Circuits	1,2	2	Explain NAND, NOR gates
18	Universal Gates & Circuits	1,2	2	Design complex circuit expressions using NAND, NOR gate
19	Class Test 2			Lecture 9 – 18
20	**Office Memos**	1,2	2	Memorandum-Meaning, Functions, Importance, uses, Procedures of writing memorandum, Notice.
21	**Written Communication**	1,2	2	Fundamentals of Business Writing, Adaptation and The Selection of Words,
22	**Written Communication**	1,2		The Planning of Letters, Selecting Formats, Characteristics of Business Letter, Classification of Business Letter,
23	**Written Communication**	1,2	2	purpose of Various Letter, Functions of First, Middle and Last Paragraph, Inquiry Letter and reply to inquiry letter,
24	**Written Communication**	1,2	2	Order placement, Acknowledgement And decline

				Letter, Persuasive Letter ,Complaint Letter,
25	**Written Communication**	3	3	Collection letter, Claim Letter, Sales Letter,
26	**Written Communication**	3	3	Congratulation Letter, Regret Letter,
27	Class Test 3			Lecture 1 – 24
28	**Written Communication**	3	3	Writing CV or Resume, Application for job/Cover letter, Joining letter, Resignation Letter, Winning interview.
29	**Fundamentals of Report Writing**	3	3	Definition; Characteristics and Types of Reports; Determining the Report Purpose, Determining the Factors,
30	**Fundamentals of Report Writing**	3	3	Gathering The Information Needed, Interpreting The Findings, Organizing The Report information, Writing The Report,
31	**Fundamentals of Report Writing**	3	3	Collaborative Report Writing. An overview of Report structure; Characteristics of The Shorter Report; Forms of Shorter Reports; Special Reports Forms;
32	**Fundamentals of Report Writing**	3	3	Organization and Content of the longer Reports, the prefatory parts, The Report proper, Structural Coherence Helpers,
33	**Fundamentals of Report Writing**	3	3	The Long Analytical Report, Planning the Graphics, placing The Graphics in The Report.
34	Assignment and Presentation 1	3	3	
35	**Short Essay writing on Topics relating to Business World**	3	3	Using Social media to connect to business world
36	**Technology-Enabled Communication**	3	3	Tools for Constructing Messages, Tools for Presenting Messages, Tools for Transmitting Messages, Tools for Collaboration.
37	Assignment and Presentation 2	3	3	
38	Class Test 4			Lecture 32 – 38
39	Review Class, Problem Solutions and Suggestion	3	3	Lecture 1 – 12
40	Review Class, Problem Solutions and Suggestion			Lecture 13 – 24

41	Review Class, Problem Solutions and Suggestion			Lecture 25 – 36
42	Review Class, Problem Solutions and Suggestion			Lecture 1-36

Reading Reference:

1. Lesikar, R., & Flatley, M. (2002). Basic business communication. Boston: McGraw-Hill/Irwin.
2. Murphy, H. (2008). Effective Business Communication. Tata McGraw Hill Education Private Limited.
3. Guffey, M. (2018). Business Communication. Toronto: Cengage.

Assessment Methods:

Category	Marks %
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Integrated Design Project /Capstone Project (Part 1)/ Capstone Project (Part 2)

Course Code: CSE-301/CSE-401/CSE-402, Credit: 3.0, Total hours: 42

Rationale:

Culminating demonstration of skills and knowledge achieved to date to apply and solve real life problems solvable through computer technology.

Objective:

To apply technical knowledge and skills for further research and design of computer system at professional engineering scale.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Develop systems' requirement specification from top-level customer requirements.

2. Analyze and compare design alternatives, at the system and subsystem levels, and use measures of performance or other criteria to rank alternatives.
3. Plan and organize an engineering design project using tools such as Gantt charts to develop a work breakdown structure, develop a schedule including milestones, and estimate effort and costs incorporating the ethical, financial and environmental issues.
4. Develop a design concept and elaborate it through to a detailed design by decomposing a system concept into component subsystems, identifying the subsystem requirements and applicable standards, and defining interfaces between the subsystems.
5. Build prototypes of key subsystems.

Course Content:

Knowledge Acquisition:

Introduction and brief with software, Discussion and submission Project Proposal, Learning Version Control System: GitHub, Interface design, Database Connectivity: Sqlite, Json, Cloud, Firebase database, Google Api, Sensor.

Implementation:

Idea Submission, Objective, Methodology, Literature Review, High Level Design, Low Level Design, Evaluation and feedback, Design & Partial Implementation (Prototype/Demo).

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
Develop system requirements from top-level customer requirements.	√	√				√						
Analyze and compare design alternatives, at the system and subsystem levels, and use measures of performance or other criteria to rank alternatives.		√	√		√							
Plan and organize an engineering design project using tools such as Gantt charts to develop a work breakdown structure, develop a schedule including milestones, and estimate effort and costs.				√		√						
Develop a design concept and elaborate it through to a detailed design by decomposing a system concept into component subsystems, identifying the subsystem requirements and applicable standards, and defining interfaces between the subsystems			√	√								
Build prototypes of key subsystems.					√				√			

Lecture Schedule:

Week	Topics	Remarks
1	Introduction and brief with software and Tools	
2	Learning Version Control System: GitHub	
3	Interface Design, activity and navigation.	
4	Database Connectivity: Sqlite, Json, Cloud* Firebase database	
5	Google Api	
6	Sensor	
7	Database Integration with the application	
8-9	Topic Selection and Project Plan	
10-11	Objective, Methodology, Literature Review	
12-14	Design & Partial Implementation (Prototype/Demo)	

Reading Reference:

1. Android Programming: The Big Nerd Ranch Guide (3rd Edition) (Big Nerd Ranch Guides) 3rd Edition
2. Professional Android, Reto Meier, Ian Lake; 4th Edition
3. Website: https://www.tutorialspoint.com/android/

Assessment Methods:

Category	Marks (%)
Project	40
Quiz	20
Viva/Presentation	10
Class Participation	10
Home Assignment /Report	10
Class performance/observation	10
Total	**100**

LEVEL-4 TERM-II

Course Title: Artificial Intelligence
Course Code: CSE 421, Credit: 3.0, Total hours: 42

Course Description:

1. To introduce to the basic concepts of Artificial Intelligence, with illustrations of current state of the art research and applications;

2. To provide a basic exposition to the goals and methods of Artificial Intelligence;

3. To recognize the characteristics of AI that make it useful to real-world problems;

4. To identify the type of an AI problem (search inference, decision making under uncertainty, game theory etc.);

5. To describe the strengths and limitations of various state-space search algorithms, and choose the appropriate algorithm;

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

Course Outcome Description		Level of Domain				PO
		C	P	A	S	
CO1	**Exhibit** strong familiarity with a number of important AI techniques, including in particular search, knowledge representation, and planning and constraint management.			A2		PO1, PO6
CO2	**Interpret** the modern view of AI as the study of agents that receive percepts from the environment and perform actions		P4			PO2, PO7
CO3	**Assess** critically the techniques presented and **apply** them to real world problems.	C3			S2	PO3, PO4
CO4	**Develop** self-learning and research skills to tackle a topic of interest on his/her own or as part of a team.	C6				PO9, P10

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	M					M						
CO2		M					S					
CO3			S	S								
CO4									M	M		

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1.	Introduction and Intelligent systems, What Is AI, The Foundations of Artificial Intelligence	1	1	Idea about Artificial Intelligence and its purpose
2.	The History of Artificial Intelligence, Applications of A.I.	1	1	Recalling the history and discuss various applications of AI
3.	Agents and Environments, Good Behavior: The Concept of Rationality	1	1	Understanding agents and difference between Rational and Irrational Behavior

4.	The Nature of Environments, The Structure of Agents	1, 2	1	How agents perceive the environment
5.	Review			Lecture 1 – 4
6.	Practice			Lecture 1 – 4
7.	Class Test 1			Lecture 1 - 4
8.	How the components of agent programs work	2	1	Workflow of various types of agents
9.	Solving Problems by Searching, Study and analysis of Various searching algorithms	2	1	Importance of Searching in AI
10.	Implementation of Depth-first search Problem Solving Agents, Searching for Solutions	2	1	Application of DFS in AI fields
11.	Uninformed Search Strategies: Breadth-first search, Uniform-cost search, Depth-first search	1, 2	1, 2	Discussion of BFS and difference between BFS, DFS. Importance of Uniform Cost Search
12.	Review			Lecture 8 - 11
13.	Practice			Lecture 8 - 11
14.	Depth-limited search, Iterative deepening depth-first search, Bidirectional search	2	2	Usage and application of DLS, IDS and Bidirectional search
15.	Informed (Heuristic) Search Strategies: Greedy Best First search	3	2	How Heuristic functions are utilized by AI agents
16.	A* search: Minimizing the total estimated solution cost, Conditions for optimality: Admissibility and consistency, Optimality of A*	3	1, 2	Optimal Path Finding using A*
17.	Memory-bounded heuristic search, Heuristic Functions	2, 3	Internet	Classification of Heuristic Search and Heuristic functions
18.	Review			Lecture 14 - 17
19.	Practice			Lecture 14 - 17
20.	Class Test 2			Lecture 14 - 17
21.	Generating admissible heuristics from sub problems: Pattern databases, Learning heuristics from experience	3	2, Internet	Detailed analysis of Heuristic Functions and methods to create them
22.	Local Search Algorithms and Optimization Problems: Hill-climbing search, Simulated annealing	4	1, Internet	Local Search Algorithms to find solution for AI related problems
23.	Local beam search, Genetic algorithms, Local Search in Continuous Spaces	4	Internet	Application of AI in Genetic Research and solving Continuous Space related problems
24.	Searching with Non-deterministic Actions: AND-OR search trees, Searching with Partial Observations	4	1, Internet	Utilizing different search trees for Non-Deterministic actions
25.	Review			Lecture 21 - 24
26.	Practice			Lecture 21 - 24
27.	Adversarial Search and Constraint Satisfaction Problems, Study of	3	1, 3	Finding best route for 1versus1 gaming.

	minimax algorithm			
28.	Adversarial Search: Games, Optimal Decisions in Games, The minimax algorithm, Optimal decisions in multiplayer games	4	1, 2, 3	Applying different adversarial search in games
29.	Alpha--Beta Pruning, Move ordering, Imperfect Real-Time Decisions	4	2, 3	Pruning trees for better performance/time complexity and space complexity
30.	Class Test 3			Lecture 27 - 29
31.	Partially Observable Games Constraint Satisfaction Problems: Defining Constraint Satisfaction Problems, Variations on the CSP formalism	2	1	Discussion and application of CSP and its different variations
32.	Stochastic Games, Evaluation functions for games of chance	2		Developing Evaluation functions for game solving algorithms
33.	Constraint Propagation: Inference in CSPs, Backtracking Search for CSPs, Local Search for CSPs	4	1	More details and insight of CSP
34.	Alpha-beta pruning and CSP, Implementation aspects of minimax algorithm and CSP	4	1	Merging Alpha-Beta pruning with CSP and Minimax Algorithm
35.	Review			Lecture 31 - 34
36.	Practice			Lecture 31 - 34
37.	Class Test 4			Lecture 31 - 34
38.	Acting under Uncertainty, Basic Probability Notation, Inference Using Full Joint Distributions, Bayes' Rule and Its Use, Representing Knowledge in an Uncertain Domain	4	1, 4	Discussion on Uncertainty and its several effects in AI
39.	Learning from Examples: Forms of Learning, Supervised Learning, Learning Decision Trees, The decision tree representation, Expressiveness of decision trees, Inducing decision trees from examples	4	1, 4	Supervised Learning techniques and application of Decision Tree in Supervised Learning
40.	Natural Language Processing: Language Models, Steps in NLP, Syntactic Analysis (Parsing), Semantic interpretation, Discourse and pragmatic Processing, Text Classification. Discourse and pragmatic	4	1	Utilizing AI in NLP and different building steps of NLP
41.	Problem Solving			Lecture 1 - 40
42.	Overall Course Discussion			Lecture 1 - 40

Reading Reference:

1. Artificial Intelligence: A Modern Approach by Peter and Norvig ISBN-0-13- 1038052
2. Artificial Intelligence by Elaine Rich, Kevin Knight and Nair ISBN-978-0-07- 008770-5, TMH
3. Prolog Programming for A.I. by Bratko, TMH
4. Artificial Intelligence and Intelligent Systems by Padhy, Oxforfd University Press

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Artificial Intelligence Sessional

Course Code: CSE 422, Credit: 1.0, Total hours: 28

Rationale:

Hands on orientation with AI programming, intelligent agents along with how to representation, planning, learning and perception of knowledge of agents.

Objective:

1. To have general understanding of major concepts and approaches in knowledge representation, planning, learning, robotics and other AI areas.
2. To develop programming skills for AI applications and explore traditional AI techniques and algorithms.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Have general understanding of major concepts and approaches in knowledge representation, planning, learning, robotics and other AI areas.
2. Develop programming skills for AI applications.
3. Exposure to traditional AI techniques and algorithms.

Course Contents:

Introduction to Intelligent Machines, State Mapping, A* Search implementation, Local search Algorithm, Adversarial Search, Constraint Satisfaction Problem Solving.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes (CO) of the Course	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
Have general understanding of major concepts and approaches in knowledge representation, planning, learning, robotics							√					
Develop programming skills for AI applications.								√				
Exposure to traditional AI techniques and algorithms.											√	

Lecture Schedule:

Week	Lecture	Topics
1-2	Lec 1 Lec 2 Lec 3 Lec 4	Orientation with AI practical areas
3-4	Lec 5 Lec 6 Lec 7 Lec 8	Assignment on State Mapping
5-6	Lec 9 Lec 10 Lec 11 Lec 12	Assignment on Search Algorithm
7-8	Lec 13 Lec 14 Lec 15 Lec 16	Assignment on Local Search
9-10	Lec 17 Lec 18 Lec 19 Lec 20	Assignment on Adversarial Search
11-12	Lec 21 Lec 22 Lec 23 Lec 24	Assignment on Constraint Satisfaction Problem
13-14	Lec 25 Lec 26 Lec 27 Lec 28	Quiz/Viva

Assessment Methods:

Category	(100%)
Tasks	50
Quiz/Viva	10
Class Participation	10

Observation	10
Class performance	20
Total	**100**

Course Title: Sociology for Engineers
Course Code: GED 421, Credit: 2.0, Total hours: 28

Rationale:

Systematic study of social behavior and human groups. It focuses on the influence of social relationships upon people's attitudes and behavior and on how societies are established and changed. This course provides students with both methodologies and knowledge of the study of critical social issues ranging in scope from family to global.

Objective:

1. To correctly identify critical social issues through a systematic study of social behavior and social change.
2. To demonstrate comprehension of roles and functions of various social institutions, relationships, several sociological theories and apply them to explain social phenomena or situations.
3. To demonstrate interest in taking part in social activities and use sociological imagination to explain their life experience in a broader social context.

Course Outcomes (CO):

Upon completion of the course, the student will be able to demonstrate knowledge of the following topics:

1. Correctly identify causes of critical social issues through a systematic study of social behavior and social change.
2. Demonstrate comprehension of roles and functions of various social institutions and relationships among them.
3. Demonstrate understanding of several sociological theories and apply them to explain social phenomena or situations.
4. Demonstrate interest in taking part in social activities.

Course Content:

Nature, scope and perspectives of sociology; stages of social research and research methods; culture and civilization; socialization and personality development; globalization; media and individual; social organization and social problem; social stratification; industrial revolution, capitalism and socialism; work and economic life; environment and human activities; climate change and global risk; population and human society; urbanization and city development; social change and technology;

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes (CO) of the Course	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
Correctly identify causes of critical social issues through a systematic study of social behavior and social change.				√								
Demonstrate comprehension of roles and functions of various social institutions and relationships among them.		√										
Demonstrate understanding of several sociological theories and apply them to explain social phenomena or situations.								√				
Demonstrate interest in taking part in social activities.										√		

Lecture Schedule:

Week	Lecture	Topics	Class Test
1	Lec 1 Lec 2	Nature, Scope And Perspectives Of Sociology;	**Class Test 1**
2	Lec 3 Lec 4	Stages Of Social Research And Research Methods	
3	Lec 5 Lec 6	Culture And Civilization	
4	Lec 7 Lec 8	Socialization And Personality Development	**Class Test 2**
5	Lec 9 Lec 10	Globalization	
6	Lec 11 Lec 12	Media And Individual	
7	Lec 13 Lec 14	Social Organization And Social Problem	
8	Lec 15 Lec 16	Social Stratification	**Class Test 3**
9	Lec 17 Lec 18	Industrial Revolution	
10	Lec 19 Lec 20	Capitalism And Socialism	

11	Lec 21 Lec 22	Work And Economic Life Environment And Human Activities	**Class Test 4**
12	Lec 23 Lec 24	Climate Change And Global Risk	
13	Lec 25 Lec 26	Population And Human Society	
14	Lec 27 Lec 28	Urbanization And City Development Social Change And Technology	

Text and Ref Books:

1. "Sociology" – Richard T. Schaefer and Robert P. Lamm, MacGraw-Hill, N.O., 2002)
2. "Social Problems" - LeRoy W. Barnes, The Dushkin Publishing Group, Guilford, CT, 2000
3. "Sociology – Primary Principles" - CN Shankar Rao
4. "Sociology – A Guide to Problems & Literature" – Bottomore
5. "Sociology " – Samuel Koening

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Engineering Ethics
Course Code: GED 423, Credit: 2.0, Total hours: 28

Course Description:

1. Definition and scopes of Ethics, different branches of ethics, social change and the emergence of new technologies
2. History and development of engineering ethics, Science and technology necessity and application.
3. Study of ethics in engineering, applied ethics in engineering, Human qualities of an engineer., Obligation of an engineer to the clients, Attitude of an engineer to other engineers, Measures to be taken in order to improve the quality of engineering profession
4. Ethical expectation: Employers and employees, inter-professional relationship,

Professional Organization – maintaining a commitment of ethical standards. Desired characteristics of a professional code. Institutionalization of ethical conduct.

5. Case Studies.
6. Historical and philosophical background of patents and other intellectual property, The U.S. Patent System: The Constitution, Congress, Patent Office (PTO), and courts, Analyzing and understanding judicial opinions, Legal fundamentals of patent protection for useful inventions
7. Design and plant patents, Legal fundamentals of copyright protection, Similarity and access, Expression vs. ideas and information, merger, Fair use of copyrighted works, Contributory copyright infringement, Critical differences between patent and copyright protection, Copyright infringement distinguished from plagiarism, Legal fundamentals of trade-secret and trademark protection, Requirements and limitations of patentability
8. The process of applying for a patent, Actions for patent infringement, Patent licensing, Non- competition agreements, Rights and obligations among co-inventors, co-authors, employers, and licensees.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				
		C	P	A	S	PO
CO 1	**Identify** ethical problems, dilemmas, and areas of responsibility in engineering practice	C1				PO1
CO 2	**Develop** an idea to solve an ethical case in the field of engineering.			A4	CT	PO8

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO 11	PO 12
CO1	M											
CO2								S				

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Introduction and scope of Ethics	1	2	About Ethics and Technology and its effect on the society
2	History of Engineering Ethics	1	2	Details about the Ethics history
3	Empact of Science and Technology in the society	1	2	Change in the society
4	Ethics in engineering	1	2	Engineering ethics and quality of an engineer
5	Attitude to other engineers	1	2, slide 2	Obligations and responsibilities of an engineer
6	Ethical Theories	1	2, slide 3	Lawrence Kohlberg's theory of moral reasoning
7	Applied ethics	1	2	Definition, stages and implementation of applied ethics
8	Class Test - 01			Lecture 4-7
9	Ethical expectation	1	2	Organization culture and types of the companies
10	Whistle-blowing	1	Slide 4	Definition and types of whistle blowing, organizational disobedience
11	Inter-professional relationship	1	Slide 4	Maintaining a commitment of ethical standards
12	Professional Ethics and virtues	1	slide	Right ethics, Virtue ethics & Duty Ethics
13	Codes of Ethics	1	Slide 5	Eight essential roles of codes of ethics, limitations of codes of ethics
14	Case Studies	2	Slide 6	Discussion on some cases of engineering ethics
15	Case Studies	2	Slide 6	Discussion on some cases of engineering ethics
16	Case Studies	2	2	Analyzing some cases of engineering ethics
17	Case Studies	2	2	Analyzing some cases of engineering ethics
18	Class Test- 02			Lecture 14-16
19	Patent	1	Slide 7	Definition and type of Patent

20	Comparative study	1	Slide 7	Difference between Copyrights, Patent & Trade Secret
21	Patent Application	1	Slide 7	Procedure for filing an application in Bangladesh
22	The U.S. Patent System	1	Slide 8	The Constitution, Congress, Patent Office (PTO), courts, Analyzing and understanding judicial opinions
23	Types of Patent	1	3	Design Patent, Plant Patent & Utility Patent
24	Patent Licensing	1	3	Actions for Patent rules violations
25	Review Class			
26	Review Class			

Reading Reference:

1. Create or Perish: The Case for Inventions and Patents - Robert HRines
2. Engineering Ethics – Carles E Haris
3. System Safety Engineering and Management - Harold E. Roland , Brian Moriarty

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Elective 2

Course Title: Computer Graphics
Course Code: CSE 413, Credit: 3.0, Total hours: 42

Course Description:

1. Study of Standard graphics primitives; Graphics hardware; Graphics pipeline; Coordinate convention;

2. Modeling transformation; Viewing transformation; Projection transformation;
3. Scan conversion; Polygons and polygon meshes; Curves and surfaces; Hidden lines and surface removal;
4. Fractals; Ray tracing; Light models; Color models; Graphics programming.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Explain** the underlying principles of different graphical transformations.	C2				PO1
CO 2	**Identify** the necessary steps of rasterization pipeline.	C2			CT	PO1
CO 3	**Compare** and **select** suitable modeling algorithms based on their advantage and disadvantages.	C3			CT	PO2
CO 4	**Propose** and **justify** different approaches for designing graphical models.			A3		PO9

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	S											
CO2	S											
CO3	S											
CO4			M									

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Introduction to computer graphics and rasterization pipeline	1	1	Recognize basic steps of rasterzation pipeline
2	Basic concepts of vector operations for computer graphics	1	2	Describe vector operations needed for computer graphics
3	Representation of Lines and Planes	1	2	Convert one representation to another
4	Intersection derivation of lines and planes	1	2	Solve problems to derive line and plane intersection
5	Transformation and homogenous co-ordinates	1,2	1,3	Identify modeling transformation.

6	Translation and scaling	1	1,3	Explain and solve translation and scaling problems
7	**Review**			Lecture 1 - 6
8	Class Test 1			Lecture 1 - 6
9	Rotation, Shear and Mirror reflection	1	1,3	Explain and solve rotation, shear and mirror reflection problems
10	Construction of composite transformations	1	1, 3	Construct composite transformation from simple ones
11	Vector alignment and Rodrigues formula	1	1	Solve alignment problems
12	Problem Solving			Lecture 9 – 11
13	Taxonomy of projection transformation.	1,2	1	Recognize different types of projection transformation
14	Perspective projection	1,2	1, 3	Identify and explain perspective projection models
15	Parallel projection	1,2	1, 3	Identify and explain parallel projection models
16	Viewing transformation	1,2	1	Construct viewing transformation matrices
17	Problem Solving			Lecture 13 – 16
18	Review			Lecture 9 – 16
19	Class Test 2			Lecture 9 – 16
20	Hidden surface removal algorithms: back face culling, painter's algorithm, z-buffer	1,3	1	Compare and apply algorithms for different sceneriosCompare and apply algorithms for different scenarios
21	Scan-line algorithm, BSP tree, depth-sort algorithm	3	1	Compare and apply algorithms for different scenarios
22	Problem Solving			Lecture 20 – 21
23	Clipping: Cohen-Sutherland algorithm	3	1	Explain the algorithm
24	Cyrus-Beck and Sutherland-Hodgman algorithm	3	1	Explain and compare with previous algorithms
25	Anti-aliasing	1	1	Recognize the process of anti-aliasing
26	Problem Solving			Lecture 23 – 25
27	Review			Lecture 20 – 25
28	Class Test 3			Lecture 20 – 25
29	Principles of ray-tracing	1	1	Describe the basic principles

30	Ray-plane and ray-sphere intersection	1	1	Identify and solve intersection problems
31	Illumination	1,3	1	Explain and compare different illumination models
32	**Shading**	1,3	1	Explain and compare different shading models
33	Curve and textures	1	1	Describe different curves and texture
34	Fractals	1	1	Recognize and construct complex fractals from simple ones
35	Project Assignment and Presentation	4		
36	Introduction to scan conversion	1	1	Recognize simple scan conversion algorithms
37	Midpoint line algorithm	3	1	Solve line scan conversion problems
38	Midpoint circle algorithm	3	1	Solve circle scan conversion problems
39	Problem Solving			Lecture 29 – 34 and Lecturer 36 – 38
40	Review			Lecture 29 – 34 and Lecturer 36 – 38
41	Class Test			Lecture 29 – 34 and Lecturer 36 – 38
42	Review Class, Problem Solutions and Suggestion			Lecture 1 – 41

Reading Reference:

1. Computer Graphics Principle and Practice -- James D Foley, Van Dam
2. Computer Graphics Using Open GL – F S Hill J R
3. Schaum's Computer Graphics - Zhigang Xiang, Roy Plastock

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: VLSI
Course Code: CSE 411, Credit: 3.0, Total hours: 42

Course Description:

1. VLSI Design Methodology: Top-down Design Approach, Technology Trends and Design Automation Algorithms;
2. Introduction to CMOS Inverters and Basic Gates; CMOS Fabrication Process and Layout; CMOS Circuit Characteristics and Performance Estimation; Buffer Circuit Design;
3. Introduction Bi-CMOS Circuits; Complex CMOS Gates;
4. CMOS Building Blocks - Adder, Comparator, Multiplier, Counter, and Shifter; Data Path and Memory structures.
5. Design Methodology and Tools; Hardware modeling - Hardware Modeling Languages, Logic Networks, State Diagrams, Data-flow and Sequencing Graphs, Behavioral Optimization; Floor Planning and Architecture Design; ASIC design using FPGA and PLD.

Course Outcome (CO) Matrix:
At the end of the course, the students should be able to:

		Level of Domain				
		C	P	A	S	PO
CO 1	**Acquire** qualitative knowledge about the fabrication process of integrated circuit using MOS transistors	C2				PO1
CO 2	**Draw** the layout of any logic circuit which helps to understand and estimate parasitic of logic circuit.	C2				PO2
CO 3	**Design** different types of logic gates using CMOS inverter and A**nalyze** their transfer characteristics.	C3			CT	PO3
CO 4	**Design** simple logic circuits using PLA, PAL, FPGA	C3			CT	PO3

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak											
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	S											
CO2	S	S										
CO3		S	S									
CO4		S	S									

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
1	Introduction to VLSI	1	3	Learn about the history of different types of integration circuit.
2	Top-down Design Approach and Technology Trends	1	3	Relate different types of design approaches and trends.
3	MOS devices and Basic Circuits.: Mos structure and conduction.	1	3	Identify various kinds of Transistor and know about the metal oxide semiconductor.
4	MOS devices and Basic Circuits.: conduction.	1	3	Learn about the Structure, conditions of MOS circuit.
5	MOS devices and Basic Circuits.: Threshold voltage	1	3	Learn about the Structure, conditions of MOS circuit.
6	MOS devices and Basic Circuits.: Ids vs Vds characteristic for NMOS Devices.	1	3	Learn about current flow of NMOS circuit
7	MOS devices and Basic Circuits.: NMOS inverter	1,2	3	Define about the characteristics of NMOS inverter with circuit diagram.
8	MOS devices and Basic Circuits.: NMOS inverter	1,2	3	Define about the characteristics of NMOS inverter with circuit diagram.
9	MOS devices and Basic Circuits.: CMOS inverter	1,2	3	Have an understanding of the characteristics of CMOS inverter circuit construction.
10	MOS devices and Basic Circuits.: Pass transistor	1,2	3,4	Have an understanding of the characteristics of NMOS, CMOS pass transistor circuit construction.
11	MOS devices and Basic Circuits.: Pass transistor	1,2	3,4	Have an understanding of the characteristics of NMOS, CMOS pass transistor circuit construction.
12	MOS devices and Basic Circuits.: Buffer Circuits Design	1,2	3,4	Demonstrate buffer circuit to
13	Review			Lecture 1 – 12
14	Class test 1			Lecture 4- 12
15	Fabrication & Design: fabrication process of Mos wafer	1,2	3	Learn the fabrication process of basic Mos wafer.

16	Fabrication & Design: fabrication process of Mos wafer	1,2	3	Learn the fabrication process of basic Mos wafer.
17	Fabrication & Design: NMOS inverter fabrication	1,2	3	Learn the fabrication process of NMOS inverter.
18	Fabrication & Design: NMOS inverter fabrication	1,2	3	Learn the fabrication process of NMOS inverter.
19	Fabrication & Design: CMOS processing	1,2	3	Learn the fabrication process of CMOS n-well and p-well.
20	Geometric Layout & Stick Diagram	2	3,4	Be able to design static CMOS combinational and sequential logic at the transistor level, including mask layout.
21	**Drawing geometric layout and stick diagram of NMOS and CMOS inverter.**	2	3,4	Be able to design static CMOS combinational and sequential logic at the transistor level, including mask layout.
22	Review			Lecture 15 – 21
23	Class test 2			Lecture 15 -19
24	Logical Circuit Design: 2-input NAND gate	1,2,3	3,4	Be able to design CMOS combinational and sequential logic at the transistor level of NAND gate, including mask layout
25	Logical Circuit Design: 2- input NOR gate	1,2,3	3,4	Be able to design static CMOS combinational and sequential logic at the transistor level, including mask layout.
26	Drawing of complex logic function	1,2,3	3,4	Be able to design static CMOS combinational and sequential logic at the transistor level, including mask layout.
27	Drawing of complex logic function	1,2,3	3,,4	Be able to design static CMOS combinational and sequential logic at the transistor level, including mask layout.
28	Drawing of complex logic function	1,2,3	3,4	Be able to design static CMOS combinational and sequential logic at the transistor level, including mask layout.
29	Review			Lecture 24 – 26
30	Class test			Lecture 20-21, 24– 26
31	CMOS Building Blocks – Parity Generator,	1,2,3	4	Design functional units including adders, multipliers, Parity Generator, Shifter
32	CMOS Building Blocks –Multiplexer	1,2,3	44	Design functional units including adders, multipliers, Parity Generator, Shifter

33	CMOS Building Blocks – Adder, Multiplier,	1,2,3		Design functional units including adders, multipliers, Parity Generator, Shifter
34	CMOS Building Blocks –Shifter.	1,2,3	4	Design functional units including adders, multipliers, Parity Generator, Shifter
35	Class test 4			Lecture 31 – 34
36	Data-flow and Sequencing Graphs	1,2	4	To introduce the concepts and techniques of modern integrated circuit design and testing
37	Behavioral Optimization	1,2	4	To introduce the concepts and techniques of modern integrated circuit design and testing
38	Floor Planning and Architecture Design	1,2	4	To introduce the concepts and techniques of modern integrated circuit design and testing
39	Floor Planning and Architecture Design	1,2	4	To introduce the concepts and techniques of modern integrated circuit design and testing
40	PLA & PAL	3	4	Design functional units including adders, logic gate, and PLAs
41	Review			Lecture 36– 39
42	Review Class, Problem Solutions and Suggestion			Lecture 1-40

Reading Reference:

1. Modern VLSI Design – System-on-Chip Design - Wayne Wolf.
2. CMOS VLSI Design- A Circuit and System Perspective - Neil H.E. Weste, David Harris, and Ayan Banerjee.
3. Design of VLSI Systems — A Practical Introduction - Linda E. M. Brackenbury
4. Basic VLSI Design - Douglas A. Pucknell and Kamran Eshraghian

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Elective 3

Course Title: Data Mining
Course Code: CSE 441, Credit: 2.0, Total hours: 28

Course Description:

1. Introduction; Data preprocessing; Data mining primitives, languages and systems;
2. Descriptive data mining: characterization and comparison;
3. Association analysis;
4. Predictive Data Mining: Classification and prediction;
5. Descriptive data mining model.
6. Mining complex types of data; Data warehousing and OLAP technology for data mining
7. Applications and trends in data mining.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Comprehend** the basic concepts and characteristics of Data Warehouse fundamentals, Data Mining Principles	C2				PO1
CO 2	**Distinguish** different data mining techniques like classification, prediction, clustering and association rule mining	C2			CT	PO1
CO 3	**Identify** appropriate data mining algorithms to solve real world problems	C1			CT	PO2
CO 4	**Design** data warehouse with dimensional modeling and apply OLAP operations	C6			TW, CT	PO3

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

Mapping of CO and Program Outcomes:

COs	Program Outcomes (POs) (Appendix 1) S = Strong, M = Medium and W = Weak

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
CO1	S											
CO2	S											
CO3		S										
CO4			M									

Lecture Outline:

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/Activities
1	KDD, Kinds of Data, Data Characterization, Data Discrimination Classification, Regression, Clustering, Outlier	1	1	Learning through the Class lecture regarding the topics
2	Feature Vector, Supervised Learning, Unsupervised Learning, Semi Supervised Learning Active Learning, Lazy Learning, Attribute	2	1	Learning through the Class lecture regarding the topics
3	Central Tendency, Measure of Dispersion,	3	1	Learning through the Class
	Data Visualization, Histogram			lecture regarding the topics And Assignment on Class up to now
4	Similarity/Dissimilarity	1, 3	1	Learning through the Class lecture regarding the topics
5	Similarity/Dissimilarity	1	1	Learning through the Class lecture regarding the topics
6	Scalar Plot, Attribute of measuring the quality of data, major tasks in Data Pre-Processing	1	1	Learning through the Class lecture regarding the topics And Assignment on Class up to now
7	Revise			Class 1 – 6
8	Class Test and VIVA 01			Class 1 – 6

9	Data Cleaning Process, Handling of Noisy Data, Entity Identification, Redundancy Problem	1, 2	1	Learning through the Class lecture regarding the topics
10	Correlation-Nominal Attributes, Correlation Coefficient for Numeric Data, Correlation vs. Covariance, Data Value Conflict	1, 2	1	Learning through the Class lecture regarding the topics
11	Correlation-Nominal Attributes, Correlation Coefficient for Numeric Data, Correlation vs. Covariance, Data Value Conflict	1, 2	1	Learning through the Class lecture regarding the topics
12	Histogram and Sampling, Data Reduction Algorithms Wavelet and PCA	1, 2	1	Learning through the Class lecture regarding the topics
13	Histogram and Sampling, Data Reduction Algorithms Wavelet and PCA	1, 2	1	Learning through the Class lecture regarding the topics
14	Data Normalization, Attribute Subset Selection, Data Transformation, Concept Hierarchy.	1,2,4	1	Learning through the Class lecture regarding the topics And Assignment on Class up to now
15	Data warehouse vs. Operational Data	4	1	Learning through the Class
	Bases, Need for separate Data warehouse, 3 tier model of Data warehouse			lecture regarding the topics
16	Models of Data warehouse, Face Table, Dimension Table, Meta Data Repository,	4	1	Learning through the Class lecture regarding the topics
17	Data Cube Measures of Computation, Data model for Data Warehouse	4	1	Learning through the Class lecture regarding the topics And Assignment on Class up to now
18	Revise			Up to Class 17
19	Class Test and VIVA 02			Up to Class 17
20	OLAP Operations, Concept Hierarchies, 4 Views of Data Warehouse	4	1	Learning through the Class lecture regarding the

				topics
21	Data Warehouse Design process, Types of Information Processing,	4	1	Learning through the Class lecture regarding the topics
22	Bitmap Indexing ROLAP, MOLAP, HOLAP,	3, 4	1	Learning through the Class lecture regarding the topics And Assignment on Class up to now
23	Data generalization by Attribute-Oriented Induction	3, 4	1	Learning through the Class lecture regarding the topics And Assignment on Class up to now
24	Apriori Algorithm, Support, Confidence	3, 4	1	Learning through the Class lecture regarding the topics
25	FP Algorithm and FP Tree,	3, 4	1	Learning through the Class lecture regarding the topics
26	Overview between these FP Algorithm and FP Tree for solving real life problems	3, 4	1	Learning through the Class lecture regarding the topics And Assignment on Class up to now
27	Revise			Up to Class 26
28	Class Test and VIVA 03			Up to Class 26
29	Revise			Up to All
30	Classification, Naïve Bayes	2,3	1	Learning through the Class lecture regarding the topics
31	ANN, Activation Function, Single Layer Perceptron	2,3	1	Learning through the Class lecture regarding the Topics
32	BPNN SVM [Basic Concept]	2,3	1	Learning through the Class lecture regarding the topics

33	Decision Tree, ID3,	2,3	1	Learning through the Class lecture regarding the topics
34	C4.5, CART	2,3	1	Learning through the Class lecture regarding the topics
35	Matrices for Evaluating , Classifier Performance, Parameters	2,3	1	Learning through the Class lecture regarding the topics And Assignment on Class up to now
36	K means neighboring algorithm, Different Methods	2,3	1	Learning through the Class lecture regarding the topics
37	K means neighboring algorithm, Different Methods	2,3	1	Learning through the Class lecture regarding the topics
38	Revise			Up to Class 37
39	Class Test and VIVA 04			Up to Class 37
40	Revise			Up to All
41	Revise			Up to All
42	Revise			Up to All

Reading Reference:

1. Data Mining - Concepts and Techniques (3rd Ed) - Jiawei Han, Micheline Kamber, Jian Pei

Assessment Methods:

Category	**Marks %**
Class Participation/ Observation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

Course Title: Data Mining Sessional
Course Code: CSE 442, Credit: 1.0, Total hours: 14

Rationale:

Implementing data analysis methods, visualization and decision making concepts of data ware housing and data mining.

Objective:

1. Gain a working knowledge of the strengths and limitations of modern data mining methods (algorithms).
2. Learn to identify problems that can profitably be addressed via data mining methods.
3. Learn how to set up data for data mining experiments.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

1. Gain a working knowledge of the strengths and limitations of modern data mining methods (algorithms).
2. Learn to identify problems that can profitably be addressed via data mining methods.
3. Learn how to set up data for data mining experiments.

Course Content:

Data warehousing and OLAP technology, Data preprocessing, Data visualization techniques, Data Cube computation and multidimensional data analysis, Association analysis; Classification and prediction; Cluster analysis; Mining complex types of data.

Mapping of Course Outcomes (CO) and Program Outcomes:

Course Outcomes(CO) of the Course	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
Gain a working knowledge of the strengths and limitations of modern data mining methods (algorithms).	√											
Learn to identify problems that can profitably be addressed via data mining methods.		√										
Learn how to set up data for data mining experiments.			√									

Lecture Outline:

Week	Lecture	Topics	Remarks
1-2	Lec 1 Lec 2 Lec 3 Lec 4	Orientation lab Practical on Data warehousing	
3-4	Lec 5 Lec 6 Lec 7 Lec 8	Implementing Data preprocessing techniques	Assignment
5-6	Lec 9 Lec 10 Lec 11 Lec 12	Implementing Data visualization techniques Implementing association analysis	Assignment
7-8	Lec 13 Lec 14 Lec 15 Lec 16	Practice on Classification and prediction;	Lab Test 1
9-10	Lec 17 Lec 18 Lec 19 Lec 20	Practice on Cluster analysis	Assignment
11-12	Lec 21 Lec 22 Lec 23 Lec 24	Final Lab Test	
13-14	Lec 25 Lec 26 Lec 27 Lec 28	Final Quiz Viva	

Reading Reference:

1. Data Mining: Concepts and Techniques - Jiawei Han, MichelineKamber, and Jian Pei, (Morgan Kaufmann)
2. Data Mining and Data Warehousing - Bharat BhushanAgarwal, SumitPrakashTayal
3. Data Warehousing, Data Mining, and OLAP - Alex Berson and Stephen J. Smith

Assessment Methods:

Category	Marks %
Lab test	40
Quiz	20
Viva	10
Class Performance	10
Report	10

Class Assessment	10
Total	**100**

APPENDIX A

Equivalence Table

	Old Course (2015-2021)			**New Course (2021-20**)**		
Ser.	**Course Code**	**Course Title**	**Cr.**	**Course Code**	**Course Title**	**Cr.**
1.	HUM-101	English	2.00	ENG 111	English Reading and Speaking	2
2.	-	-	-	ENG 121	English Writing and Listening	3
3.	-	-	-	HUM 211	Bangla Language	2
4.	-	-	-	GED 121	History of the Emergence of Bangladesh	2
5.	-	-	-	GED 211	Finance and Investment	3
6.	-	-	-	GED 221	Engineering Economics	3
7.	-	-	-	GED 411	Business Communications	3
8.	HUM-103	Sociology	2.00	GED 421	Engineers for Society, Environment and Sustainability	2
9.	CSE-417	Engineering Ethics and Management	3.00	GED 423	Engineering Ethics	2
10.	CHEM-101	Chemistry	3.00	CHE 111	Chemistry	3
11.	-	-	-	CHE 112	Chemistry Sessional	1
12.	PHY-103	Physics	3.00	PHY 121	Mechanics, Wave, Heat and Thermodynamics	3
13.	PHY-104	Physics Sessional	0.75	PHY 124	Basic Physics Sessional	1
14.	-	-	-	PHY 211	Electromagnetism, Optics and Modern Physics	3
15.	MATH-141	Mathematics-I (Differential Calculus and Integral Calculus)	3.00	MATH 111	Differential and Integral Calculus	3
16.	MATH-143	Mathematics-II (Ordinary and Partial Differential Equations and Coordinate Geometry)	3.00	MATH 121	Coordinate Geometry and Vector Analysis	3
17.	MATH-245	Mathematics-III (Vector Analysis, Matrices and Fourier Analysis)	3.00	MATH 211	Linear Algebra and Complex Analysis	3
18.	CSE-313	Applied Statistics for Computer Science	3.00	MATH 221	Probability and Statistics	3
19.	-	-	-	CE 311	Engineering Drawing	1
20.	CE-150	Engineering Drawing & CAD Sessional	1.50	CE 312	Engineering Drawing & CAD Sessional	1
21.	EEE-163	Electrical Circuit Analysis	3.00	EEE 111	Electrical Circuit Analysis	3
22.	EEE-164	Electrical Circuit Analysis Sessional	1.50	EEE 112	Electrical Circuit Analysis Sessional	1.5
23.	EEE-169	Electronic Devices and Circuits	3.00	EEE 121	Electronic Devices and Circuits	3
24.	EEE-170	Electronic Devices and Circuits Sessional	1.50	EEE 122	Electronic Devices and Circuits Sessional	1.5

25.	CSE-102	Computer Fundamentals and Basic Programming Sessional	1.50	CSE 112	Introduction to Computing Sessional	1.5
26.	CSE-103	Discrete Mathematics	3.00	CSE 113	Discrete Mathematics	3
27.	CSE-105	Structured Programming Language	3.00	CSE 121	Structured Programming Language	3
28.	CSE-106	Structured Programming Language Sessional	1.50	CSE 122	Structured Programming Language Sessional	1.5
29.	CSE-203	Data Structures	3.00	CSE 211	Data Structures	3
30.	CSE-204	Data Structures Sessional	1.50	CSE 212	Data Structures Sessional	1.5
31.	CSE-205	Object Oriented Programming Language	3.00	CSE 213	Object Oriented Programming Language	3
32.	CSE-206	Object Oriented Programming Language Sessional-I (C++)	1.50	CSE 214	Object Oriented Programming Language Sessional	1.5
33.	CSE-215	Algorithms	3.00	CSE 221	Computer Algorithms and Complexity	3
34.	CSE-216	Algorithms Sessional	1.50	CSE 222	Computer Algorithms and Complexity Sessional	1.5
35.	CSE-201	Digital Logic Design	3.00	CSE 223	Digital Logic Design	3
36.	CSE-202	Digital Logic Design Sessional	1.50	CSE 224	Digital Logic Design Sessional	1.5
37.	CSE-311	Numerical Analysis	3.00	CSE 225	Numerical Methods	3
38.	CSE-312	Numerical Analysis Sessional	0.75	CSE 226	Numerical Methods Sessional	1
39.	CSE-307	Operating System	3.00	CSE 311	Operating System	3
40.	CSE-308	Operating System Sessional	0.75	CSE 312	Operating System Sessional	1
41.	CSE-301	Database Management Systems	3.00	CSE 313	Database Management System	3
42.	CSE-302	Database Management Systems Sessional	1.50	CSE 314	Database Management System Sessional	1.5
43.	CSE-213	Computer Architecture	3.00	CSE 315	Computer Architecture and Design	3
44.	CSE-319	Software Engineering	3.00	CSE 317	Software Engineering and Design Patterns	3
45.	CSE-322	Software Development Sessional	1.50	CSE 318	Software Engineering and Design Patterns Sessional	1
46.	CSE-217	Theory of Computation	3.00	CSE 319	Theory of Computation	3
47.	CSE-317	Data Communication	3.00	CSE 321	Data Communications	3
48.	CSE-318	Data Communication Sessional	0.75	CSE 322	Data Communications Sessional	1
49.	CSE- 410	Software Development for Web Apps	1.50	CSE 323	Web Technologies	2
50.	-	-	-	CSE 324	Web Technologies Sessional	1.5
51.	CSE-305	Microprocessors and Micro-controller	3.00	CSE 325	Microprocessor, Assembly Language and Computer Interfacing	3
52.	CSE-306	Microprocessors and Micro-controller Sessional	0.75	CSE 326	Microprocessor, Assembly Language and Computer Interfacing Sessional	1.5
53.	CSE-309	Computer Network	3.00	CSE 327	Computer Networks	3

54.	CSE-310	Computer Network Sessional	1.50	CSE 328	Computer Networks Sessional	1.5
55.	-	-	-	CSE 329	Management Information System	2
56.	-	-	-	CSE 301	Integrated Design Project	2
57.	-	-	-	CSE 411	Machine Learning	2
58.	-	-	-	CSE 412	Machine Learning Sessional	1
59.	CSE-303	Compiler	3.00	CSE 413	Compiler Design and Construction	3
60.	CSE-304	Compiler Sessional	0.75	CSE 414	Compiler Design and Construction Sessional	1
61.	-	-	-	CSE 415	Research Methodology and Complex Engineering Activities	2
62.	CSE-403	Artificial Intelligence	3.00	CSE 421	Artificial Intelligence	3
63.	CSE-404	Artificial Intelligence Sessional	0.75	CSE 422	Artificial Intelligence Sessional	1
64.	CSE-400	Project and Thesis	1.50	CSE 401	Capstone Project (Part 1)	2
65.	CSE-400	Project and Thesis*	3.00	CSE 402	Capstone Project (Part 2)	4
66.	CSE-4XO	Option-I	3.00	CSE ***	Elective 1	3
67.	CSE-4XO	Option-II	3.00	CSE ***	Elective 2	2
68.	CSE-4XE	Option-II Sessional	0.75	CSE ***	Elective 2 Sessional	1
69.	-	-	-	CSE ***	Elective 3	3
70.	CSE-101	Computer Fundamentals and Basic Programming	3.00	-	-	-
71.	HUM-102	English Sessional	1.50	-	-	-
72.	ME-181	Basic Mechanical Engineering	2.00	-	-	-
73.	EEE-269	Electrical Drives and Instrumentation	3.00	-	-	-
74.	EEE-270	Electrical Drives and Instrumentation Sessional	0.75	-	-	-
75.	CSE-210	Assembly Language Programming Sessional	1.50	-	-	-
76.	CSE-211	Digital Electronics and Pulse Technique	3.00	-	-	-
77.	CSE-212	Digital Electronics and Pulse Technique Sessional	0.75	-	-	-
78.	CSE-220	Object Oriented Programming language Sessional-II (Java)	0.75	-	-	-
79.	MATH-247	Mathematics-IV (Complex Variable and Laplace Transform)	3.00	-	-	-
80.	CSE-315	Digital System Design	3.00	-	-	-
81.	CSE-316	Digital System Design Sessional	0.75	-	-	-
82.	CSE-350	Industrial Training	1.00	-	-	-
83.	CSE-401	System Analysis, Design and Development	3.00	-	-	-
84.	CSE-402	System Analysis, Design and Development Sessional	0.75	-	-	-
85.	CSE-405	Computer Interfacing	3.00	-	-	-
86.	CSE-406	Computer Interfacing Sessional	0.75	-	-	-
87.	CSE-407	Queueing Systems	3.00	-	-	-
88.	CSE-411	VLSI Design	3.00	-	-	-
89.	CSE-412	VLSI Design Sessional	0.75	-	-	-
90.	CSE-413	Computer Graphics	3.00	-	-	-

91.	CSE-414	Computer Graphics Sessional	0.75	-	-	-
92.	HUM-415	Financial and managerial Accounting	2.00	-	-	-

APPENDIX B
CATEGORY OF COURSES

UGC Suggestion

Type	**Credit Hours (in percentage of total credit hours)**	**Remarks**
Language & General Education	12-15%	Compulsory: English – One Course Bengali – One Course
Basic Science	8-10%	Compulsory: Physics – One Course & Lab Chemistry – One Course
Mathematics	8-10%	Topics: Differential and Integral Calculus, Probability and Statistics, Complex Variables, Vector Analysis, Differential Equations, Coordinate Geometry, Linear Algebra.
Other Engineering	8-10%	Introduction to Electrical Engineering, Electronic Devices and circuits & Pulse Techniques, Electrical Drives and Instrumentation, Engineering Drawing
Core Subjects	40-50%	Programming, Hardware System, Logics and Algorithms, Network Systems, Software System and Engineering, Computer and System Security etc.
Elective Subjects	12-15%	Computing Theory, Communication and Networking Systems, Data Science, Software Engineering, Hardware, ICT.

BAIUST Proposal

Type	**Credit Hours (in percentage of total credit hours)**	**Remarks**
Language Courses	5-6% (07Cr)	As per UGC Recommendation
General Education Courses	11% (15Cr)	
Basic Science Courses	8% (11Cr)	As per UGC Recommendation
Mathematics Courses	9% (12Cr)	As per UGC Recommendation
Other Engineering Courses	8% (11Cr)	As per UGC Recommendation
Program Core Courses	57% (89Cr)	External Subject Expert rearranged some elective courses as core courses based on that subject's demand.
Elective Courses	6% (09Cr)	
Total	**154 Credit**	